

AUTOM8 – Autonomous Stair-Climbing Delivery Robot

Comprehensive Technical Documentation

Independent Post-Graduation Technical Documentation and Engineering Analysis

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Document Overview

This document presents the comprehensive technical documentation of AUTOM8 – Autonomous Stair-Climbing Delivery Robot, originally completed as a Bachelor of Technology Final Year Team Project.

Following my graduation in June 2025, I independently undertook the task of preparing this comprehensive engineering documentation. This work was completed solely by me and extends beyond the original academic project report. Its purpose is to systematically document the engineering knowledge, design methodology, prototype implementation, and technical understanding gained throughout the project.

The documentation has been significantly expanded through post-graduation technical analysis, engineering calculations, repeated prototype evaluation, practical observations, and detailed performance assessment. During this process, I analyzed the robot's behaviour under different operating conditions, documented experimental results, identified design limitations, evaluated engineering trade-offs, and proposed practical future improvements based on the findings.

This document includes the project's background and motivation, literature review, complete system architecture, mechanical design, electrical and control systems, stair-climbing mechanism, detailed explanation of every subsystem and component, prototype implementation, testing procedures, experimental results, engineering calculations and derivations, cost analysis, identified limitations, future development possibilities, and reflections on areas where I require additional technical knowledge for further improvement.

While the original prototype was developed as a team project, this comprehensive technical documentation, post-graduation engineering analysis, additional calculations, expanded explanations, performance evaluation, limitation assessment, and future development study were independently prepared by me after graduation as part of my professional portfolio and continuous engineering learning.

ABSTRACT

AUTOM8 — Campus Delivery Robot

Proof-of-Concept Prototype

B. Tech Robotics and Automation Engineering | Parul University, Vadodara

AUTOM8 is a stair-climbing campus delivery robot designed, fabricated, and tested as a final year B. Tech project in Robotics and Automation Engineering at Parul University, Vadodara. The project addresses a practical gap in Indian institutional environments: very limited low-cost, stair-capable delivery robots currently exist for multi-floor campus deployment. The total prototype cost is approximately ₹33,000 (€370), funded entirely by the student team. AUTOM8 is presented as a proof-of-concept prototype demonstrating the mechanical and control feasibility of the proposed approach.

The central technical contribution is a sequential gantry lift mechanism applied to a three-body segmented aluminium extrusion chassis. The robot consists of three independent boxes — front, middle, and back — each 20 cm wide and 36 cm tall, fabricated from 20×20 mm T-slot and V-slot aluminium profiles. The three boxes are connected by four GT2 belt-driven V-slot carriage assemblies driven by four 100 RPM Johnson DC gantry motors mounted on the middle box. During stair climbing, the three boxes are lifted onto successive stair treads in sequence through a six-phase control loop. The compatible stair geometry range is tread depth 280–400 mm and riser height 150–220 mm, covering the full range specified by the National Building Code of India (NBC IS 1644), with reference to IS standards) for educational buildings.

The stair-climbing trigger logic was developed through three successive stages: timing-based control, IR sensor triggering, and HC-SR04 ultrasonic sensor triggering. IR sensors were found unsuitable due to sensitivity to ambient lighting. The final prototype implementation uses timing-based control — since stair dimensions are fixed and uniform throughout the campus environment, precise timing sequences reliably trigger each phase without sensor dependency. HC-SR04 ultrasonic triggering was also fully developed and validated, and can be activated when the robot encounters staircases of unknown or varying dimensions outside the standard campus environment. The final implementation successfully completed the six-phase stair-climbing sequence across multiple test runs. Observed limitations include sensor response delay at phase transitions, middle box tilt during the constraint-reaction lift phase, and GT2 belt elongation after repeated cycles.

Eight 200 RPM Johnson DC wheel motors provide flat terrain drive. Navigation is implemented as pre-programmed encoder-based path execution. An MPU-6050 IMU provides real-time heading correction during flat terrain motion. Obstacle detection uses three HC-SR04 sensors on the front box for flat terrain navigation, halting the robot and activating a buzzer when an obstruction is detected within range, resuming automatically once the path is clear. Additionally, three HC-SR04 sensors at the bottom of each box serve as stair detection triggers for the gantry sequence. All control logic runs on an Arduino Mega 2560, powered by a custom 3S4P lithium-ion battery pack at 11.1 V nominal and 8800 mAh capacity.

Design calculations confirm a gantry motor torque safety factor of not less than $7.5\times$ across all lifting phases and voltage conditions (12 V, 10 V, 9 V). Payload capacity is stated as around 7 kg for all operations, governed by the GT2 belt tensile working load limit. All subsystems - stair climbing, obstacle detection, heading correction, and encoder-based navigation - were individually developed and validated. A fully integrated end-to-end deployment test is identified as the immediate next step. A prototype web interface was also designed to demonstrate the intended ordering workflow.

Proposed future work includes closed-loop gantry position control, MPU-6050 yaw integration during stair climbing, dual encoder for improved odometry, upgraded motor drivers, a payload compartment enclosure, and an autonomous docking charging station. While the immediate deployment context is the academic campus, the mechanical architecture is intentionally generalisable to other multi-floor indoor environments whose staircase dimensions fall within the validated NBC compatibility range.

Keywords: *Campus delivery robot, stair-climbing mechanism, sequential gantry lift, three-body segmented chassis, HC-SR04 ultrasonic sensing, timing-based control, encoder-based navigation, aluminium extrusion, Arduino Mega, GT2 belt drive, reactive obstacle detection and halt, proof-of-concept prototype.*

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CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

The integration of robotic systems into everyday logistics has accelerated considerably over the past decade. Robotic platforms are now routinely deployed in warehouses, airports, and controlled commercial environments to reduce manual handling, improve delivery consistency, and lower operational costs. Yet within Indian academic institutions — where the daily movement of parcels, library materials, food, and administrative documents between multiple floors is a persistent operational burden — automation has seen virtually no practical adoption. This gap arises not from lack of demand, but from the combined effect of two barriers: the high cost of commercially available robotic platforms, and their near-universal inability to navigate staircases.

Ground-level delivery robots, including globally recognised platforms such as Starship Technologies, are designed exclusively for flat terrain and cannot ascend or descend stairs [9]. At an approximate unit cost of USD 5,500 — equivalent to around ₹4,60,000 — even these ground-level systems are beyond the procurement budgets of most Indian educational institutions. At the other end of the capability spectrum, legged robotic platforms capable of stair navigation, such as those developed by Boston Dynamics, are priced at tens of thousands of dollars, making them entirely impractical for institutional deployment in the Indian context. Within India, commercially available service robots — including food delivery platforms such as the BellaBot (approximately ₹9,50,000) and the Milagrow StarButler (approximately ₹6,99,000) — are similarly restricted to single-floor operation [10]. None of the commercially available indoor delivery robots in the Indian market currently offer stair-climbing capability.

The technical challenge of stair climbing for wheeled delivery robots remains an active and unresolved area of research globally. At CES 2024, MOBINN — a stair-climbing delivery robot backed by Hyundai — attracted significant attention specifically because it attempted to address what was widely described as one of the fundamental unsolved limitations of delivery robots: the inability to navigate stairs [11]. That a well-resourced industrial research effort was still working on this problem in 2024 reflects the genuine mechanical and control complexity involved. For the Indian institutional market, where elevator infrastructure is inconsistent, buildings frequently lack accessibility provisions, and procurement budgets are constrained, the absence of a low-cost stair-capable robot represents a real and addressable gap.

It is from this context that the AUTOM8 project was conceived. The robot was designed, fabricated, and tested entirely from student-funded resources, with a total prototype cost of approximately ₹33,000 (€370). This budget constraint was treated not as a limitation to be apologised for, but as a core design requirement. A solution that cannot be realistically adopted by the institutions that need it most is not a useful solution. The central aim of the project is to demonstrate that a mechanically sound, stair-capable delivery robot can be built at a cost that makes it genuinely deployable in Indian institutional environments.

AUTOM8 is a three-body wheeled robot with a gantry-based lifting mechanism, designed to climb staircases conforming to the dimensional requirements of the National Building Code of India [12]. The robot incorporates ultrasonic sensors for stair detection and obstacle detection, a rotary encoder for distance-based path execution, and an MPU-6050 inertial measurement unit for real-time heading correction during flat terrain motion. The stair-climbing trigger logic was developed through three successive stages: timing-based control, IR sensor triggering, and HC-SR04 ultrasonic sensor triggering. The final prototype implementation uses timing-based control for the fixed campus staircase environment; the ultrasonic-based triggering approach was also fully developed and is available for deployment on staircases of unknown or varying dimensions. Each of these subsystems was individually developed and validated. While a fully integrated end-to-end deployment prototype is identified as future work, the core stair-climbing mechanism — the primary technical contribution of this project — has been physically demonstrated.

Although the immediate application context of AUTOM8 is the academic campus, the mechanical architecture is designed with adaptability in mind. The staircase geometry encountered in hospitals, restaurants, hotels, and corporate office buildings falls within the same NBC dimensional standards as institutional staircases. Where stair dimensions vary between environments, the ultrasonic-based triggering system can be activated without requiring fundamental reprogramming of the lifting sequence. With appropriate upgrades to the navigation system, payload enclosure, and sensor suite — made possible with additional funding — the same platform is applicable across a range of multi-floor indoor environments. This scalability is a deliberate design consideration, not a retrospective observation.

1.2 Problem Statement

Multi-floor delivery operations within Indian academic institutions rely predominantly on manual effort. The recurring movement of items between floors — across buildings that frequently lack reliable elevator access — represents an avoidable workload on institutional staff. Commercially available autonomous ground-level robots cannot navigate staircases and are therefore restricted to single-floor

operation. Existing stair-capable robotic research platforms are mechanically complex and economically inaccessible for institutional deployment at scale.

This project addresses the following specific problem: the absence of a low-cost, stair-capable delivery robot that can operate across the multiple floors of a standard Indian campus building using staircases conforming to NBC IS 1644 dimensional specifications, without dependency on elevator access, permanent infrastructure modification, or budgets beyond the reach of Indian educational institutions.

1.3 Aim and Objectives

The aim of this project is to design, fabricate, and validate a proof-of-concept delivery robot capable of navigating multi-floor campus environments through stair climbing, incorporating real-time heading correction, obstacle detection and halt, and encoder-based path execution, within a self-funded budget constraint.

The specific objectives of the project are as follows:

- To design and fabricate a three-body modular robot frame capable of supporting a calculated payload capacity while maintaining structural integrity throughout the stair-climbing sequence.
- To design and implement a gantry-based lifting mechanism that independently elevates each body segment to the height of a standard stair riser using DC motors and belt-driven linear rails.
- To develop and validate a stair detection and sequential triggering algorithm, with timing-based control as the primary implementation for fixed-dimension campus staircases, and HC-SR04 ultrasonic sensor triggering as the developed alternative for variable staircase environments.
- To implement real-time heading correction using an MPU-6050 inertial measurement unit, continuously adjusting motor output to maintain straight-line motion during flat terrain navigation.
- To implement encoder-based pre-programmed path execution for flat terrain navigation between defined delivery waypoints.
- To integrate an ultrasonic obstacle detection system that halts robot motion and activates an audible alert upon detecting an obstruction within a defined range, and resumes automatically once the path is clear.
- To validate the stair-climbing mechanism against the dimensional specifications of the National Building Code of India (NBC IS 1644) and confirm compatibility with standard institutional staircase geometries.

- To demonstrate all subsystems within a self-funded prototype budget of approximately ₹33,000 (€370), establishing a reference cost point for low-budget institutional robotics development.

1.4 Scope of the Project

This project covers the mechanical design, fabrication, electrical integration, sensor implementation, and control firmware of a proof-of-concept stair-climbing delivery robot. The primary test environment is Parul University, Vadodara, with staircases conforming to NBC IS 1644 specifications. The validated stair compatibility range is tread depth 280–400 mm and riser height 150–220 mm, covering the full spectrum of standard institutional staircases in India.

The following are within the scope of this prototype: mechanical frame fabrication; gantry mechanism design and physical stair-climbing demonstration; timing-based and ultrasonic-based stair detection and sequential trigger logic; real-time heading correction using MPU-6050; encoder-based fixed-route path execution; and ultrasonic obstacle detection with halt and buzzer alert. All subsystems were individually developed and validated. Structural payload capacity was validated through torque and load calculations. It is noted that due to the nature of incremental development and budget constraints, the subsystems were tested independently; a fully integrated end-to-end test is identified as an immediate next step.

The following are outside the scope of this prototype and are identified as future development work: fully autonomous dynamic path planning, simultaneous localisation and mapping (SLAM), a functional mobile application for order placement and delivery tracking, and integration with campus network infrastructure. These components were conceptually designed and are described in **Chapter 7** as proposed future work pending further funding and development.

While the immediate deployment context is an academic campus, the mechanical design is intentionally generalisable to other multi-floor indoor environments. Hospitals, restaurants, hotels, and corporate office buildings employ staircases within the same NBC dimensional range. Future adaptation of the AUTOM8 platform to these environments is anticipated to require modifications primarily to the navigation software and payload interface, leaving the core mechanical and lifting architecture unchanged.

1.5 Report Organisation

This report is organised into eight chapters as follows:

Chapter	Content
Chapter 1	Introduction — Background and motivation, problem statement, objectives, scope, and report organisation.
Chapter 2	Literature Review — Survey of existing delivery and stair-climbing robots, identification of the research gap, and positioning of AUTOM8 relative to existing work.
Chapter 3	Mechanical Design — Frame design, material selection, weight analysis, gantry mechanism design, and wheel and motor selection.
Chapter 4	Electrical and Control System — Sensor suite, motor drivers, battery and power architecture, Arduino Mega 2560 pin configuration, and control firmware overview.
Chapter 5	Stair Climbing Mechanism — Phase sequence design, geometric stair compatibility analysis, NBC compliance, timing-based and ultrasonic-based trigger logic.
Chapter 6	Prototype Implementation and Testing — Hardware assembly, subsystem test results, observed behaviour, known limitations, and payload analysis.
Chapter 7	Proposed Future Work — Fully autonomous navigation, mobile application, SLAM integration, multi-environment adaptation, upgraded sensor suite.
Chapter 8	Conclusion — Summary of contributions, reflection on objectives achieved, and closing remarks.

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CHAPTER 2

LITERATURE REVIEW

2.1 Introduction to the Review

The development of AUTOM8 draws from three areas of existing work: indoor and campus delivery robotics, stair-climbing mechanism design, and sensor-based navigation for mobile platforms. Each area has been studied with reasonable depth in the literature, but their intersection — a robot that is simultaneously low-cost, mechanically capable of stair negotiation, and practically deployable in an Indian institutional setting — has received limited direct attention. This chapter reviews the most relevant published work across these domains, notes what each system achieved and where its limitations lie, and identifies the specific gap that the AUTOM8 prototype was designed to address.

2.2 Campus and Indoor Delivery Robots

Research into delivery robots for indoor and campus environments has grown steadily over the past decade. Two systems are of particular relevance to this project.

Maan, Madiwale, and Bishnoi (2021) presented Xenia, a robot developed for food delivery and sanitisation in hospital environments [1]. The system was motivated by the need to reduce human-to-human contact during the COVID-19 pandemic, and employed ultrasonic sensors for obstacle detection, MATLAB/Simulink for path planning, and a structured payload compartment for carrying food and sanitisation materials. Within hospital corridors, the robot demonstrated functional navigation and delivery capability. Its limitation is clear: the system was designed entirely for single-floor operation. The authors acknowledged that multi-floor deployment would require elevator integration, which was left as future work. Xenia is relevant to AUTOM8 primarily because it validates the hospital and indoor delivery context as a genuine application domain, and because it illustrates that even purpose-built delivery robots from published research stop short of addressing the staircase problem.

A more technically ambitious approach to campus delivery was presented by Gujarathi et al. (2021), who developed an autonomous outdoor delivery robot for the VNIT campus [2]. The system used GPS, LiDAR, and an inertial measurement unit with extended Kalman filter fusion for localisation, the A* algorithm for route planning, and an ENet-based semantic segmentation pipeline for obstacle and road

detection. The resulting platform achieved a meaningful level of outdoor navigation autonomy. Despite this, the system shares the fundamental limitation of Xenia: it operates on a single navigable plane and makes no provision for floor transitions. Additionally, the hardware and computational requirements of the system — LiDAR, GPU-accelerated deep learning, and ROS — place it well outside the cost range of a self-funded student project. AUTOM8 draws a deliberate contrast with this approach by prioritising sensor simplicity and cost accessibility over navigation sophistication, while focusing its technical contribution on the stair-climbing capability that both of these systems lack.

2.3 Stair-Climbing Mechanisms for Mobile Robots

Stair climbing imposes simultaneous mechanical demands on a robot that are difficult to satisfy with a single locomotion architecture. The robot must support its weight on flat ground, elevate its contact points across a vertical riser, maintain stability through the transition, and recover to normal operation on the upper surface. Different design approaches handle these demands in fundamentally different ways, each with distinct trade-offs.

A comprehensive review of stair-climbing robots published in IEEE Access (2023) categorised existing approaches into legged, tracked, wheel-legged hybrid, and purpose-built wheeled mechanisms [3]. The survey found that legged robots, while mechanically capable, carry high energy consumption, complexity, and cost that make them impractical for delivery applications. Tracked systems offer stability on uneven terrain but are poorly suited to smooth indoor floors and impose high rolling resistance during flat operation. Wheel-legged hybrids have attracted research interest but have not achieved meaningful market adoption in the delivery sector. The survey concluded that no widely adopted cost-effective solution had emerged as of the review date — a finding that directly motivates the approach taken in AUTOM8.

A closely related system in terms of mechanical concept is sTetro, a modular stair-climbing robot developed for autonomous staircase cleaning [4]. The sTetro design is notable because it shares AUTOM8's fundamental architectural decision: the robot body is divided into three independent cuboid segments, with interconnecting mechanisms that allow each segment to be raised or repositioned independently during stair negotiation. This three-body approach addresses the same core problem that AUTOM8 targets — how to transition a multi-segment robot across a stair riser without requiring continuous locomotion reconfiguration. In sTetro's case, the interconnecting mechanism is a vertical conveyor belt system, and the robot uses an RGB-D camera and Jetson Nano processor for perception and control. Its application is cleaning rather than delivery, and its computational and sensor requirements are considerably higher than those of AUTOM8. Nevertheless, the convergence of two independently

developed systems on a three-body segmented architecture for stair climbing provides meaningful validation that this structural approach is mechanically sound. The primary difference in AUTOM8 is the use of a gantry-based linear lifting mechanism — assembled from commercially available GT2 belt, V-slot rails, and standard DC motors — rather than a conveyor belt interconnect, which reduces fabrication complexity and cost substantially.

More recently, Kim, Chung, and Cho (2024) presented MOBINN, a stair-climbing delivery robot backed by Hyundai, which used novel flexible wheels to absorb the geometric discontinuity of stair risers [5]. The system drew attention at CES 2024 as an example of the stair-climbing limitation being taken seriously at an industrial research level. The flexible wheel design is innovative but requires custom-manufactured components and a sophisticated control system that are not reproducible at student project scale. MOBINN is relevant here not as a design model but as recent evidence that the problem AUTOM8 addresses — stair-capable low-cost indoor delivery — is considered a genuine open challenge even by well-resourced industrial research teams.

The mechanism adopted in AUTOM8 — a sequential gantry lift in which each of the three body segments is elevated independently by a dedicated motor-driven linear actuator — was chosen specifically because it decouples stair climbing from flat terrain locomotion. The wheels are optimised for flat terrain and play no role during stair negotiation. The gantry system handles all vertical displacement. This separation of function is a deliberate design decision that trades climbing speed for mechanical simplicity and component accessibility. The stair-climbing mechanism was validated through physical prototype testing; the observed behaviour, including a temporary tilt during middle body segment elevation due to centre of mass shift, is documented in **Chapter 6**. The robot self-corrects as the sequence completes and contact is re-established on the upper stair surface.

2.4 Sensor Systems for Mobile Robot Navigation and Detection

The sensor architecture of AUTOM8 comprises six HC-SR04 ultrasonic sensors — three positioned for stair detection as sequential triggers for the gantry mechanism, and three positioned for obstacle detection and halt during flat terrain navigation — a rotary encoder for distance-based path execution, and an MPU-6050 inertial measurement unit for real-time heading correction during flat terrain motion. It is important to note that due to the nature of the prototype, these systems were tested and validated independently rather than simultaneously. In particular, the MPU-6050 heading correction system is deactivated during stair climbing, as the physical tilt of the robot body during gantry operation would produce erroneous orientation readings. Each sensor system was selected based on its established role in the mobile robotics literature and its availability within the project's budget constraints.

The HC-SR04 ultrasonic sensor has been widely used in low-cost mobile robotics for obstacle and proximity detection. A study presented at IEEE SCEECs (2023) evaluated HC-SR04-based obstacle detection on an Arduino-controlled platform and confirmed its practical suitability for indoor environments where obstacles are predominantly large solid objects [6]. The authors noted that while ultrasonic sensing is susceptible to specular reflection from very smooth surfaces and lacks the resolution of LiDAR, it provides a reliable and cost-effective solution for indoor detection tasks. In AUTOM8, the HC-SR04 sensors serve two distinct roles: three sensors detect stair risers as proximity triggers for the gantry sequence, while three additional sensors detect obstacles during flat terrain navigation and trigger an immediate halt with buzzer alert. The use of the same sensor type for both functions was a deliberate cost and simplification decision.

The combination of an MPU-6050 IMU and a rotary encoder for motion control was evaluated in published work by Sagita et al. (2024) in *Control Systems and Optimization Letters*, which implemented this sensor pairing on a wheeled mobile robot for orientation maintenance and distance tracking [7]. The study confirmed that this combination is adequate for short-range fixed-route navigation without requiring full sensor fusion pipelines. In AUTOM8, the MPU-6050 is used exclusively during flat terrain segments to maintain straight-line heading. Its deactivation during stair climbing is a known limitation of the current prototype and is identified as an area for improvement in **Chapter 7**.

The use of wheel encoder odometry as the primary distance measurement method for indoor navigation is well supported in the literature. Mohamed et al. (2019) reviewed odometry approaches for autonomous navigation across multiple platforms in *IEEE Access* and confirmed that wheel encoder odometry remains the practical standard for short-range indoor tasks where cost, simplicity, and real-time performance are priorities [8]. Cumulative drift over longer distances is a known limitation of encoder-only odometry, which is why the navigation implementation in AUTOM8 is restricted to pre-measured fixed routes within a defined campus area rather than dynamic path planning.

2.5 Comparative Analysis of Reviewed Systems

Table 2.1 presents a structured comparison of the reviewed systems against AUTOM8 across parameters relevant to the design objectives of this project. The table is intended to show where AUTOM8 sits relative to existing work, not to overstate its capabilities. The limitations column for AUTOM8 reflects the current prototype status honestly.

Table 2.1: Comparison of reviewed delivery and stair-climbing robot systems

System	Stair Capable	Approx. Cost	Environment	Navigation Type	Key Limitation
Xenia [1]	No	Not disclosed	Hospital indoor	MATLAB-SIMULINK waypoint	Single floor only; elevator dependency for multi-floor
VNIT Robot [2]	No	High — LiDAR, GPU	Outdoor campus	GPS, A*, EKF, deep learning	Ground level only; high cost; GPU dependency
Starship Tech.	No	~₹4,60,000/unit	Outdoor pavement	Proprietary autonomous	Flat terrain only; cost prohibitive for Indian institutions
sTetro [4]	Yes	High — Jetson Nano, RGB-D	Indoor staircase	RGB-D camera, conveyor belt mechanism	Cleaning only; high cost; complex interconnect
MOBINN [5]	Yes	Very high — industrial	Indoor/outdoor	Industrial autonomous	Custom flexible wheels; industrial cost; not reproducible at student level
AUTOM8 (This work)	Yes (validated)	~₹33,000 (€370) prototype	Indoor campus; adaptable	Encoder + MPU-6050 (independent modes)	Subsystems tested independently; MPU-6050 inactive during stair climbing; no full integration test completed

2.6 Research Gap and Positioning of AUTOM8

The review presented in this chapter points to a consistent and well-defined gap. Delivery robots designed for indoor and campus environments have made meaningful progress in navigation and obstacle detection, but remain universally restricted to flat, single-floor operation. Stair-climbing robots exist in the literature, but they rely on either expensive specialised hardware — as with MOBINN — or complex interconnect mechanisms designed for non-delivery applications — as with sTetro. No published system reviewed here combines stair-climbing capability with a prototype cost accessible to a self-funded undergraduate project in the Indian context.

To the best of the authors' knowledge, the following specific gaps remain unaddressed in the published literature at the time of this project. First, no low-cost stair-climbing delivery robot has been developed and validated against the staircase dimensional standards prescribed by the National Building Code of India. Second, the three-body segmented architecture — while independently arriving at a similar concept to sTetro — has not previously been applied in the context of delivery with a gantry-based linear lift mechanism assembled from off-the-shelf components. Third, the use of ultrasonic proximity sensing as the trigger mechanism for an adaptive sequential stair-climbing sequence — without vision systems, depth cameras, or prior knowledge of stair geometry — has not been documented for a delivery-oriented platform in the reviewed literature.

AUTOM8 does not claim to fully resolve these gaps. What the prototype demonstrates is that a three-body gantry-based stair-climbing mechanism can be physically realised at low cost, that sensor-based sequential triggering using HC-SR04 sensors is a viable approach for stair detection, and that the resulting operating envelope is geometrically compatible with NBC IS 1644 staircase specifications. The navigation and integration aspects of the system — encoder-based path execution and MPU-6050 heading correction — are validated as independent subsystems. Their combined operation with the stair-climbing mechanism remains as identified future work, as does the development of a fully autonomous navigation stack.

The broader relevance of this work extends beyond the campus context. The staircase geometry found in hospitals, restaurants, hotels, and corporate office buildings in India falls within the same NBC dimensional range as institutional staircases. The mechanical architecture of AUTOM8 is therefore applicable, with appropriate modifications to the navigation and payload systems, to any multi-floor indoor environment where stair access is the primary floor transition method. This generalisability is a design intention, not a claim of demonstrated deployment.

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CHAPTER 3

MECHANICAL DESIGN

3.1 Overview of Mechanical Architecture

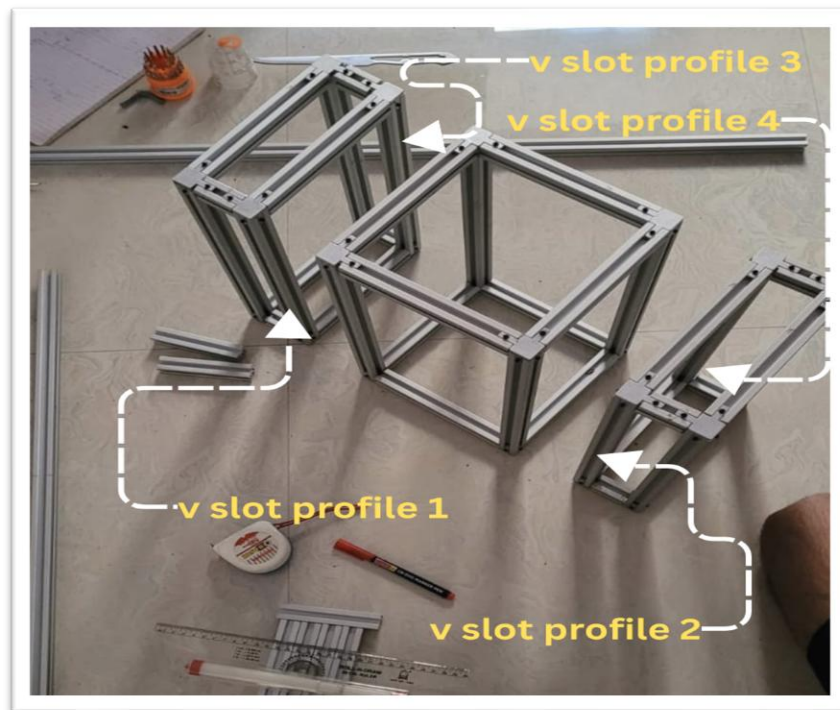
The mechanical design of AUTOM8 is shaped by a single governing constraint: the robot must be capable of negotiating a staircase riser while also functioning as a stable, wheeled delivery platform on flat terrain. These two requirements pull in opposite directions. A compact, low-profile robot is well suited to flat terrain navigation but cannot clear a riser without some form of active lifting. A robot with large-diameter wheels or leg-like appendages can manage stairs passively but performs poorly on smooth indoor floors and becomes difficult to fabricate at low cost.

The design response to this constraint is a segmented three-body architecture in which flat terrain locomotion and stair negotiation are handled by entirely separate mechanisms. Each of the three body sections — referred to throughout this report as the front box, middle box, and back box — is a self-contained structural unit mounted on its own pair of wheels. The boxes are not rigidly joined to one another. Instead, they are connected exclusively through the gantry mechanism: aluminium gantry plates fixed to the middle box carry POM V-wheels that engage with V-slot rails mounted on the inner faces of the front and back boxes. This arrangement allows either the outer boxes or the middle box itself to be raised vertically along these rails, depending on which sections are resting on a fixed surface and which are free to move. During **Phase 1** of the stair-climbing sequence, the front box is lifted while the middle and back boxes remain grounded. In **Phase 3**, all four gantry motors run simultaneously with reversed drive direction; at this point both outer boxes are resting stably on stair surfaces, so their rails act as fixed columns — the reaction force transfers through the carriage plates and raises the middle box upward between them. This dual-direction lifting capability — outer boxes rising against a stationary middle box, and the middle box rising against fixed outer boxes — is the core mechanical principle of the sequential stair-climbing architecture.

The choice of 20×20mm aluminium extrusion profile as the primary structural material was made on the basis of cost, availability, and ease of assembly. Aluminium extrusion profiles are widely available from Indian suppliers, require no welding or machining for basic assembly, and offer a favourable strength-to-weight ratio for a ground-based mobile robot at this scale. The T-slot channels on each face of the profile accept M5 T-nuts and bolts, allowing components to be mounted at any position along the length without drilling. V-slot profiles, used for the gantry rails on the front and back boxes,

include a precision-machined groove on each face that accepts the POM V-wheels of the gantry carriage plates with minimal play.

Figure 3.1 — Three-box chassis overview showing front, middle, and back boxes on flat ground



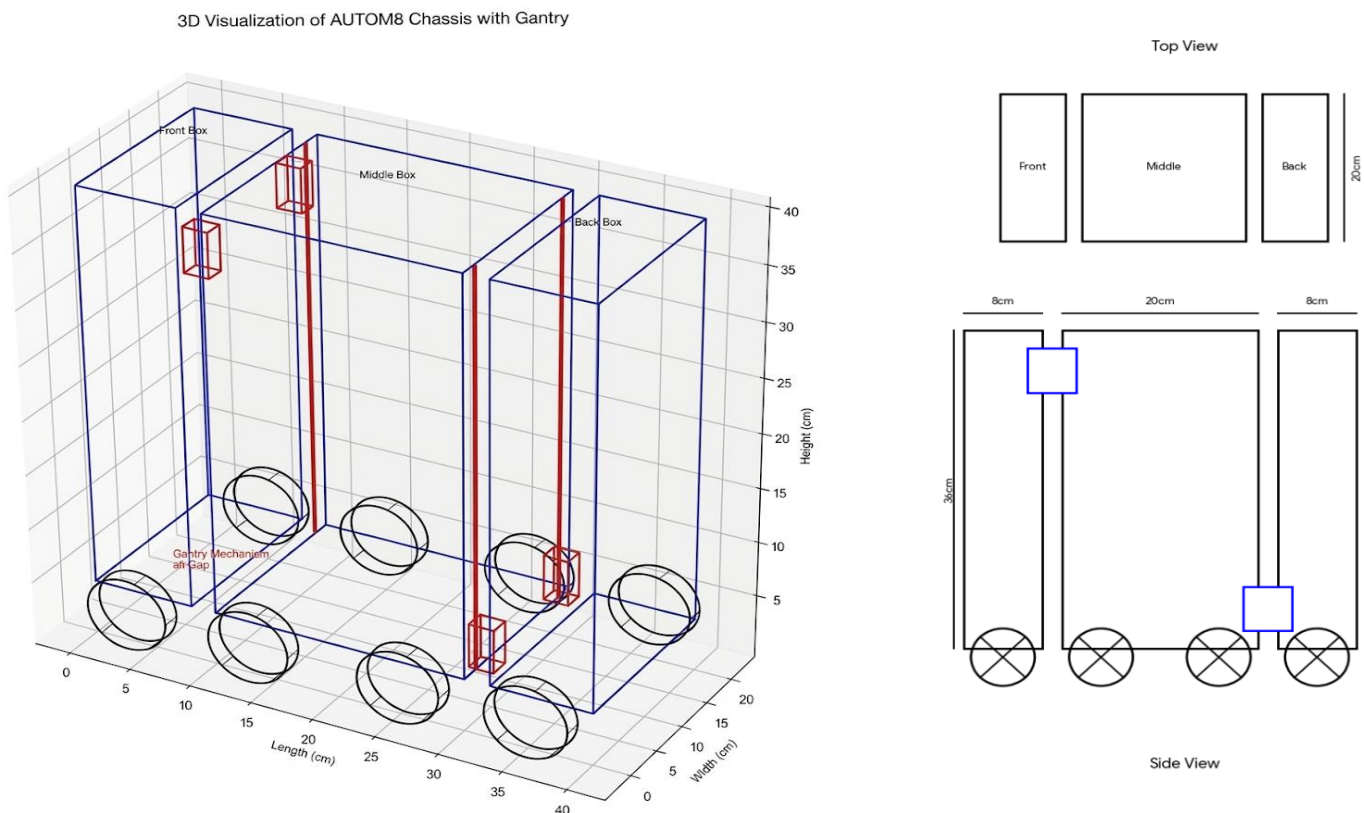
3.2 Structural Frame

The structural frame of AUTOM8 consists of three independent box assemblies, each built from 20×20mm aluminium profiles joined by three-way corner joints and inner bracket connectors secured with M5 grub screws. No welding is used anywhere in the structural assembly. All connections are mechanical, allowing the frame to be disassembled, modified, and reassembled without specialist tooling.

The front box and back box are structurally identical. Each measures 8cm in length, 20cm in width, and 36cm in height. Two vertical members on the inner face of each outer box are V-slot profiles; these serve as the running rails for the gantry carriage. The remaining thirteen profile members in each outer box are standard T-slot profiles. The front and back boxes each carry two drive wheels, one on each lateral side, and house two 200 RPM wheel motors secured by steel U-clamp mounts bolted directly to the bottom horizontal profile of the box.

The middle box is the structural and functional core of the robot. It measures 20cm in length, 20cm in width, and 36cm in height — the same height as the outer boxes, ensuring a level platform across all three sections when the robot is on flat ground. The middle box carries four drive wheels, two on each lateral side, driven by four 200 RPM motors. It also carries all four 100 RPM gantry motors and the gantry carriage plates that connect to both outer boxes. The two front-facing gantry motors are mounted on the upper portion of the middle box front face, positioned high so the belt path aligns with the upper section of the front box V-slot rails. The two rear-facing gantry motors, by contrast, are mounted on the lower portion of the middle box rear face, positioned low to align with the back box V-slot rails. This asymmetric mounting reflects the different geometric constraints of the front and back lifting paths within the assembled frame. The middle box carries the majority of the electronic components: the Arduino Mega, battery pack, four of the six L298N motor driver modules (the gantry drivers and the two middle wheel drivers), the MPU-6050 IMU, buzzer, buck converter, and both power switches. The remaining two motor driver modules — one controlling the front box wheel motors and one controlling the back box wheel motors — are mounted inside their respective boxes, keeping high-current connections short. This distribution makes the middle box the heaviest section of the robot.

Figure 3.2 — Structural frame showing aluminium extrusion assembly with V-slot and T-slot



A clearance gap of approximately 5mm is maintained between the face of the middle box and the inner face of each outer box. This gap is a direct consequence of the gantry plate geometry: the POM V-wheels on the carriage plates hold the outer box profiles at a fixed standoff distance from the carriage plate surface. The gap serves a functional purpose — it ensures that the outer boxes do not contact the middle box during vertical travel, which would generate friction and resist the lifting motion. The gap was not a designed parameter but an outcome of the component geometry, and is consistent across both inter-box connections.

3.3 Gantry Lifting Mechanism

The gantry mechanism is the central mechanical innovation of AUTOM8. It is responsible for raising and lowering the front and back boxes over stair risers during the stair-climbing sequence. Understanding its operation requires distinguishing clearly between what moves and what remains stationary during actuation.

The gantry carriage plates — four in total, two engaging the front box rails and two engaging the back box rails — are fixed rigidly to the middle box. They do not move. Each carriage plate is a 3mm black anodised 6061-T5 aluminium plate carrying four high-strength POM V-wheels with 625ZZ chrome steel bearings. The V-wheels engage the V-slot groove on the inner face of the adjacent outer box vertical profiles. When the gantry motor drives the belt, it is the outer box — with its V-slot rails — that travels vertically upward or downward, while the carriage plate and the middle box remain in place. This distinction is important: the gantry does not lift a carriage; it moves an entire outer box section along fixed rails that are themselves part of the outer box structure.

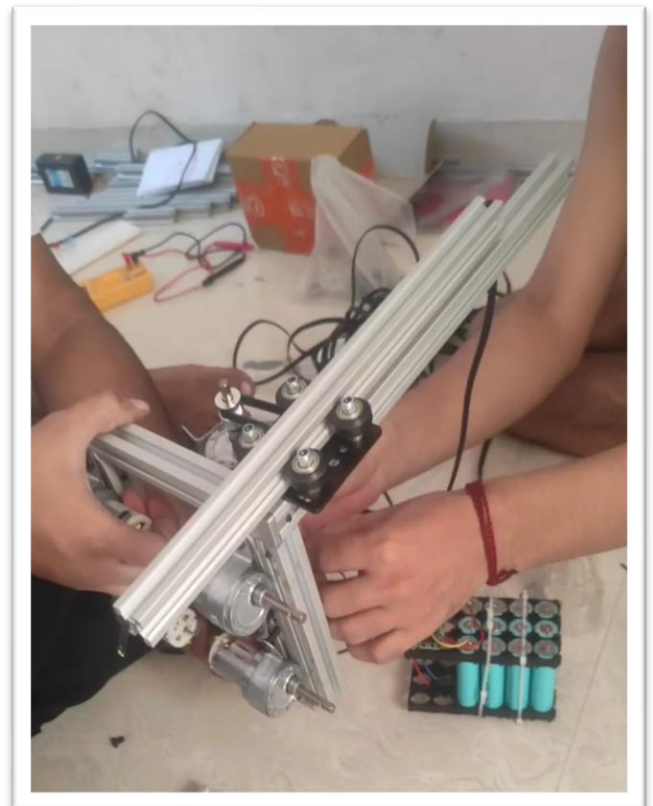
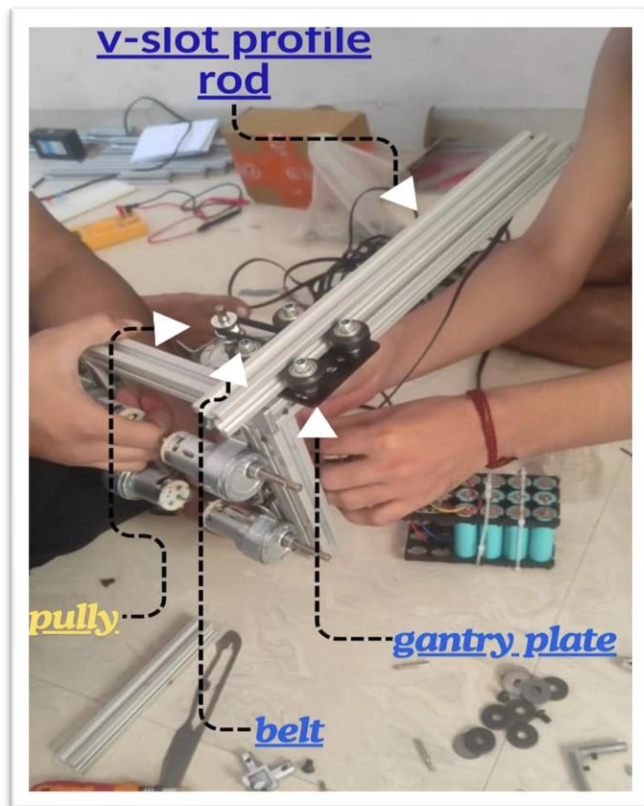
The linear actuation is provided by a GT2 rubber timing belt with a 2mm pitch and 6mm width. The belt is routed inside the inner groove of the V-slot profile on the outer box. At the lower end of the rail, the belt is anchored by a bolt that fixes it to the profile. The belt runs upward through the groove and loops around a 20-tooth GT2 aluminium pulley with a 5mm bore, mounted directly on the D-type shaft of the 100 RPM gantry motor. The motor is mounted on the middle box frame. When the motor shaft rotates in the lifting direction, the pulley pulls the belt upward; since the belt is anchored at the bottom of the V-slot rail, the rail — and the entire outer box attached to it — rises. Reversing the motor direction lowers the box back to the ground plane.

Four gantry motors are used in total. The two front-facing motors on the upper portion of the middle box drive the front box rails, and the two rear-facing motors on the lower portion of the middle box drive the back box rails. Motors on the same side are wired to rotate in opposite physical directions but drive the belt in the same linear direction, as the left and right rails of a given outer box require

mirror-image rotation to produce the same lifting motion. This is handled by the physical wiring of the motor terminals rather than by firmware logic.

A critical and often overlooked aspect of the gantry mechanism is its ability to lift the middle box itself. When both outer boxes are resting stably on stair surfaces — as they are at the start of **Phase 3** of the stair-climbing sequence — their V-slot rails become fixed vertical columns. Running all four gantry motors simultaneously in the reverse direction causes the motor pulleys to drive the belts downward relative to the rails. Since the belts are anchored at the bottom of each rail and the rails cannot move, the reaction force is transmitted through the carriage plates into the middle box frame. The middle box consequently rises vertically between the two fixed outer boxes, climbing the V-slot rails on both sides simultaneously. Once the stair detection sensor on the middle box reads greater than 3cm — indicating the middle box has risen past the riser face — the gantry motors continue briefly to cover the remaining wheel radius distance (34mm), then stop. The middle box then rests at the upper stair surface level. This three-way lifting capability (front box up, back box up, and middle box up) is what enables the robot to negotiate a full stair step as a sequential three-phase operation.

Figure 3.3 — Gantry mechanism showing GT2 belt, 20-tooth pulley, V-slot profile, and gantry plate.



The gantry lifting speed is determined by the motor speed and pulley geometry. At the rated output of 100 RPM with a 20-tooth GT2 pulley at 2mm pitch, the belt linear speed is $100 \times 20 \times 2 = 4000\text{mm}$ per minute, or approximately 66.7mm per second at full rated voltage. Actual gantry actuation duration per phase was determined during prototype testing and is recorded in **Chapter 6**. The linear actuation system was validated through independent testing of the V-slot rail and belt assembly prior to integration with the full robot.

3.4 Wheel and Drive System

AUTOM8 uses eight drive wheels in total, arranged as two on the front box, four on the middle box, and two on the back box. Each wheel is a 68mm diameter, 40mm wide robot wheel with a nylon hub and rubber tread, mounted directly on the 5.7mm shaft of a 200 RPM Johnson geared DC motor. No coupling or hub adapter is used — the wheel bore is secured directly to the shaft by the set screw included with the wheel. This direct-mount arrangement eliminates backlash and simplifies the drive assembly.

Each motor is held in position by a 2-inch steel U-clamp secured to the bottom horizontal profile of its respective box using M5 bolts and T-nuts in the profile channel. The U-clamp wraps around the cylindrical body of the gearbox and is tightened to hold the motor perpendicular to the lateral face of the frame, positioning the shaft and wheel outboard of the profile. This mounting method was selected because it requires no drilling, no custom brackets, and no precision machining — the clamp can be repositioned along the profile slot if wheel alignment needs adjustment.

The 200 RPM motors have a rated torque of 56.1 N-cm and a stall torque of 350 N-cm at 12V. With a wheel radius of 34mm, the rated torque at the wheel contact patch for a single motor is approximately $56.1/3.4 = 16.5\text{N}$ of traction force. Across eight motors operating simultaneously, the total available traction force significantly exceeds the rolling resistance of the 12kg robot on a smooth indoor surface.

During flat terrain navigation, the firmware drives the wheel motors between 70-80 PWM duty cycle determined during prototype testing (**refer to Calculation 6, Table 6.4**) to maintain a controlled and stable delivery speed.

During the stair-climbing sequence, wheel motor operation is managed by the firmware based on which phase is active. In phases where a box is being lifted, the wheels of that box are stationary. In phases where the robot advances toward the next trigger position, all eight wheels drive simultaneously. The wheel motor control during stair climbing is described in detail in **Chapter 5**.

3.5 Complete Weight Analysis

The complete weight of the AUTOM8 prototype was determined using three complementary methods. These methods were applied to different component categories depending on availability of measurement tools, access to physical parts, and the precision required for each item.

Method 1 — Density-Based Calculation (Aluminium Profiles)

Applied to all 20×20mm aluminium extrusion profiles. The theoretical linear density of the 6063-T5 alloy profile cross-section ($156 \text{ mm}^2 \times 2,700 \text{ kg/m}^3 = 0.421 \text{ kg/m}$) was determined from the published cross-section drawing of the profile. A conservative margin of 7% was added to account for manufacturing tolerances, anodising, and the slight difference in groove geometry between T-slot and V-slot variants, giving a design value of 0.45 kg/m. Profile lengths were physically measured on the fabricated frame and multiplied by this linear density. This method was used for all structural profile weight entries in **Tables 3.1** and **3.2**.

Method 2 — Direct Physical Measurement (Connectors and Fasteners)

Applied to hardware components that could not be reliably calculated from geometry alone. Three-way corner joints, inner brackets, and M5 grub screw assemblies were each weighed individually using a digital kitchen scale (resolution: 1g). The values obtained — 45g per corner joint, 30g per inner bracket, and 1.5g per grub screw set — were confirmed across multiple samples. The Indian market die-cast corner joints were found to be noticeably heavier (45g) than the aluminium-alloy variants quoted in international supplier catalogues (18–22g), making direct physical measurement necessary rather than relying on published data.

Method 3 — Supplier Specifications (Commercial Components)

Applied to all purchased electronic and electromechanical components. Weights were taken from the product specification sheets and product listings of the Indian suppliers **Robu.in** and **Robokits India**. For components where supplier-listed weights were unavailable or appeared inconsistent, an estimated value based on physical dimensions and material was used and marked with the ~ symbol throughout the tables. The grand total was cross-checked against a physical assessment of the fully assembled robot to confirm overall plausibility.

The structural frame weight is presented first as a per-box breakdown (**Table 3.1**), since each box was fabricated and verified independently. The complete prototype weight breakdown follows as **Table 3.2**.

Table 3.1: Structural frame weight breakdown by box (profiles + joints + fasteners only)

Box Section	Profiles (g)	Joints + Brackets (g)	Grub Screws (g)	Total (g)
Front Box (8 × 20 × 36 cm)	1,071	585	75	1,731
Middle Box (20 × 20 × 36 cm)	1,539	660	90	2,289
Back Box (8 × 20 × 36 cm)	1,071	585	75	1,731
Total Structural Frame	3,681	1,830	240	5,751

The structural frame accounts for approximately 48 percent of the total prototype weight, reflecting the relatively dense aluminium extrusion construction. The motor system — gantry and wheel motors combined — accounts for a further 21 percent. The battery system at 735g represents approximately 6 percent of total weight, which is consistent with typical lithium-ion battery pack proportions for a 12V mobile robot in this power class. **(from Table 3.2)**

The weight distribution across the three boxes is uneven by design. The middle box carries the four gantry motors, four-wheel motors, the battery, the Arduino, four of the six motor driver modules, and the majority of the sensor electronics. The front and back boxes each carry their own two-wheel motors and one dedicated motor driver module. This centre-heavy distribution is mechanically favourable for flat terrain stability, as the heaviest section has the most wheel contact — four wheels versus two on each outer box. However, during the stair-climbing sequence, when either outer box is elevated and its wheels leave the ground, the shift in the supported weight distribution produces observable tilting of the robot body. This behaviour is documented in **Chapter 6**.

Table 3.2: Complete weight breakdown of AUTOM8 prototype

Component	Quantity	Unit Weight (g)	Total (g)
Structural aluminium frame (all boxes)	—	—	5,751
100 RPM gantry motors	4	210	840
200 RPM wheel motors	8	210	1,680
Drive wheels (68mm)	8	100	800
Battery system (cells + BMS + holders + wiring)	—	—	735
L298N motor driver modules	6	35	210
Arduino Mega 2560	1	37	37
GT2 belt + pulleys	—	—	~90
HC-SR04 ultrasonic sensors ×6 (3 stair detection + 3 obstacle detection)	6	9	54
MPU-6050 IMU	1	~10	~10
Magnetic encoder	1	~8	~8
Buzzer	1	~5	~5
LM2596S buck converter	1	~10	~10
Steel U-clamp motor mounts	12	30	360
V-slot mini gantry plate sets	4	110	440
Rocker switches	2	~26	~53
Wiring (Dupont + 18AWG + connectors)	—	—	~100
M5 fasteners	—	—	~100
Base boards (internal mounting plates)	—	—	~260

Cable management (zip ties, tape, rubber tube)	—	—	~65
Miscellaneous hardware	—	—	~300
Decorative elements + sensor mounts	—	—	~75
Aluminium profile ballast (back box weight equalisation, ~24cm T-slot)	—	~110	~110
TOTAL (prototype, unloaded)	—	—	~12,108g ≈ 12.1 kg

3.5.1 Per-Box Component Weight Breakdown

The following three tables present a complete component-by-component weight breakdown for each box. This per-box format provides direct visibility into load distribution across the chassis and serves as reference data for the structural and dynamics analysis in Chapter 6.

Front Box — ~2,717g ≈ 2.72 kg

The front box carries the two forward drive motors, three obstacle detection sensors on the top-front face, one stair detection sensor on the bottom-front face, the magnetic encoder, and all decorative display elements. Motor Driver 1 is mounted inside the front box to minimise cable length to the front wheel motors.

Table 3.3: Front box — complete component weight breakdown

Component	Qty × Unit (g)	Total (g)	Total (kg)
Structural frame (profiles + joints + fasteners)	1 × 1,731	1,731	1.731
Wheel motors (200 RPM)	2 × 210	420	0.420
Drive wheels (68mm)	2 × 100	200	0.200
Motor mounts — U-clamps	2 × 30	60	0.060
Motor driver L298N (Driver 1)	1 × 35	35	0.035

HC-SR04 sensor (stair detection, bottom-front ×1)	1 × 9	9	0.009
HC-SR04 sensors (obstacle detection, top-front ×3)	3 × 9	27	0.027
Cardboard HC-SR04 sensor mount	1 × ~20	~20	~0.020
Magnetic Hall effect encoder	1 × ~8	~8	~0.008
Base board (8×20cm)	1 × ~30	~30	~0.030
Wooden display sheet + stickers	1 × ~55	~55	~0.055
Wiring share	—	~20	~0.020
M5 fasteners share	—	~17	~0.017
Miscellaneous hardware share	—	~60	~0.060
Zip ties + tape share	—	~15	~0.015
FRONT BOX TOTAL	—	~2,717	~2.72 kg

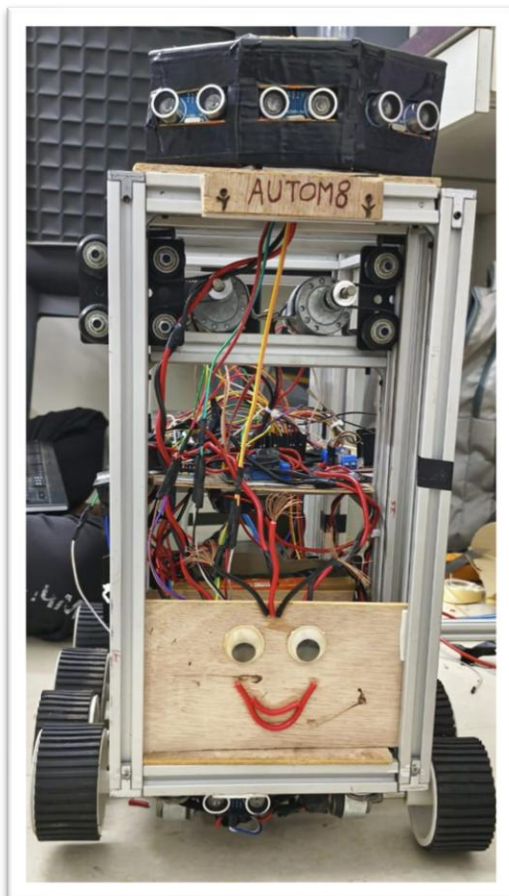


Figure 3.4 — Front box assembled view showing wheel motors, HC-SR04 obstacle detection sensors (top-front), and stair detection sensor (bottom-front).

(The buzzer response with sensor testing video can be accessed by clicking on the image)

Middle Box — ~6,623g $\approx$ 6.62 kg

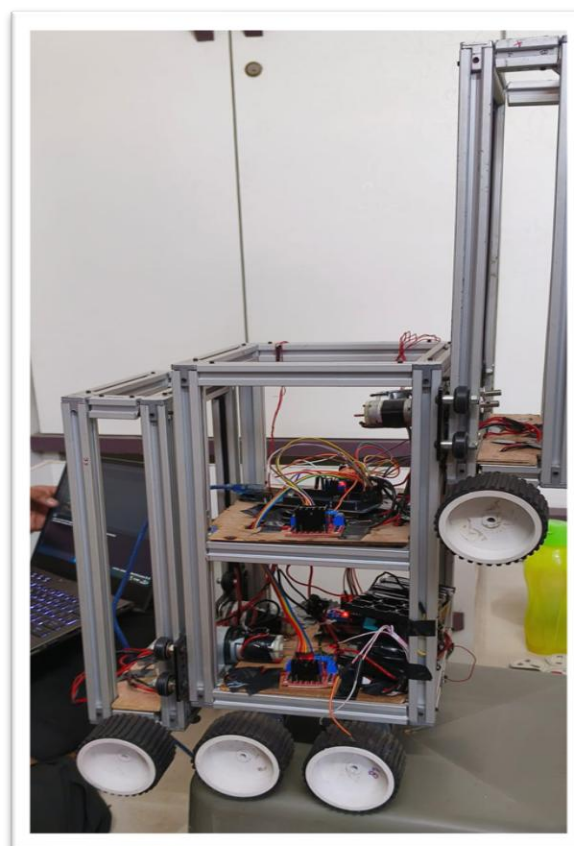
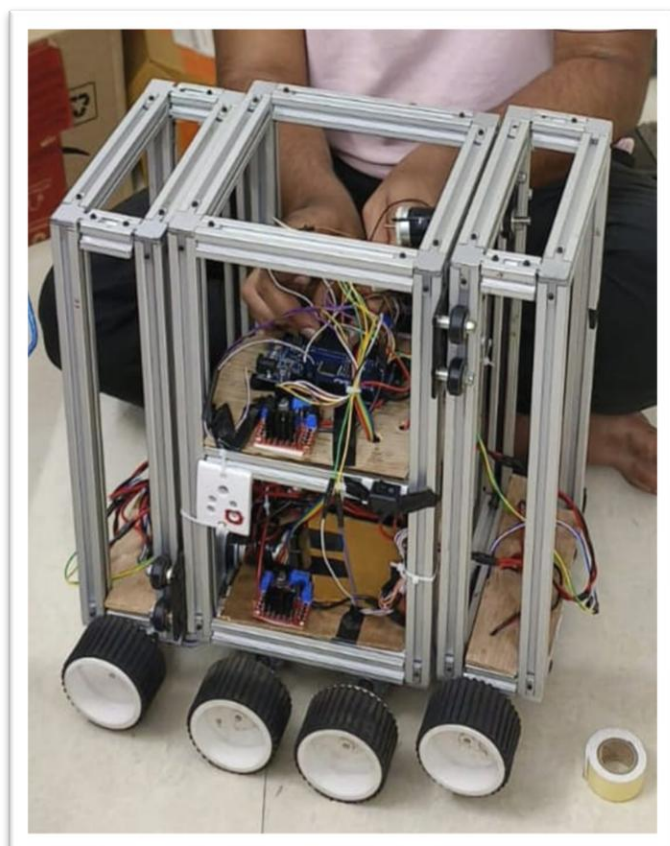
The middle box is the structural and electronic core of AUTOM8, accounting for approximately 55% of total robot weight. It carries all four gantry motors (two upper front-facing, two lower rear-facing), four-wheel motors, the complete electronics bay across two internal compartments, all four gantry carriage plate assemblies, and one stair detection sensor on the bottom-front face.

Table 3.4: Middle box — complete component weight breakdown

Component	Qty $\times$ Unit (g)	Total (g)	Total (kg)
Structural frame (profiles + joints + fasteners)	1 $\times$ 2,289	2,289	2.289
Gantry motors (100 RPM) — 2 upper front + 2 lower rear	4 $\times$ 210	840	0.840
Wheel motors (200 RPM)	4 $\times$ 210	840	0.840
Drive wheels (68mm)	4 $\times$ 100	400	0.400
Motor mounts — U-clamps (4 gantry + 4 wheel)	8 $\times$ 30	240	0.240
V-slot gantry plate sets (carriage)	4 $\times$ 110	440	0.440
GT2 timing belt + pulleys	1 set $\times$ ~90	~90	~0.090
HC-SR04 sensor (stair detection, bottom-front $\times$ 1)	1 $\times$ 9	9	0.009
Arduino Mega 2560	1 $\times$ 37	37	0.037
Motor drivers L298N (Driver 3–4 wheel + 5–6 gantry)	4 $\times$ 35	140	0.140
Battery system (3S4P cells + BMS + holders + assembly)	1 $\times$ 735	735	0.735
Buck converter LM2596S	1 $\times$ ~10	~10	~0.010
MPU-6050 IMU	1 $\times$ ~10	~10	~0.010
Buzzer (piezo 5V)	1 $\times$ ~5	~5	~0.005

Logic switch (5V line)	1×3	3	0.003
Main power switch (12V line)	$1 \times \sim 50$	~ 50	~ 0.050
Base boards $\times 2$ (20 $\times$ 20cm)	$2 \times \sim 100$	~ 200	~ 0.200
Wiring share	—	~ 60	~ 0.060
M5 fasteners share	—	~ 66	~ 0.066
Miscellaneous hardware share	—	~ 180	~ 0.180
Zip ties + tape share	—	~ 35	~ 0.035
MIDDLE BOX TOTAL	—	$\sim 6,623$	~ 6.62 kg

Figure 3.5 and 3.6— Middle box overview showing gantry motors (upper front and lower rear), carriage plates, and internal electronics compartment showing.



Back Box — $\sim 2,707\text{g} \approx 2.71\text{ kg}$

The back box is structurally identical to the front box. It carries the two rear drive motors and Motor Driver 2. To equalise the weight distribution between the front and back boxes — the front box carries approximately 110g of additional components (sensors, encoder, display elements) — short sections of 20×20mm aluminium T-slot profile totalling approximately 24cm were added to the back box as ballast, bringing both outer boxes to approximately the same total weight.

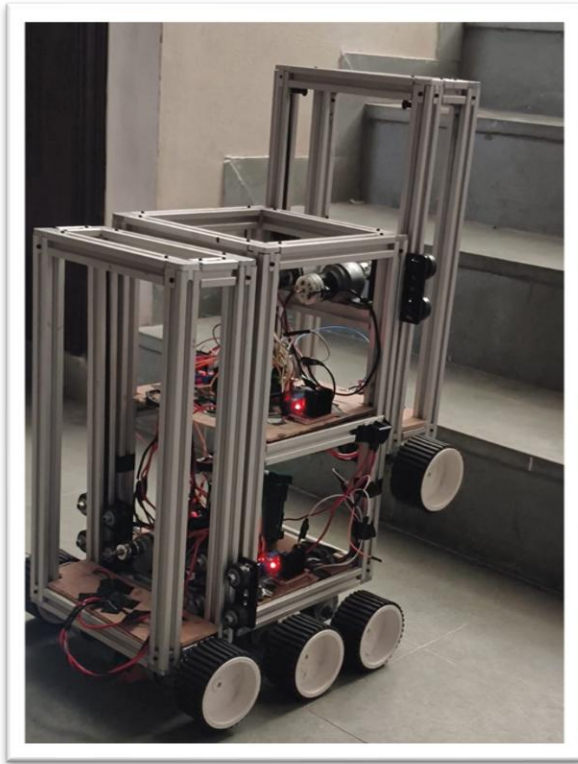
Table 3.5: Back box — complete component weight breakdown

Component	Qty × Unit (g)	Total (g)	Total (kg)
Structural frame (profiles + joints + fasteners)	1 × 1,731	1,731	1.731
Wheel motors (200 RPM)	2 × 210	420	0.420
Drive wheels (68mm)	2 × 100	200	0.200
Motor mounts — U-clamps	2 × 30	60	0.060
Motor driver L298N (Driver 2)	1 × 35	35	0.035
HC-SR04 sensor (stair detection, bottom-front ×1)	1 × 9	9	0.009
Base board (8×20cm)	1 × ~ 30	~ 30	~ 0.030
Aluminium profile ballast ($\sim 24\text{cm}$ T-slot, weight equalisation)	—	~ 110	~ 0.110
Wiring share	—	~ 20	~ 0.020
M5 fasteners share	—	~ 17	~ 0.017
Miscellaneous hardware share	—	~ 60	~ 0.060
Zip ties + tape share	—	~ 15	~ 0.015
BACK BOX TOTAL	—	$\sim 2,707$	$\sim 2.71\text{ kg}$

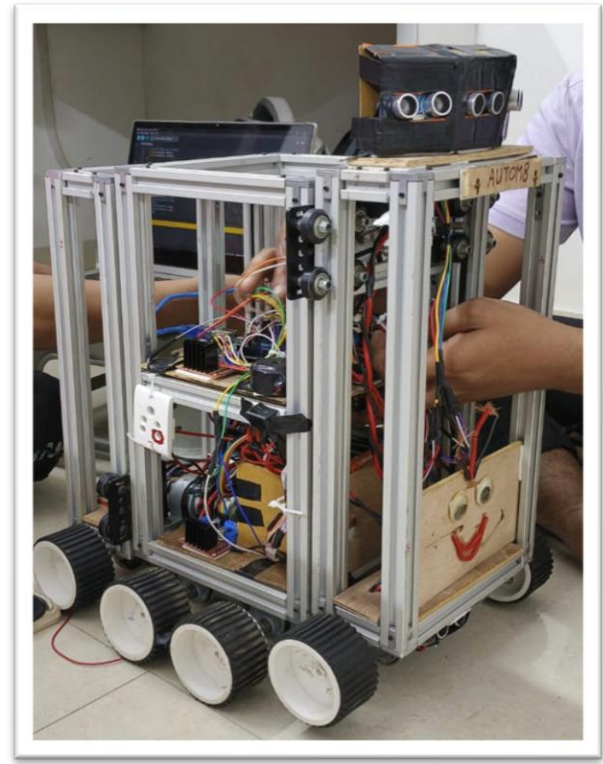
Figure 3.7 — Back box assembled view showing back box (testing image)

Figure 3.8 — Fully assembled prototype showing all three boxes, wheel arrangement, and external wiring

(fig 3.7)



(fig 3.8)



3.6 Fabrication Process

The fabrication of the AUTOM8 prototype involved three main processes: profile cutting, mechanical assembly, and battery pack soldering.

Aluminium profiles were cut to the required lengths using a manual hacksaw and a mitre guide. Each profile length was measured against the design dimensions and marked before cutting. Cut ends were deburred using a file to ensure clean seating in the corner joints. No CNC machining or power tools were used for profile preparation. The cutting process introduces a dimensional tolerance of approximately $\pm 1\text{mm}$ per cut, which is acceptable for a prototype assembly where all connections are made through adjustable T-slot joints rather than precision interfaces.

Mechanical assembly was carried out using the standard T-slot hardware — three-way corner joints at all structural corners, inner brackets at intermediate connection points, and M5 grub screws to secure T-nuts in the profile channels. The gantry plate sets were assembled according to the manufacturer specification, with eccentric spacers used to tension the V-wheels against the V-slot rails to eliminate play without introducing excessive friction. Motor mounts were positioned and tightened with the wheel in place to ensure correct lateral alignment before the clamp screws were fully torqued.

The battery pack was assembled by soldering twelve Orange ICR18650-22C cells into a 3S4P configuration using nickel strip interconnects and a soldering iron. Cells were arranged in two 3×4 plastic holders. The BMS board was connected to the cell terminals following the standard 3S wiring sequence. All solder joints were inspected visually and tested for continuity before the pack was placed into the robot. The assembled pack produces a nominal voltage of 11.1V and a capacity of 8800mAh.

3.7 Design Limitations

Several mechanical limitations of the current prototype are acknowledged and documented here for transparency.

The payload compartment was designed as an open platform on the top surface of the middle box. No enclosure, locking mechanism, or load cell was fabricated or tested in the prototype. Payload capacity was estimated through motor torque calculations and is described in **Calculation 4**. Physical testing with a load was not carried out.

The wiring in the prototype is routed externally along the frame profiles and managed with zip ties and cable clips. No wiring enclosure or conduit is used. This is adequate for a controlled prototype environment but would require significant improvement before deployment in a real campus setting.

Observable tilting of the robot body during middle box elevation was recorded during stair-climbing tests. This is a direct consequence of the weight distribution described in **Section 3.5** — when the middle box is the only section with wheel contact, and its centre of mass is not centered over the contact patch, the robot tilts. The robot self-corrects as the sequence completes and contact is re-established on the upper stair surface. This behaviour is discussed further in **Chapter 6**.

The inter-box gap of approximately 5mm was not measured with a precision instrument during this project and should be verified against the actual assembled robot before any future mechanical analysis relies on this value.

The gantry mechanism does not include end-stop switches or position sensors. Box elevation is controlled entirely by timed motor operation. If the motor runs for longer than required — due to a firmware error or unexpected load — there is no mechanical stop to prevent over-travel. This is identified as a priority safety improvement for any future iteration of the design.

CHAPTER 4

ELECTRICAL AND CONTROL SYSTEM

4.1 Overview of Electrical Architecture

The electrical system of AUTOM8 is designed around a centralised 12V lithium-ion battery supply that powers all twelve motors through six L298N dual H-bridge motor driver modules. A separate 5V regulated rail, derived from the main battery via a step-down buck converter, powers all logic-level components — the Arduino Mega 2560 microcontroller, the sensor suite, and the buzzer. Two independent power switches allow the main drive circuit and the logic circuit to be isolated independently, providing a basic level of operational safety during testing and maintenance.

The Arduino Mega 2560 serves as the sole microcontroller, receiving sensor inputs and issuing PWM motor control signals to all six driver modules. Three HC-SR04 ultrasonic sensors mounted at the bottom-front of each box detect stair risers and trigger the stair-climbing phase sequence. Three additional HC-SR04 sensors mounted at the top-front of the front box provide obstacle detection and halt during flat terrain navigation. An MPU-6050 inertial measurement unit provides heading correction on flat ground, and a single magnetic Hall effect encoder on the front wheel motor provides distance estimation for pre-programmed path execution.

4.2 Power System

4.2.1 Battery Pack

The robot is powered by a custom-assembled 3S4P lithium-ion battery pack built from twelve Orange ICR18650-22C cells. The three-series configuration provides a nominal output voltage of 11.1V (approximately 12V), which directly drives the motor drivers and, via the buck converter, the 5V logic rail. The four-parallel configuration provides a total pack capacity of 8800mAh, sufficient for extended operation at prototype testing durations.

Table 4.1: Battery pack specifications

Cell Model	Orange ICR18650-22C
Cell Nominal Voltage	3.7V
Cell Capacity	2200mAh
Configuration	3S4P (3 series × 4 parallel)
Pack Nominal Voltage	11.1V (≈12V)
Pack Total Capacity	8800mAh
Total Cells	12
Total Pack Weight	735g (cells + BMS + assembly)

4.2.2 Battery Management System

A 3S 12V 25A lithium battery protection board (BMS) is integrated into the battery pack assembly. The BMS provides overcharge, over-discharge, and short-circuit protection for all twelve cells. Its continuous operating current rating of 25A covers the combined draw of all twelve motors under normal load, while the instantaneous rating of 35–40A provides headroom for startup current spikes.

Table 4.2: BMS specifications

Model	3S 12V 25A 18650 Lithium Battery Protection Board
SKU	43646 (Robu.in)
Operating Current	0–25A continuous
Instantaneous Current	35–40A
Balance Current	40–50mA
Weight	20g

4.2.3 5V Regulated Power Rail — LM2596S Buck Converter

An LM2596S step-down DC-DC buck converter steps the 11.1V battery voltage down to a stable 5V supply for all logic-level components. The converter operates at 92% conversion efficiency and is rated at 2A continuous output (3A maximum), which comfortably covers the combined current draw of the Arduino Mega, six HC-SR04 sensors, the MPU-6050, the encoder, and the buzzer. The output voltage is adjustable via an onboard trimmer and was set to 5.0V during initial assembly.

Table 4.3: Buck converter specifications

Model	LM2596S DC-DC Buck Converter
SKU	11548 (Robu.in)
Input Voltage	3–40V
Output Voltage	Set to 5V (adjustable 1.5–35V)
Output Current	2A rated, 3A maximum
Switching Frequency	150 kHz
Conversion Efficiency	92%
Weight	~10g

4.2.4 Power Switches

Two independent rocker switches are used to control the two power rails. Switch 1 is a 6–16A rated domestic SPST rocker switch that controls the main 12V line between the battery and all motor driver modules. Switch 2 is a smaller 6A rated SPST rocker switch that controls the 5V logic line between the buck converter and the Arduino and sensor suite. This arrangement allows the logic circuit to be powered while the drive circuit remains off, which was useful during programming and sensor calibration.

4.3 Microcontroller — Arduino Mega 2560

The Arduino Mega 2560 (ATmega2560) was selected as the microcontroller for AUTOM8. The primary reason for this choice was the pin count requirement: controlling six L298N motor driver modules

requires a minimum twelve PWM-capable digital output pins (two per driver), plus additional digital pins for direction control and sensor inputs. The Arduino Uno, with only 14 digital I/O pins and 6 PWM outputs, was insufficient. The Mega's 54 digital I/O pins and 15 PWM outputs provided sufficient capacity for all motor control, sensor, and communication requirements.

Table 4.4: Arduino Mega 2560 specifications

Model	Arduino Mega 2560 Rev3
SKU	A000067
Microcontroller	ATmega2560
Clock Speed	16 MHz
Digital I/O Pins	54 (15 PWM-capable)
Analog Input Pins	16
Communication	UART ×4, I2C, SPI
Operating Voltage	5V I/O
Input Voltage	7–12V
Current per I/O Pin	20mA maximum
Dimensions	101.5 × 53.3mm
Weight	37g

4.4 Motor Driver System — L298N

Six L298N dual H-bridge motor driver modules are used to control all twelve motors. Each L298N module has two independent channels, each capable of driving one motor bidirectionally at up to 2A continuous. PWM signals from the Arduino Mega control motor speed on each channel; direction is set via separate digital HIGH/LOW signals to the IN1/IN2 pins of each channel. Six modules provide exactly twelve channels, one per motor.

4.4.1 Driver Module Specifications

Table 4.5: L298N motor driver specifications

Model	L298N 2A Based Motor Driver Module
SKU	43821 (Robu.in)
Driver IC	L298N Double H-Bridge
Operating Voltage	5–35V
Current per Channel	2A continuous
Channels per Module	2
Total Modules	6 (12 channels total)
Dimensions	44 × 44 × 28mm
Weight	35g each

4.4.2 Driver Assignment

Each driver module is assigned to a specific pair of motors. Drivers 1 and 2 are mounted inside the front and back boxes respectively, keeping high-current wiring to the wheel motors in those boxes as short as possible. Drivers 3 and 4 are mounted in the lower compartment of the middle box and control the four middle wheel motors. Driver 5 is mounted on the upper portion of the middle box front face and controls the two front-facing gantry motors. Driver 6 is mounted on the lower portion of the middle box rear face and controls the two rear-facing gantry motors.

Table 4.6: Motor driver assignments

Driver	Location	Motors Controlled	Motor Type
Driver 1	Front box	2 × front wheel motors	200 RPM
Driver 2	Back box	2 × back wheel motors	200 RPM
Driver 3	Middle box — lower	2 × middle wheel motors (left)	200 RPM
Driver 4	Middle box — lower	2 × middle wheel motors (right)	200 RPM

Driver 5	Middle box — upper (front face)	2 × front gantry motors	100 RPM
Driver 6	Middle box — lower (rear face)	2 × rear gantry motors	100 RPM

4.5 Sensor System

AUTOM8 uses four distinct sensing sub-systems, each serving a specific function in the overall control architecture: stair detection, obstacle detection, heading correction, and distance estimation. Each sub-system operates independently and interfaces with the Arduino Mega via dedicated digital or I2C connections.

4.5.1 HC-SR04 Ultrasonic Sensors — Stair Detection

Three HC-SR04 ultrasonic sensors are mounted at the bottom-front of each box — one per box — facing forward and slightly downward. In the sensor-triggered stair climbing approach, these sensors detect the vertical face of a stair riser as the robot approaches. When a sensor reading falls below the trigger threshold of 3cm, the Arduino Mega registers a stair detection event and initiates the corresponding phase of the gantry lift sequence. In this robot timing-based approach used for the final presentation.

The 3cm trigger distance was selected to ensure reliable riser detection across the full compatible stair geometry range (tread 280–400mm, riser 150–220mm) while avoiding false triggers from surface irregularities.

Table 4.7: HC-SR04 stair detection sensor specifications

Model	HC-SR04 Ultrasonic Range Finder
SKU	2444 (Robu.in)
Operating Voltage	5V
Average Current	2mA
Frequency	40 kHz
Sensing Range	2–400cm

Sensing Angle	15°
Trigger Distance	3cm (stair detection threshold)
Quantity	3 (one per box — front, middle, back)
Placement	Bottom-front of each box, forward-facing
Weight	9g each

4.5.2 HC-SR04 Ultrasonic Sensors — Obstacle Detection

Three additional HC-SR04 sensors are mounted at the top-front of the front box for obstacle detection during flat terrain navigation. One sensor faces directly forward, one is angled left, and one is angled right, providing a wide forward coverage zone. When any of the three sensors detects an object within the configured detection range, the Arduino Mega immediately stops all wheel motors and activates the buzzer. The robot resumes forward motion automatically once the obstacle clears the detection zone.

The obstacle detection trigger distance was set during prototype testing. The configured value is 25 cm.

Table 4.8: HC-SR04 obstacle detection sensor placement

Sensor	Mounting Position	Orientation	Function
HC-SR04 — 1	Top-front, front box	Forward-facing (centre)	Forward obstacle detection
HC-SR04 — 2	Top-front, front box	Angled left	Left-side obstacle detection
HC-SR04 — 3	Top-front, front box	Angled right	Right-side obstacle detection

4.5.3 MPU-6050 Inertial Measurement Unit

The MPU-6050 six-axis IMU (3-axis accelerometer + 3-axis gyroscope) is mounted in the middle box and communicates with the Arduino Mega via the I2C protocol. During flat terrain navigation, the

MPU-6050 monitors the robot's heading angle. If the robot drifts from its intended heading — due to uneven floor surface, motor speed variation, or external disturbance — the Arduino Mega applies a corrective adjustment by reducing the PWM on the faster side until the heading error falls within the acceptable threshold.

The MPU-6050 is deliberately disabled during the stair-climbing sequence. During stair climbing, intentional tilting of each box occurs as part of the gantry lift phases. If the IMU were active during these transitions, it would misinterpret the deliberate tilt as a heading error and attempt to apply corrections that would interfere with the climbing sequence. This selective activation is implemented in firmware and represents a deliberate design decision.

Table 4.9: MPU-6050 IMU specifications

Model	MPU-6050 3-Axis Accelerometer and Gyro Sensor
SKU	2846 (Robu.in)
Communication	I2C Protocol
Gyroscope Range	$\pm 250 / 500 / 1000 / 2000^\circ/\text{s}$ (selectable)
Accelerometer Range	$\pm 2, \pm 4, \pm 8, \pm 16\text{g}$ (selectable)
ADC Resolution	16-bit
Operating Voltage	3–5V DC
Dimensions	$20 \times 16\text{mm}$
Active During	Flat terrain navigation only
Disabled During	Stair climbing sequence (firmware-controlled)
Weight	$\sim 10\text{g}$

4.5.4 Magnetic Hall Effect Encoder

A single circular magnetic Hall effect encoder is mounted on the shaft of the front-left wheel motor. The encoder generates a digital pulse signal as the motor shaft rotates. The Arduino Mega counts these pulses to estimate the distance the robot has travelled on flat ground. This distance estimate is used in conjunction with pre-programmed turn commands to execute the fixed delivery route. Only one encoder was fitted due to cost and wiring complexity constraints at the prototype stage. The 200 RPM wheel motors have an internal gear ratio of 90:1, and the encoder is mounted on the motor output shaft. The pulse count per wheel revolution is 1,260 counts, based on the encoder specification.

The encoder is disabled during the stair-climbing sequence. Under the varying motor loads of the gantry phases, encoder pulse counts become unreliable and are not used for positional control during stair climbing. Re-enabling occurs automatically in firmware when flat terrain navigation resumes.

Table 4.10: Magnetic encoder specifications

Type	Circular Magnetic Hall Effect Encoder Board
Mounting	Front-left wheel motor shaft
Operating Voltage	3.3–5V
Output	Digital pulse signal
Motor Gear Ratio	90:1 (200 RPM motor)
Counts per Wheel Rev	1,260 counts * <i>"Based on the encoder's 14 pulses per revolution and the motor gear ratio of 90:1: $14 \times 90 = 1,260$ counts per wheel revolution"</i>
Quantity	1 (single encoder, prototype stage)
Active During	Flat terrain navigation only
Disabled During	Stair climbing sequence (firmware-controlled)
Weight	~8g

4.5.5 Active Piezo Buzzer

A 5V active piezo buzzer is mounted in the middle box and connected to a digital output pin of the Arduino Mega. The buzzer is triggered by the obstacle detection system: when any of the three forward-facing HC-SR04 sensors detects an obstacle within the detection range, the Arduino activates the buzzer simultaneously with stopping the wheel motors. The buzzer continues to sound while the obstacle remains within range and stops automatically once the path is clear. This provides an audible alert to nearby personnel that the robot has detected an obstruction.

4.6 Control Logic

All control logic for AUTOM8 runs on the Arduino Mega 2560. The firmware implements three distinct operating modes: flat terrain navigation, stair climbing, and obstacle detection and halt. Transitions between modes are triggered by sensor events. The system uses open-loop PWM motor control — there is no closed-loop speed feedback during either flat terrain navigation or stair climbing. This is a documented limitation of the prototype stage.

4.6.1 Flat Terrain Navigation

During flat terrain navigation, all eight-wheel motors are driven at 70-80 percent PWM through their respective L298N drivers. The MPU-6050 heading correction system is active, continuously monitoring the robot's yaw angle. If the heading drifts, the Arduino reduces the PWM on the faster side until the heading error is corrected, then restores equal PWM to both sides.

Distance tracking is performed using pulse counts from the single Hall effect encoder on the front wheel motor. The pre-programmed delivery route is stored as a sequence of distance targets and turn commands. When the encoder count reaches a programmed distance target, the Arduino executes the next command in the sequence — either continuing straight, executing a turn, or stopping at a delivery point.

4.6.2 Stair Climbing Phase Sequence

The stair climbing sequence is triggered when the front box stair detection sensor registers a reading below 3cm, indicating the robot has reached a stair riser. The sequence consists of six phases executed in order. Both the MPU-6050 and the encoder are disabled for the duration of the stair climbing sequence and re-enabled when all three boxes have been lifted onto the upper stair level.

The gantry motors are driven at different PWM during all lift phases. Each lift phase is controlled by the stair detection sensor on the box being lifted. The gantry motors start when the sensor reads $\leq 3\text{cm}$ and continue until the sensor reads $> 3\text{cm}$, confirming the box has cleared the riser face. Because the sensor is mounted at the bottom of the profile and the wheel sits 34mm below it, the gantry continues briefly after

the >3cm reading to cover this wheel radius offset, ensuring the wheel rises fully above the upper tread surface. The motors then stop and the robot translates forward, allowing the lifted box to descend naturally onto the upper tread. The wheel motors used during translation phases are driven at 70-80 percent PWM. The full six-phase stair climbing sequence is described in detail in **Chapter 5**, which covers the stair climbing mechanism exclusively.

4.6.3 Reactive Obstacle Detection and Halt

The obstacle detection system operates continuously during flat terrain navigation. The three forward-facing HC-SR04 sensors on the front box are polled in each control loop iteration. If any sensor returns a reading below the obstacle detection threshold of 25 cm, the Arduino immediately sets all eight wheel motor PWM outputs to zero, halting the robot. The buzzer is activated simultaneously. The robot remains stationary and the buzzer continues until all three sensors return readings above the threshold. Forward motion then resumes automatically without any manual intervention. This system is correctly classified as reactive obstacle detection and halt — the robot detects obstacles in real time and stops, but does not perform path planning or obstacle circumnavigation.

4.7 Design Limitations

Several limitations of the electrical and control system were identified and documented during prototype testing. These are acknowledged as areas for improvement in future iterations.

The most significant limitation is the open-loop motor control architecture. Without closed-loop speed feedback, the robot cannot compensate for variations in motor performance under different load conditions. During the stair-climbing sequence, unequal loading on the gantry motors can cause slight differential lift rates between the left and right sides of each box, resulting in observable tilt during phase transitions.

The single encoder provides only an approximation of travelled distance. With only one wheel monitored and no slip detection, accumulated distance error increases over longer routes. The measured accuracy was not formally quantified at prototype stage and is acknowledged as a limitation for future development.

The L298N motor driver modules are rated at 2A per channel continuous. Under stall conditions, the 100 RPM gantry motors can draw significantly higher current momentarily. While the BMS instantaneous rating of 35–40A provides protection at the battery level, thermal monitoring of the driver modules was not implemented in the prototype. Driver module temperature was observed to rise during extended stair climbing test cycles.

Exposed wiring at prototype stage creates a risk of shorts and intermittent connections under vibration. A production version would require proper cable management, strain relief, and enclosed electronics enclosures.

CHAPTER 5

STAIR CLIMBING MECHANISM

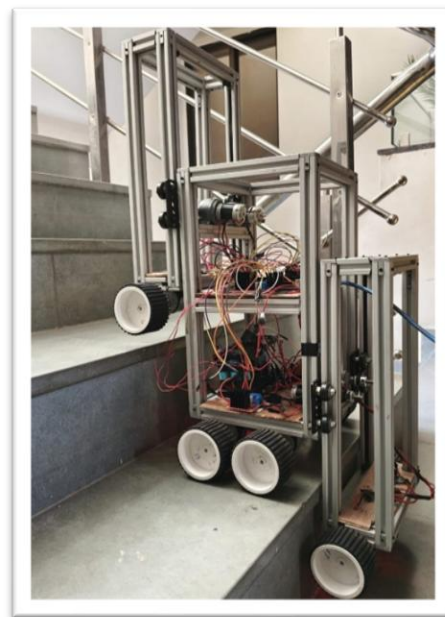
5.1 Overview

The stair climbing capability is the central technical contribution of this project. The mechanism uses a six-phase sequential gantry lift approach. Each phase either lifts one section of the three-box chassis onto the next stair step, or translates the chassis forward to position the next section at the riser. The three sections — front box, middle box, and back box — are lifted in sequence, each using the other two as anchors, until the entire chassis has advanced by one stair step. This process repeats for each subsequent step.

The mechanism was designed to be compatible with the stair geometry found in Indian educational and institutional buildings. The National Building Code (NBC) of India specifies a minimum tread depth of 300mm and maximum riser height of 150mm for educational buildings. AUTOM8 was designed to accommodate a wider range — tread 280–400mm and riser 150–220mm — to cover variation in existing campus stair construction that may not strictly comply with the NBC standard.

Figure 5.1 — AUTOM8 stair climbing — full sequence overview on test staircase timing-based control testing videos

(The two videos can be accessed by clicking on the images)



5.2 Compatible Stair Geometry

The compatible stair geometry is determined by the physical dimensions of the chassis — specifically the spacing between the three boxes, the gantry lift height range, and the detection trigger distance used in the sensor-triggered control approach. The following table summarises the confirmed compatibility range.

Table 5.1: Compatible stair geometry specification

Minimum Tread Depth	280mm
Maximum Tread Depth	400mm
Minimum Riser Height	150mm
Maximum Riser Height	220mm
Ideal Tread (design)	290–300mm (matches NBC educational standard)
Stair Detection	HC-SR04 ultrasonic, 3cm trigger distance
Total Robot Length	370mm (front 80 + middle 200 + back 80 + 2 × 5mm gap)

5.2.1 Geometry Basis

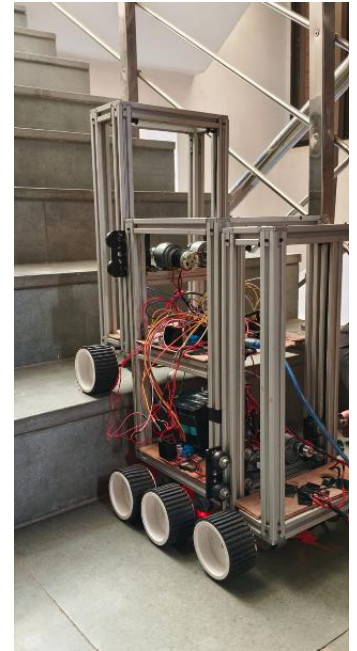
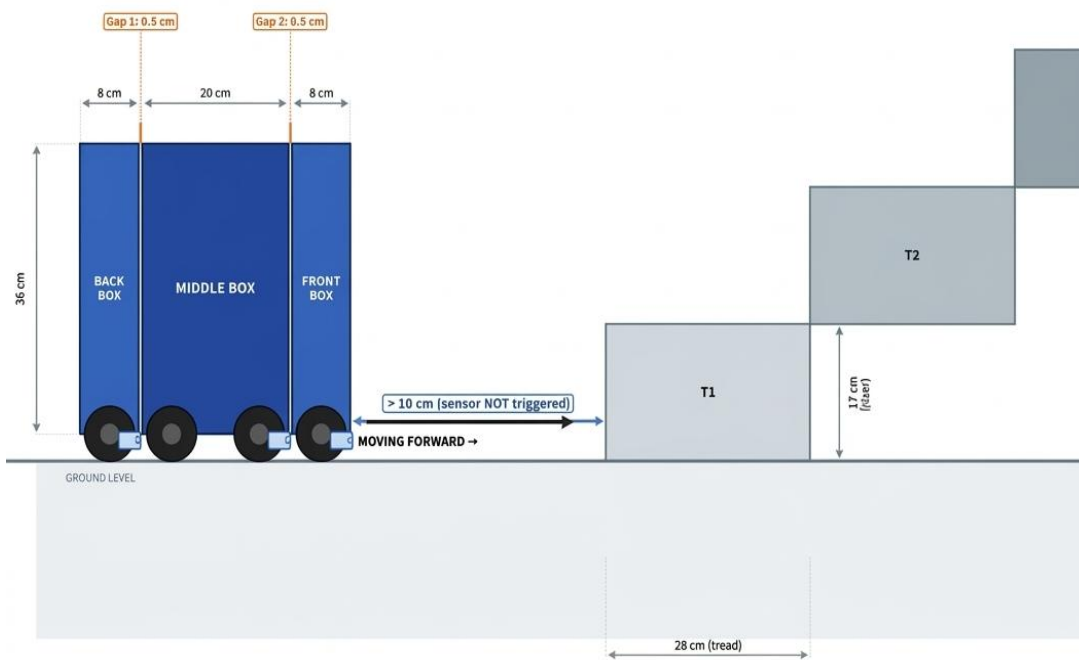
The minimum tread depth of 280mm is set by the geometry of the back box relative to the stair riser when the front box has been lifted. At 280mm tread, the rear wheel of the back box sits with a margin of approximately 5mm from the edge of the lower stair — the minimum acceptable clearance for stable footing. Tread depths below 280mm would place the back wheel over the edge, making the back box unstable during the lift.

The maximum tread depth of 400mm is limited by the detection geometry of the sensor-triggered approach. The three stair detection sensors are mounted at the bottom-front of each box. For the back box sensor to reliably detect the riser on the next step, the chassis must be positioned close enough for the 15° beam angle of the HC-SR04 to intersect the riser face within the 3cm trigger distance. Beyond 400mm tread, the back sensor would not detect the riser before the front box sensor had completed its phase.

The maximum riser height of 220mm is set by the gantry lift travel distance. The gantry plate carriages travel vertically along the 36cm V-slot rails. The usable lift range accounts for the carriage start

position and end-stop clearance, giving a maximum safe lift of approximately 220mm plus the wheel radius (34mm), sufficient to clear a 220mm riser.

Figure 5.2 — Stair geometry diagram showing tread, riser, and robot positioning at Phase 0



5.3 Gantry Lifting Mechanism

5.3.1 Mechanical Principle

The gantry lifting mechanism converts rotary motor output into controlled vertical linear motion using GT2 timing belts and aluminium V-slot rail carriages. Four 100 RPM Johnson DC gantry motors are mounted on the middle box — two on the upper portion of the front face (front-facing pair) and two on the lower portion of the rear face (rear-facing pair). Each motor drives a 20-tooth GT2 aluminium pulley. A GT2 rubber timing belt connects each pulley to a V-slot mini gantry plate carriage that rides along the V-slot vertical rails on the inner faces of the front and back boxes respectively.

When the front-facing gantry motors run, the belts drive the front box carriages upward relative to the middle box, lifting the front box. When the rear-facing motors run, the back box is lifted. When all four motors run simultaneously — with both outer boxes fixed on stair surfaces — the reaction force through the belts lifts the middle box upward between the two anchored outer boxes. This constraint-reaction lifting principle is described in **Section 5.3.2**.

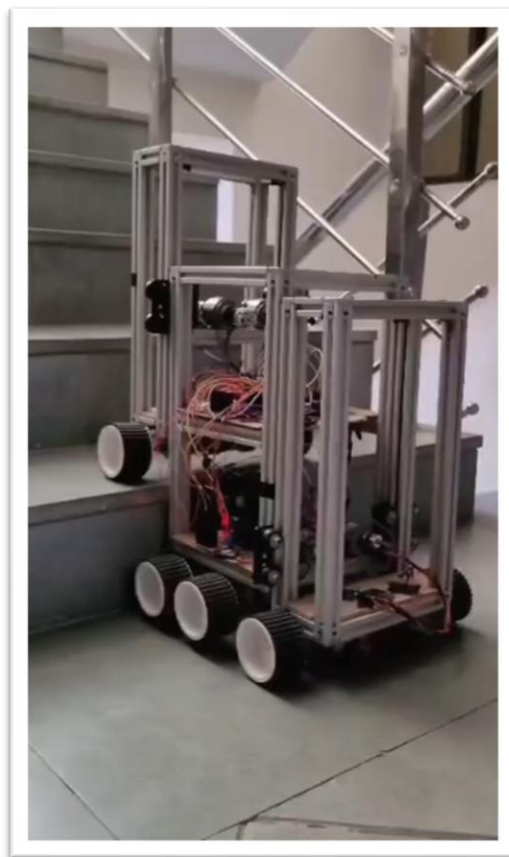
5.3.2 Constraint-Reaction Lifting Principle (Middle Box)

Lifting the middle box requires a different approach from lifting the outer boxes, because the gantry motors and their pulleys are mounted on the middle box itself. The outer boxes cannot be driven upward relative to the middle box in the same direction as the middle box lift — they are already resting on stair surfaces and are effectively fixed.

When both outer boxes are resting on stair treads and their wheels are in contact with the stair surface, they act as fixed anchors. Running all four gantry motors simultaneously causes the belts to exert force on the carriage plates, which are attached to the outer box V-slot rails. Since the outer boxes cannot move downward (they are resting on the stairs), the reaction force is transferred back through the carriage plates and middle box frame, lifting the middle box upward along the rails. The middle box rises between the two stationary outer columns until the lift phase duration completes.

Figure 5.3 — Constraint-reaction lift showing middle box rising between front and back boxes

(The video can be accessed by clicking on the image)



5.3.3 Belt and Pulley Specifications

Table 5.2: GT2 belt and pulley system specifications

Belt Type	GT2 Rubber Timing Belt
Belt Pitch	2mm
Belt Width	6mm
Belt Material	Rubber with fibreglass reinforcement
Total Belt Used	$\cong$ 160cm (from 2-metre kit)
Pulley Type	GT2 Aluminium Timing Pulley
Pulley Teeth	20T
Pulley Bore	5mm (matches 100 RPM motor shaft exactly)
Pulley Quantity	4 (one per gantry motor)
Theoretical Lift Speed	66.7mm/sec at full rated voltage
Belt + Pulley Weight	$\cong$ 90g total

5.3.4 V-Slot Rail and Carriage System

The V-slot vertical rails are the 36cm aluminium extrusion pieces on the inner faces of the front and back boxes. These rails have a V-groove profile on their inner face that serves as the track for the gantry carriage plate wheels. Four Black 2020 V-Slot Mini Gantry Plate Sets are used — two on each outer box, one on each rail. Each gantry plate set consists of a 6061-T5 anodised aluminium plate carrying four POM mini V-wheels with 625ZZ chrome steel bearings. The eccentric spacers on two of the four wheels allow tension adjustment to eliminate play in the carriage travel.

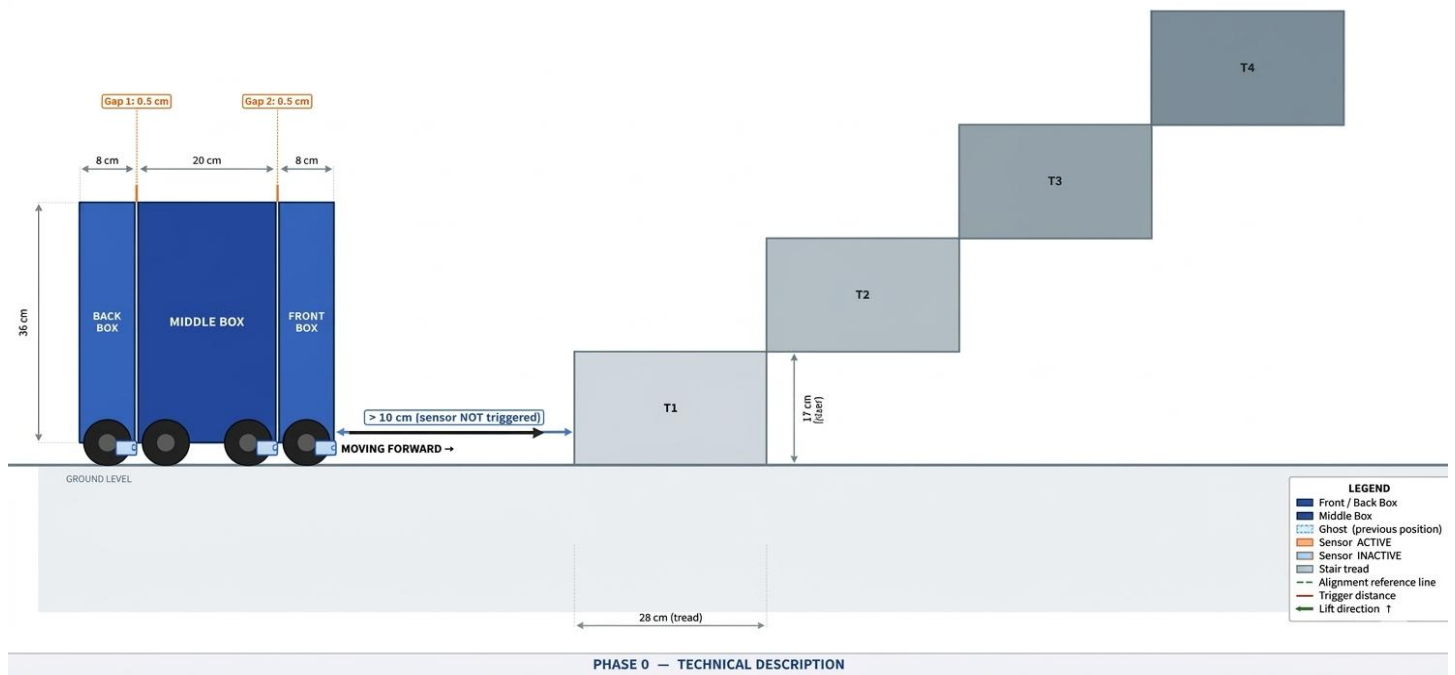
Table 5.3: V-slot gantry carriage plate specifications

Product	Black 2020 V-Slot Mini Gantry Plate Set
Source	Probots.co.in
Plate Material	6061-T5 Aluminium, 3mm, Black Anodized
Wheel Material	High-strength POM with 625ZZ chrome steel bearings
Wheels per Set	4 (2 fixed + 2 eccentric spacer for adjustment)
Compatibility	20×20mm V-slot aluminium profile
Quantity	4 sets (2 front box + 2 back box)
Weight per Set	110g

5.4 Six-Phase Stair Climbing Sequence

The complete stair climbing sequence consists of six phases executed in order. The sequence is triggered when the front box stair detection sensor registers a reading below 3cm. At that point, the Arduino Mega disables the encoder and MPU-6050, stops all wheel motors, and begins **Phase 1**. Each lift phase is terminated by sensor feedback — the gantry stops when the stair detection sensor of the box being lifted transitions from $\leq 3\text{cm}$ to $>3\text{cm}$, plus a brief continuation to cover the wheel radius offset. The sequence runs to completion before flat terrain navigation resumes. Each phase is described below in the order of execution.

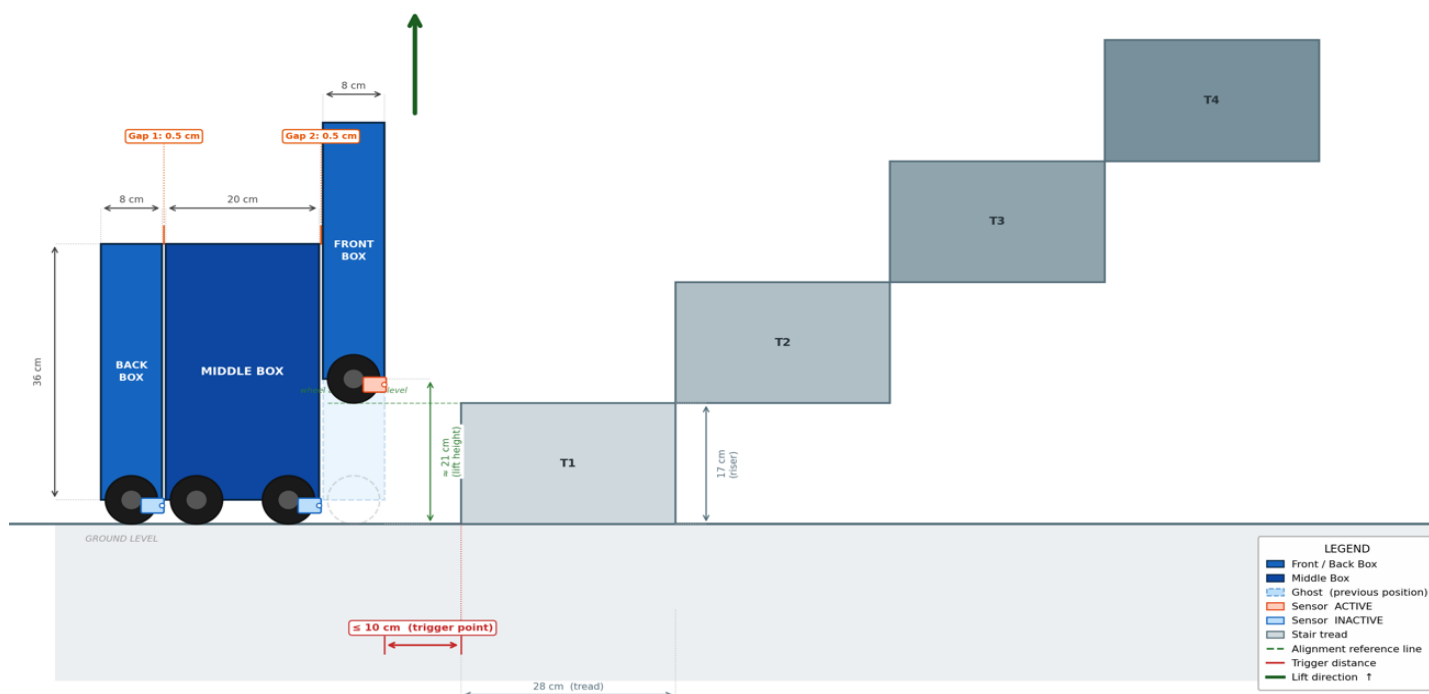
Figure 5.4 — Phase 0: robot approaching stair on flat ground, all wheels in contact



Phase 1 — Front Box Lift

Trigger (sensor-triggered approach): Front box HC-SR04 stair detection sensor reads $\leq 3\text{cm}$ from riser face. Action: Arduino activates front-facing gantry motor pair (Driver 5). Both front gantry motors run simultaneously, driving the GT2 belts that lift the front box vertically upward along the V-slot rails. The front box rises until the front box stair detection sensor reads $> 3\text{cm}$, confirming the box has cleared the riser face. The gantry motors continue briefly to cover the 34mm wheel radius offset between the sensor mounting position and the wheel centre, ensuring the front wheel rises fully above the upper tread surface. The gantry then stops. All wheel motors remain off during this phase. The middle and back boxes remain stationary on the lower stair tread.

Figure 5.5 — Phase 1: front box lifted above riser height, middle and back boxes on lower tread

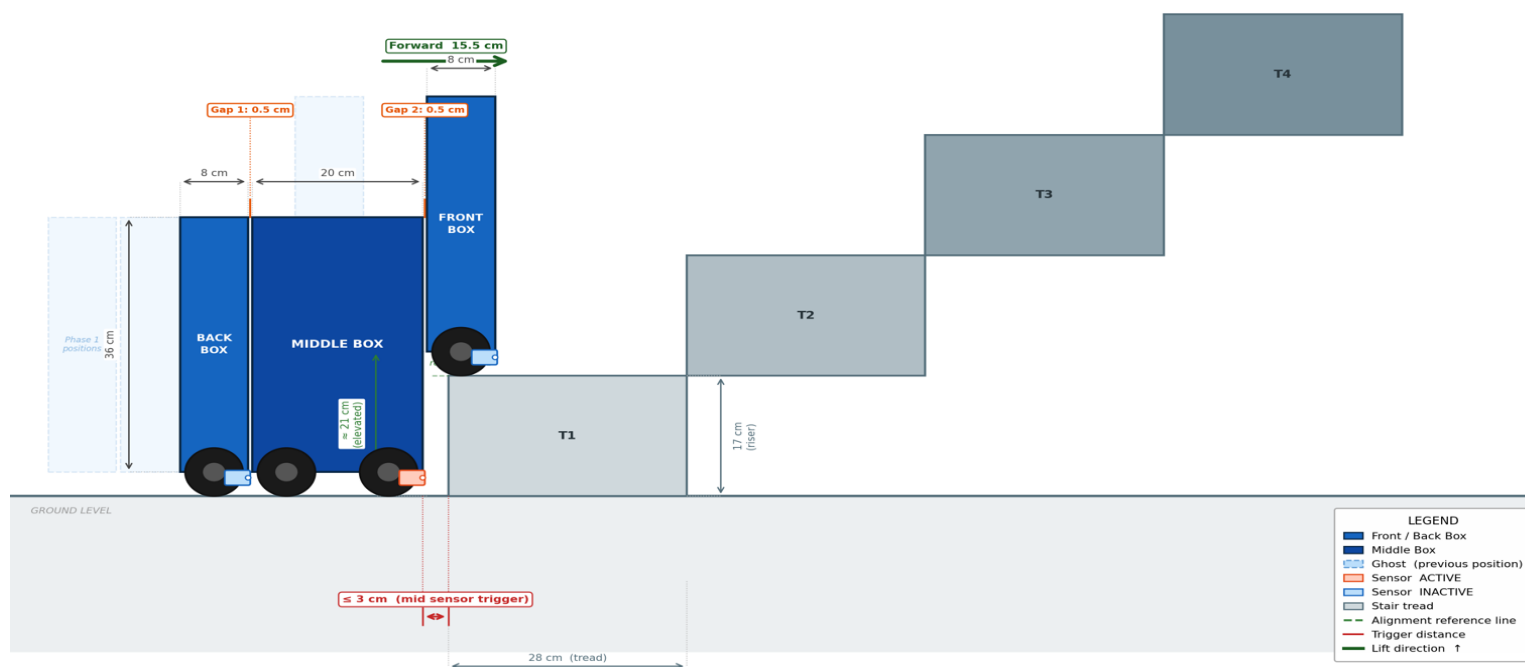


PHASE 1 – TECHNICAL DESCRIPTION

Phase 2 — Forward Translation (Front Box Advance)

Action: Wheel motors activate and drive the robot forward. The front box — now elevated — travels forward with the robot. As the robot advances, the raised front box moves over the upper tread and descends naturally onto it under its own weight. The middle box stair detection sensor now approaches the riser face as the robot continues forward.

Figure 5.6 — Phase 2: robot translating forward, front box wheels landing on upper tread

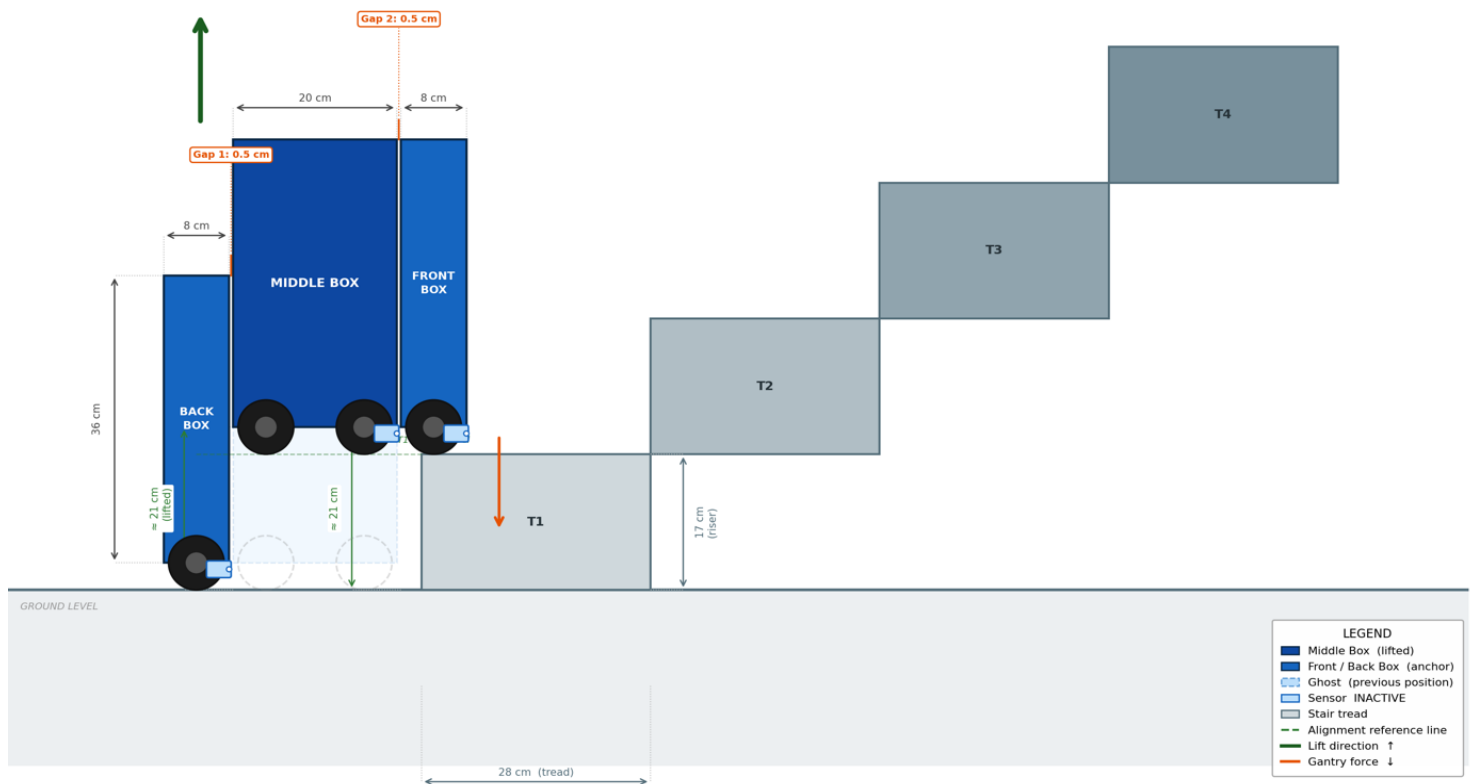


PHASE 2 — TECHNICAL DESCRIPTION

Phase 3 — Middle Box Lift

Trigger (sensor-triggered approach): Middle box HC-SR04 stair detection sensor reads $\leq 3\text{cm}$ from riser face. Action: All four gantry motors activate simultaneously (Drivers 5 and 6). With the front box resting on the upper tread and the back box still on the lower tread, both outer boxes act as fixed anchors. The constraint-reaction principle described in **Section 5.3.2** causes the middle box to rise upward between the two anchored outer boxes. The middle box rises until the middle box stair detection sensor reads $> 3\text{cm}$, confirming the middle box has cleared the riser face. The gantry motors continue briefly to cover the 34mm wheel radius offset, then stop.

Figure 5.7 — Phase 3: middle box rising via constraint-reaction, front box on upper tread, back box on lower tread

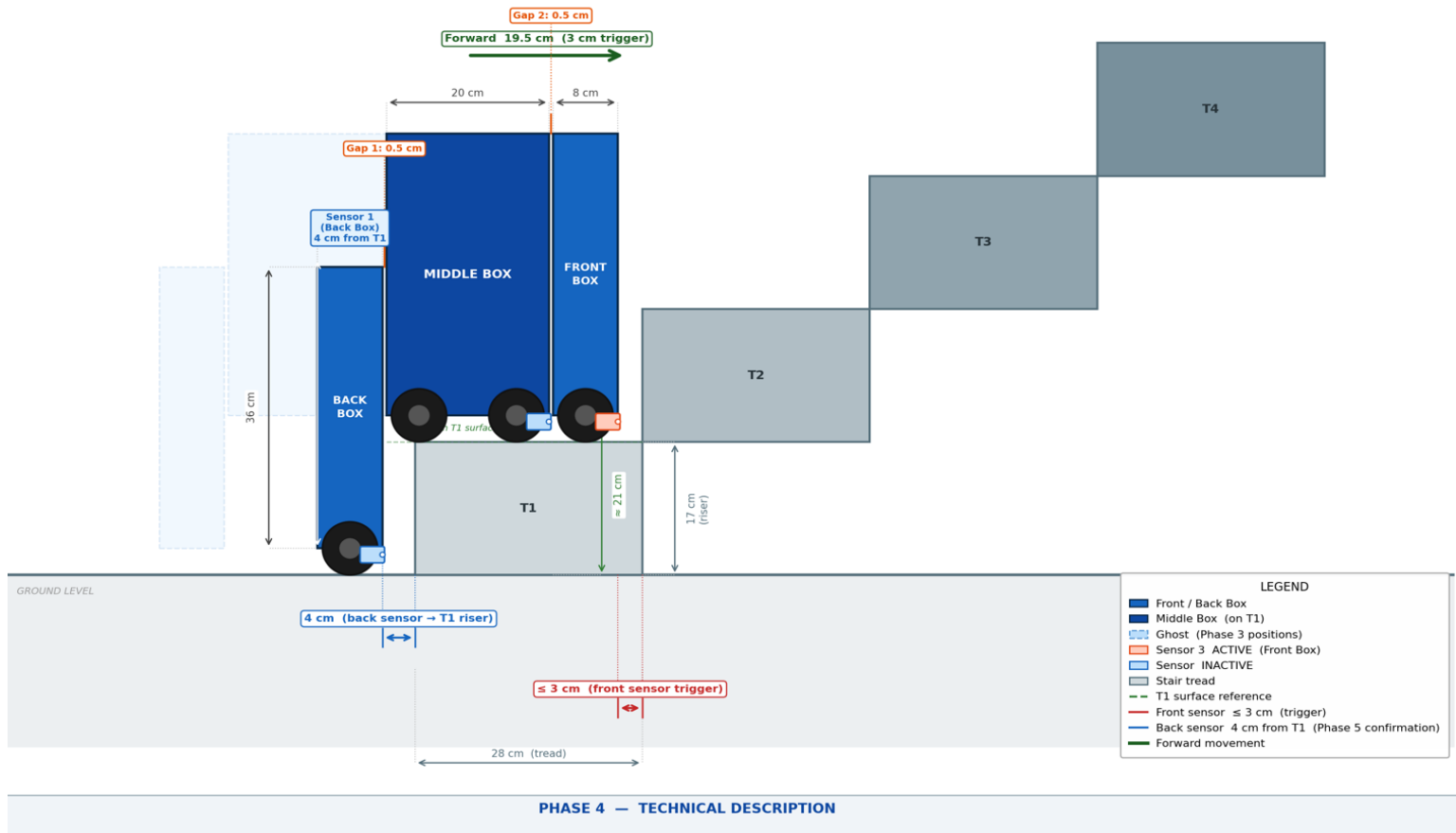


PHASE 3 — TECHNICAL DESCRIPTION

Phase 4 — Forward Translation (Middle Box Advance)

Action: Wheel motors activate. The robot translates forward. The elevated middle box moves over the upper tread and descends naturally onto it under its own weight as the robot advances. At this point the front and middle boxes are on the upper tread; the back box remains on the lower tread. The back box stair detection sensor now approaches the riser face.

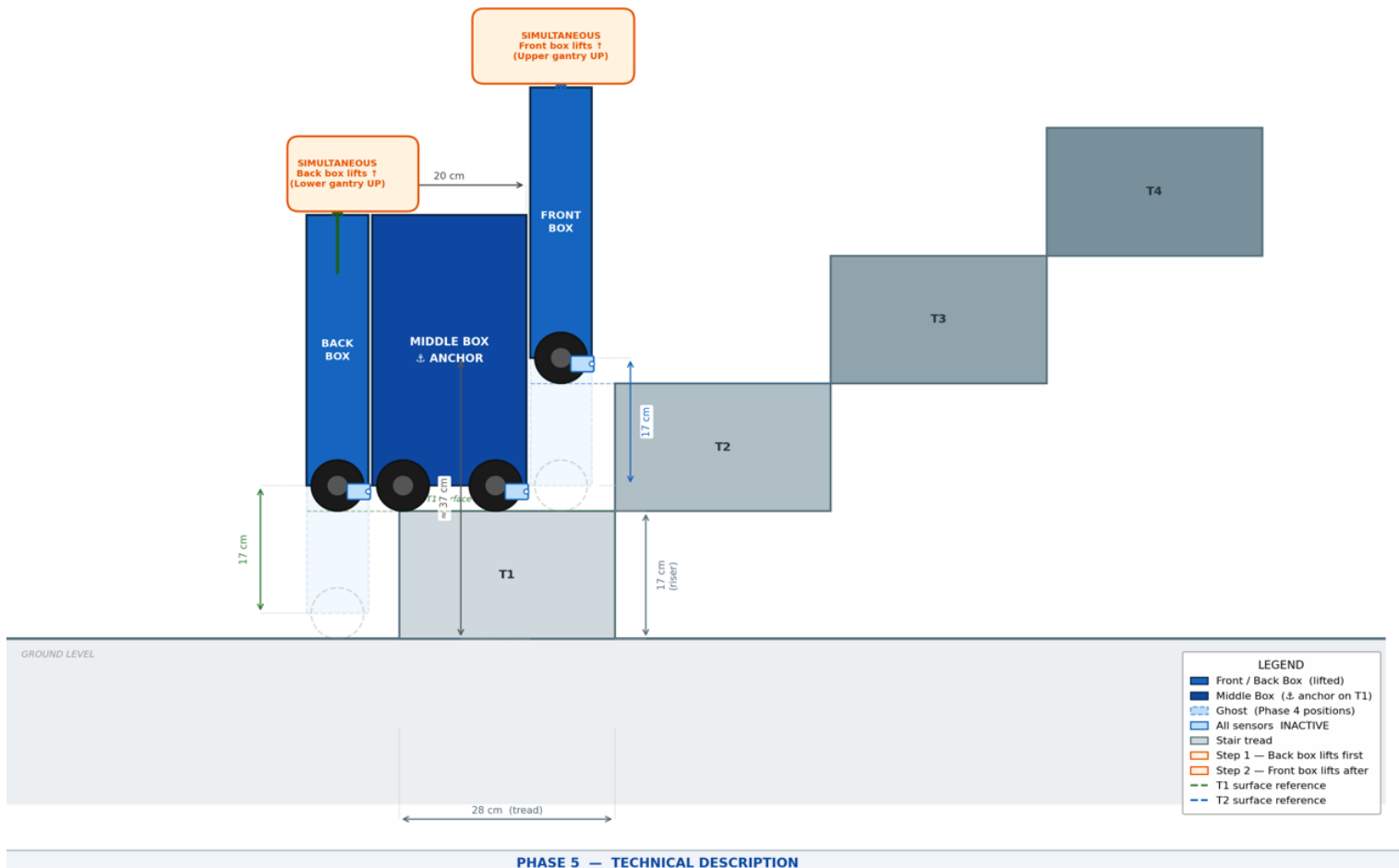
Figure 5.8 — Phase 4: middle box advancing onto upper tread, front and middle boxes on upper tread



Phase 5 — Back Box Lift

Trigger (sensor-triggered approach): Back box HC-SR04 stair detection sensor reads ≤ 3 cm from riser face. Action: Rear-facing gantry motor pair activates (Driver 6). With the front and middle boxes resting on the upper tread as anchors, the rear gantry motors lift the back box vertically. The back box rises until the back box stair detection sensor reads > 3 cm, confirming it has cleared the riser face. The gantry motors continue briefly to cover the 34 mm wheel radius offset, then stop. Here, with 280 mm tread size both front and back box sensors detect the stairs at the same time, that makes it ideal dimension for this robot.

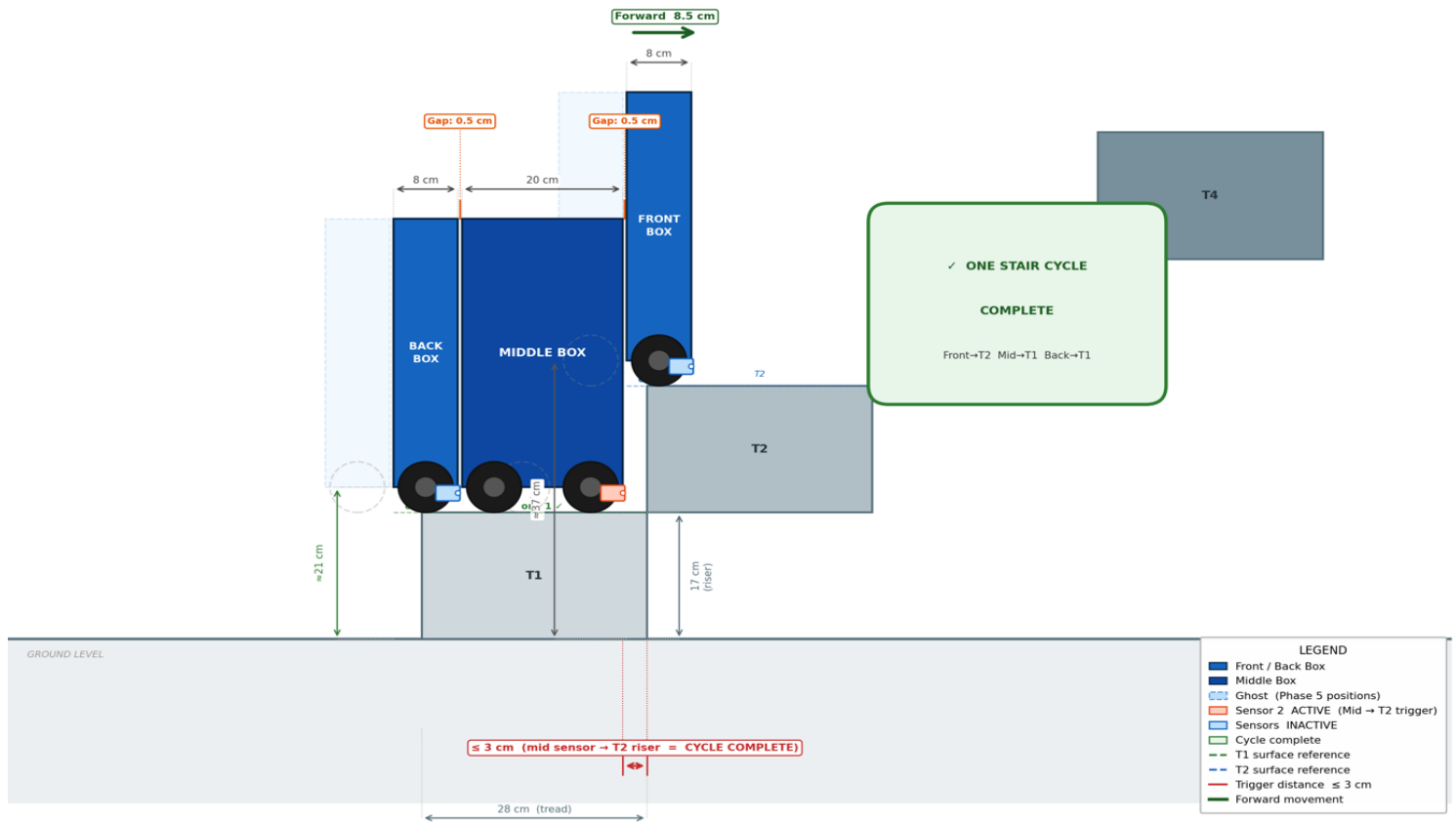
Figure 5.9 — Phase 5: back box being lifted by rear gantry motors, front and middle boxes anchored on upper tread



Phase 6 — Final Translation and Reset

Action: All wheel motors activate and drive the robot forward. The elevated back box moves over the upper tread and descends naturally onto it under its own weight. All boxes are now above lower base. The encoder and MPU-6050 are re-enabled by the Arduino firmware. Flat terrain navigation resumes. If another stair is detected immediately, the sequence restarts from **Phase 1**.

Figure 5.10 — Phase 6: all three boxes on upper tread, stair step complete, robot level



PHASE 6 — TECHNICAL DESCRIPTION

5.5 Phase Summary

The following table provides a consolidated reference for the six-phase stair climbing sequence, including the trigger condition, motors active, and outcome for each phase.

Table 5.4: Six-phase stair climbing sequence summary (general situation)

Phase	Trigger / Condition	Motors Active	Outcome
Phase 1	Front sensor $\leq 3\text{cm}$	Front gantry pair (Driver 5)	Front box lifted above riser
Phase 2	Lift complete	wheel motors	Robot advances, front box lands on upper tread
Phase 3	Mid sensor $\leq 3\text{cm}$	All 4 gantry motors (Drivers 5+6)	Middle box lifted via constraint-reaction

Phase 4	Lift complete	wheel motors	Robot advances, middle box lands on upper tread
Phase 5	Back sensor $\leq 3\text{cm}$	Rear gantry pair (Driver 6)	Back box lifted above riser
Phase 6	Lift complete	All wheel motors	Back box lands, all boxes level above base

5.6 Sensor Roles During Stair Climbing

In the sensor-triggered stair climbing approach, three HC-SR04 ultrasonic sensors — one at the bottom-front of each box — serve as the inputs to the stair climbing state machine. The MPU-6050 and encoder are both disabled during the sequence, as described in **Chapter 4**. The table below summarises the role of each sensor in the phase transitions of this approach.

Table 5.5: Stair detection sensor roles per phase

Sensor	Location	Triggers	Direction Signal
Front box sensor	Bottom-front, front box	Phase 1 start	Front gantry lift ON
Middle box sensor	Bottom-front, middle box	Phase 3 start	All gantry motors ON
Back box sensor	Bottom-front, back box	Phase 5 start	Rear gantry lift ON

5.7 Observed Behaviour and Limitations

The stair climbing sequence was tested on a physical staircase over multiple cycles. The following observations were recorded during prototype testing and are documented as known limitations at this stage of development.

5.7.1 Box Tilt During Phase Transitions

During **Phase 3** (middle box lift), slight differential lift was observed between the left and right sides of the middle box. This is attributed to minor variation in belt tension and motor loading between the left and right gantry motor pairs. The tilt was self-correcting once the lift phase completed and the box settled onto the tread. No structural damage resulted from the observed tilt range.

5.7.2 Stair Nosing Contact

At certain stair profiles, the lower edge of the front box made contact with the stair nosing (the protruding front edge of the upper tread) during **Phase 2** translation. This contact momentarily increased the resistance on the wheel motors and in some test cycles caused a brief stall. The issue is more pronounced on stairs with a pronounced nosing overhang. A chamfered leading edge on the front box bottom rail is identified as a solution for future iterations.

5.7.3 Belt Elongation Under Cyclic Loading

GT2 rubber timing belts are subject to elongation under repeated cyclic tensile loading. After extended stair climbing test cycles, a measurable increase in belt slack was observed, requiring re-tensioning via the eccentric spacers on the carriage wheels. Belt elongation is a documented limitation of rubber belt drives under high-load conditions and is acknowledged as a maintenance requirement in the current design. A steel cable or rigid rack-and-pinion system is identified as a more durable alternative for future development.

5.7.4 Open-Loop Phase Timing

Gantry lift phases are terminated by sensor feedback — the gantry stops when the stair detection sensor transitions from $\leq 3\text{cm}$ to $>3\text{cm}$, with a brief continuation to cover the 34mm wheel radius offset. This means the mechanism adapts naturally to different riser heights within the compatible range, since the sensor reading always reflects the actual box position relative to the riser face. However, the brief continuation after the $>3\text{cm}$ reading is a fixed duration calculated from the wheel radius and gantry operating speed. If the actual gantry speed varies from the value used in this calculation — due to battery voltage drop or motor load variation — the wheel alignment may be slightly off. Integration of a positional limit switch at the target height would eliminate this residual open-loop element in future iterations.

5.7.5 Traversal Speed

The overall stair climbing speed at prototype stage is slow. The total time to complete the full six-phase sequence per step was not formally measured but was observed to be significantly longer than a human stair climbing pace. This is acceptable for a proof-of-concept prototype but would require significant improvement for practical campus deployment. Increasing motor speed, reducing phase timing margins, and parallelising phases where mechanically safe are identified as the primary optimisation paths.

5.8 Prototype Validation

The stair climbing mechanism was validated through physical testing on a staircase with a tread depth of approximately 290mm and riser height of approximately 170mm — within the compatible range and representative of typical Indian educational building stair construction. The six-phase sequence was completed across four consecutive steps in multiple test runs. Video documentation of the stair climbing demonstration is maintained as supplementary evidence.

For the final prototype deployment, the stair climbing sequence is executed using pre-programmed timing and distance-based control, calibrated to the known stair dimensions of the Parul University campus. This approach was selected because the target staircase has consistent and known dimensions, and because vibration during motor operation was found to affect sensor placement stability during extended testing. The sensor-triggered mechanism described throughout this chapter remains a valid, tested, and more general approach suitable for deployment on staircases with variable or unknown dimensions.

The validation confirms that the core mechanical and control design is sound at prototype stage. The limitations identified in **Section 5.7** are characteristic of a first-generation prototype and do not invalidate the fundamental operating principle of the sequential gantry lift mechanism.

CHAPTER 6

PROTOTYPE IMPLEMENTATION AND TESTING

6.1 Overview

This chapter documents the physical construction of the AUTOM8 prototype and the testing conducted to evaluate its performance. The construction process involved fabrication of the aluminium profile chassis, assembly of the mechanical and electronic subsystems, and iterative refinement of motor mounting and wiring arrangements. The testing programme progressed through three distinct stages of stair-climbing control, each informing the next. Additional tests were conducted for obstacle detection, heading correction, and encoder-based distance estimation. Results and observations from all tests are documented in this chapter.

6.2 Prototype Construction

6.2.1 Tools Used

The following tools were used during fabrication and assembly of the AUTOM8 prototype. No CNC machining or power drilling was used at any stage — all fabrication was carried out using hand tools and workshop equipment available at the institution.

Table 6.1: Tools used during prototype construction

Tool	Purpose
Profile cutter (workshop)	Cutting 20×20mm aluminium extrusion profiles to required lengths
File / deburring tool	Smoothing cut profile ends to ensure clean seating in corner joints
Vernier caliper + measuring tape	Measuring profile lengths before cutting and verifying assembled dimensions
Hex key / Allen key set	Tightening M5 grub screws, inner bracket bolts, and motor mount fasteners
Soldering iron	Soldering motor wiring connections, battery cell interconnects, and sensor leads

Wire stripper	Preparing wire ends before soldering and connector crimping
Heat shrink tubing + electrical tape	Insulating solder joints and securing wire bundles
Multimeter	Measuring voltage, current, and continuity during electronics assembly and testing

6.2.2 Structural Frame Fabrication

Aluminium extrusion profiles were cut to the required lengths using a profile cutter in the institution workshop. Each profile length was measured against the design dimensions using a vernier caliper and marked before cutting. Cut ends were deburred with a file to ensure flat, clean faces that seat correctly in the 3-way corner joints. The T-slot and V-slot profiles were assembled into the three box frames using corner joints, inner brackets, and M5 grub screws tightened with hex keys. The 5mm inter-box gap between adjacent boxes was set during assembly by placing 5mm spacers between the boxes and tightening all fasteners with the spacers in position before removal. Maintaining this gap consistently across both sides of each interface required careful alignment and was one of the more time-consuming steps of the structural assembly.

6.2.3 Gantry Belt, Pulley, and Motor Assembly

The gantry subsystem assembly was the most time-consuming and technically demanding part of the construction process. Each 100 RPM gantry motor was mounted on the middle box frame using a steel U-clamp, with rubber padding inserted between the motor body and the clamp to dampen vibration. Achieving correct motor shaft alignment with the V-slot rail axis required multiple trial iterations before a satisfactory position was found for each motor.

The GT2 aluminium pulleys were pressed onto the 5mm D-type motor shafts and secured with the grub screw. GT2 rubber timing belts were then routed from each pulley to the corresponding gantry carriage plate on the front or back box V-slot rail. Achieving correct belt tension was a particularly challenging step — the belt must be taut enough to transmit lifting force without slipping, but not so tight as to overload the motor shaft bearings. The eccentric spacers on the carriage plate V-wheels were used for final tension adjustment. Multiple adjustment cycles were required before consistent, slip-free lifting was achieved on all four belt-pulley pairs.

6.2.4 Motor Mounting and Vibration Control

All twelve motors — four gantry and eight-wheel motors — were mounted using 2-inch steel U-clamps. The Johnson DC motors produce significant vibration under load, particularly at higher PWM duty cycles. Initial test runs revealed that this vibration was transmitting through the aluminium frame, causing rattling and instability in the sensor mounts. To address this, rubber padding was inserted between each motor body and its U-clamp before final tightening. This simple solution significantly reduced transmitted vibration. The final motor mounting arrangement for each motor was arrived at through trial and error, testing different rubber pad thicknesses and clamp torque levels until acceptable vibration levels were achieved.

Wheel alignment with the motor shaft and floor surface was a separate challenge. Each wheel had to be seated squarely on the motor shaft and positioned so that all wheels on a given box made consistent contact with the floor. Misaligned wheels caused uneven drive and increased current draw. Alignment was checked by running the robot on a flat surface and observing for drift, then adjusting individual wheel positions until straight-line travel was consistent.

6.2.5 Electronics Assembly and Wiring

All electronic components were mounted on base boards inside the three boxes. The middle box was divided into two internal compartments: the upper compartment housing the Arduino Mega 2560, gantry motor drivers, MPU-6050, and buzzer; the lower compartment housing the battery pack, buck converter, middle wheel motor drivers, and power switches. Components were secured to the base boards using PCB standoffs and zip ties.

Battery pack assembly involved arranging twelve 18650 cells in the 3S4P configuration in two plastic cell holders, connecting the cells with nickel strip interconnects, and soldering the BMS board connections. All solder joints on the battery pack were insulated with heat shrink tubing before the pack was wrapped in insulation tape for the final assembly.

Wiring the twelve motors across six L298N driver modules produced a large number of connections, which initially caused significant confusion during assembly and troubleshooting. To manage this, all wires were labelled at both ends before installation, then routed and secured with zip ties along defined paths within each box. This labelling and management system, implemented after initial wiring difficulties, made subsequent testing and fault-finding considerably more straightforward.

6.3 Stair Climbing Testing

Stair climbing testing was conducted across three stages, each using a different control approach. Each stage informed the next. All stair climbing tests were conducted on the same physical staircase. Video documentation of the testing is maintained as supplementary evidence. The final deployment uses the timing and distance-based approach from **Stage 1**, calibrated to the known stair dimensions of the campus, as this was found to be the most reliable method for the specific deployment environment.

6.3.1 Stage 1 — Timing and Measurement-Based Control

The first stair climbing trials used a purely timing and measurement-based control approach. The phase sequence was programmed with fixed time delays for each gantry lift phase and fixed encoder-counted distances for each translation phase. No sensor feedback was used to trigger phase transitions — the sequence simply executed each step for a predetermined duration before moving to the next.

The primary challenge with this approach was speed control. The wheel and gantry motor speeds were difficult to set consistently due to variation in battery voltage across the charge cycle and mechanical load variation between test runs. At speeds that were too high, the gantry would overshoot the target lift height; at speeds that were too low, the boxes would not clear the riser fully. Arriving at a set of timing values that produced reliable climbing required extensive trial-and-error iteration.

After sufficient iteration, the robot was able to climb stairs with this approach. However, observable tilting of boxes during phase transitions was present in all successful runs, and occasional collisions between the front box profile and the stair nosing occurred when translation speed was slightly above the tuned value. Despite these limitations, **Stage 1** demonstrated that the fundamental sequential gantry lift mechanism was mechanically viable.

An important observation from **Stage 1** is that for a fixed, known staircase with consistent dimensions, a timing and measurement-based approach can be a viable and simpler alternative to sensor-based control. Once tuned for a specific stair geometry, the sequence can be reliably reproduced without any sensor input. This makes the timing-based approach suitable for dedicated fixed-route deployments.

6.3.2 Stage 2 — IR Sensor-Triggered Control

Following the limitations observed in **Stage 1**, the team implemented IR proximity sensor triggering to replace the fixed timing approach. Three LM393-based IR sensors were mounted at the bottom-front of each box, and the phase sequence was modified to trigger each gantry lift phase upon detection of the stair riser at less than 5cm.

The IR sensor approach produced poor results in practice. The primary cause of failure was the

sensitivity of IR sensors to ambient lighting conditions. The stair climbing tests were conducted in an environment with variable lighting — a combination of overhead fluorescent lighting and daylight from nearby windows. Under these conditions, the IR sensors produced inconsistent readings: the ambient infrared radiation from fluorescent tubes and daylight partially saturated the receiver, causing the sensor to report false detections or fail to detect the riser reliably at the correct distance. The result was that the phase sequence triggered at incorrect points, with some phases triggering prematurely and others failing to trigger at all.

Additional limitations of IR sensors in this application include sensitivity to the surface colour and reflectivity of the stair material. Dark or matte stair surfaces absorb a higher proportion of the emitted infrared signal, reducing the reflected intensity reaching the receiver and shortening the effective detection range below the 5cm threshold. The wide 35° detection angle of the sensor also caused false triggers from the floor surface ahead of the riser in some orientations. Taken together, these factors made IR sensors unsuitable for reliable stair detection in a real-world indoor environment, and the approach was abandoned.

6.3.3 Stage 3 — *HC-SR04 Ultrasonic Sensor-Triggered Control*

Following the failure of IR sensors in **Stage 2**, HC-SR04 ultrasonic sensors were implemented for stair detection. Ultrasonic sensors operate on sound reflection rather than light, making them immune to ambient lighting conditions and surface colour variations that had affected the IR sensors. The HC-SR04 sensors produced significantly more consistent and reliable riser detection across all test conditions, and the sensor-triggered mechanism was successfully validated during testing.

With the ultrasonic sensor-triggered control, the robot successfully climbed stairs within the compatible geometry range across multiple test runs. The phase sequence executed correctly in the majority of trials, with each lift phase initiating reliably when the corresponding sensor read below 3cm and terminating correctly when the sensor transitioned above 3cm after the box cleared the riser face. However, during extended testing it was observed that motor vibration caused occasional physical displacement of the sensor mounts, affecting detection consistency. Combined with the fact that the target deployment staircase at Parul University has known and consistent dimensions, the timing and distance-based approach from **Stage 1** was adopted as the final deployment method. The sensor-triggered mechanism remains a valid and tested approach for environments where stair dimensions are variable or unknown.

The following limitations were observed during **Stage 3** testing and are documented as known constraints of the current prototype:

Sensor response delay: A notable delay was observed between the physical event of the sensor reading crossing the 3cm threshold and the Arduino Mega executing the corresponding phase transition. This delay, inherent in the polling-based sensor reading approach implemented in the firmware, caused slight positional overshoot at the point of phase transition. The effect was more pronounced at higher wheel PWM settings.

Middle box tilt: During **Phase 3**, when the middle box is lifted via the constraint-reaction mechanism, observable tilting of the front and back boxes was recorded. This tilt is attributed to minor differences in belt tension and loading between the left and right gantry motor pairs. The tilt was self-correcting — by **Phase 4**, when the middle box settled onto the upper tread, the robot returned to a level orientation and subsequent phases executed normally.

Speed and collision tradeoff: At higher wheel PWM settings, the front box profile was observed to collide with the stair nosing during **Phase 2** translation. This occurred because the sensor response delay, combined with the higher translation speed, did not allow sufficient time for the phase transition to execute before the front box reached the nosing. Reducing the wheel PWM eliminated the collision but increased the overall climbing time. A balance between speed and reliable phase execution was arrived at through testing.

Belt loosening: After approximately three to four consecutive stair climbing cycles, measurable slack developed in the GT2 timing belts. This required re-tensioning via the eccentric spacers on the gantry carriage plates before further testing. Belt elongation under cyclic tensile loading is a known characteristic of rubber timing belts and is identified as a maintenance requirement in the current design.

6.4 Obstacle Detection Testing

The obstacle detection system was tested by placing physical objects of varying sizes directly in the path of the robot during flat terrain navigation. When the forward-facing HC-SR04 sensors on the front box detected an object within the configured detection range, the robot stopped immediately and the buzzer activated simultaneously. The robot remained stationary and the buzzer continued sounding for as long as the obstacle remained within detection range. Once the obstacle was removed from the robot's path, forward motion resumed automatically without any manual intervention, confirming the reactive obstacle detection and halt behaviour described in **Chapter 4**. The system responded consistently across all obstacle tests conducted. No video documentation of this test was recorded; the results are documented based on direct observation during testing. The obstacle detection trigger distance was set during testing

to a value that provided sufficient stopping distance at the robot's operating wheel speed. The configured trigger distance is 25 cm.

6.5 Encoder Distance Estimation Testing

The magnetic Hall effect encoder mounted on the front wheel motor shaft was tested for distance estimation accuracy on flat ground. The robot was driven a known distance along a straight corridor and the distance reported by the encoder pulse count was compared against the physical measurement. The encoder produced a reasonable approximation of the actual distance travelled; however, measurable error was present in all test runs. The exact deviation was not formally quantified at prototype stage. The error is attributed to the single-encoder configuration — with only one wheel monitored, any slip or differential between wheels accumulates as distance error over longer runs.

For the purposes of the pre-programmed delivery route at prototype stage, the encoder-based distance estimation was sufficient to execute the fixed sequence of move and turn commands. The accumulated error over the test route length did not prevent successful completion of the route, though final stopping position showed some variation between runs. Improving distance estimation accuracy through a dual-encoder configuration with slip detection is identified as a priority improvement for future iterations.

6.6 MPU-6050 Heading Correction Testing

The MPU-6050 heading correction system was tested independently on flat ground. With the heading correction system active, the robot was observed to navigate in a consistently straight line, with the Arduino Mega continuously monitoring heading variation and applying corrective PWM adjustments to counteract any drift. The real-time heading monitoring was verified to be active throughout flat terrain navigation by observing the serial output from the Arduino during testing.

To evaluate the correction response, the robot was physically deflected from its heading during operation — an operator applied a lateral force to rotate the robot off its original direction. The robot responded by actively resisting the deflection and returning to its original heading direction once the external force was removed. The resistance to deflection was notably strong, resulting from two combined effects: the grip of the rubber-tyred wheels on the floor surface resisting lateral sliding, and the active PWM correction applied by the MPU-6050 heading correction firmware working to counteract the imposed rotation. This combined mechanical and active resistance was consistently observed across multiple deflection tests.

The exact angular deflection applied during the test was not measured. The test was conducted as a qualitative demonstration of the heading correction capability rather than a quantitative characterisation. Video documentation of this test was recorded and is maintained as supplementary evidence. The test confirmed that the MPU-6050 heading correction system functions as designed during flat terrain navigation.

Figure 6.1 — MPU-6050 heading correction test — robot maintaining straight heading during flat terrain navigation

(The two videos can be accessed by clicking on the images)



6.7 Battery Runtime

The 3S4P battery pack was operated through a full discharge cycle during extended prototype testing. The robot was run continuously through a combination of flat terrain navigation and stair climbing cycles until the battery management system cut off power at the low-voltage protection threshold. The total runtime was not formally measured during this test. The battery pack performed without any cell

failure, thermal event, or BMS intervention beyond the normal low-voltage cutoff across all testing sessions.

6.8 Testing Summary

The following table summarises the testing conducted on the AUTOM8 prototype, the method used, and the overall outcome of each test.

Table 6.2: Prototype testing summary

Test	Method	Outcome
Stair climbing — Stage 1	Timing + measurement-based control	Functional with tuning; tilting and occasional nosing collision observed
Stair climbing — Stage 2	IR sensor-triggered	Unreliable; ambient light and surface reflectivity caused false triggers
Stair climbing — Stage 3 (HC-SR04)	HC-SR04 ultrasonic sensor-triggered	Validated; sensor vibration displacement noted; timing/distance approach adopted for final deployment
Obstacle detection	Physical obstacle placed in path	Robot stopped and buzzer activated; resumed automatically when obstacle removed
Encoder distance estimation	Known distance comparison	Approximate; measurable error present, not formally quantified
MPU-6050 heading correction	Straight-line navigation + deflection test	Heading maintained; strong resistance to deflection confirmed
Battery runtime	Full discharge during extended testing	Full cycle completed; runtime not formally measured
Overall	Integrated prototype testing	Core mechanism validated; limitations documented for future iteration

6.9 Prototype Component Costing

The following table presents a complete component-by-component cost breakdown for the AUTOM8 prototype. All prices are as purchased during the project period (September 2023 – March 2024) from Indian suppliers. Components purchased solely for the testing phase and not present in the final assembled robot are marked accordingly. The grand total reflects the complete expenditure; the final robot cost excludes testing-only items.

Table 6.3: Complete component cost breakdown — AUTOM8 prototype

Component / Description	Qty	Unit Price	Total	Notes
Aluminium profiles + hardware (T-slot, V-slot, joints, brackets, grub screws)	1 lot	—	₹12,121	
100 RPM Johnson DC gantry motor (Robu.in)	4	₹561	₹2,244	
200 RPM Johnson DC wheel motor (Robu.in)	8	₹540	₹4,320	
Robot wheel 68mm (Robokits India)	8	₹70.80	₹566	
V-slot mini gantry plate set (Probots.co.in)	4	₹899	₹3,596	
GT2 belt + pulley kit (2m belt + 2×20T pulleys)	2	₹200	₹400	
Orange ICR18650-22C battery cell (Robu.in)	12	₹156	₹1,872	
3S 12V 25A BMS protection board (Robu.in SKU 43646)	1	₹114	₹114	
L298N motor driver module (Robu.in SKU 43821) — 6 active + 4 replacement	10	₹130	₹1,300	
Arduino Mega 2560 Rev3 (final robot)	1	₹1,500	₹1,500	
Arduino Uno (testing only — not in final robot)	1	₹500	₹500	Testing only
HC-SR04 ultrasonic sensor (Robu.in SKU 2444)	6	₹61	₹366	
IR proximity sensor LM393 (Robu.in SKU 7748)	3	₹28	₹84	Testing only — replaced by HC-SR04 in final design
MPU-6050 IMU (Robu.in SKU 2846)	1	₹173	₹173	

Magnetic Hall effect encoder	1	₹400	₹400	
LM2596S DC-DC buck converter (Robu.in SKU 11548)	1	₹100	₹100	
SPST rocker switches	2	₹50	₹100	
Metal U-clamp 2 inch (motor mounts)	12	₹40	₹480	
Battery charger (for 3S Li-ion pack)	1	₹500	₹500	
Multimeter (DT830D digital)	1	₹180	₹180	
Wire cutter	1	₹120	₹120	
Miscellaneous (wiring, zip ties, tape, fasteners, delivery charges, stationery)	1 lot	—	₹2,500	
GRAND TOTAL — Complete Prototype (including testing components)	—	—	₹33,536	
FINAL ROBOT ONLY (excluding testing-only components)	—	—	₹32,952	Excl. Arduino Uno + IR sensors

The total prototype cost of approximately ₹33,000 demonstrates that a functional stair-climbing campus delivery robot can be realised at a cost accessible to undergraduate project budgets. The single largest cost category is the aluminium extrusion structural frame at ₹12,121 (36% of total), followed by the drive and gantry motors at ₹6,564 combined (20%). The V-slot gantry plate sets at ₹3,596 represent the highest per-unit cost among the electronic and mechanical components. To the best of the authors' knowledge, the total cost compares favourably with similar stair-climbing robot prototypes reported in the literature, which typically require significantly higher expenditure due to more complex sensor and computing platforms.

CHAPTER 7

PROPOSED FUTURE WORK

7.1 Overview

The AUTOM8 prototype demonstrates the core concept of a stair-climbing campus delivery robot and validates the sequential gantry lift mechanism as a viable approach. However, the prototype stage reveals a number of limitations and unexplored directions that form the basis for future development. This chapter documents the proposed improvements and extensions identified through the prototype testing and design process. Each proposed direction is directly motivated by a specific limitation, observation, or design decision documented in the preceding chapters.

7.2 Mechanical and Structural Improvements

7.2.1 Belt Replacement — Rack and Pinion or Steel Cable

The GT2 rubber timing belts used in the current prototype were observed to elongate under cyclic tensile loading, requiring manual re-tensioning after approximately three to four stair climbing cycles. For a production-grade system intended for continuous deployment, this maintenance requirement is unacceptable. A rack-and-pinion mechanism or steel cable drive would eliminate belt elongation entirely, providing a mechanically more durable lifting drive. Both alternatives offer positive engagement with no slip and significantly longer service life under repeated loading.

7.2.2 Payload Compartment Design

The current prototype does not include a dedicated payload compartment. Delivery items would need to be placed on the middle box frame, which is not suitable for practical use. A future iteration should include a lockable payload enclosure mounted on the middle box, sized appropriately for the target delivery items — food containers, books, or small parcels. The compartment should be accessible to the client at the pickup point and to the recipient at the delivery point, with a simple locking mechanism that can be released via the accompanying web interface.

7.2.3 Weight Reduction

The current prototype weighs approximately 12kg unloaded, primarily due to the dense aluminium extrusion frame construction. The die-cast three-way corner joints alone contribute approximately 1,080g.

A future iteration could investigate lighter aluminium alloy extrusions, 3D-printed structural connectors, or carbon fibre composite elements for non-load-bearing components to reduce overall robot weight and improve payload-to-weight ratio.

7.2.4 Chamfered Front Box Leading Edge

Stair nosing collision was observed during **Phase 2** translation in testing. A chamfered or angled leading edge on the bottom rail of the front box would allow the front box to ride over the nosing profile rather than colliding with it, reducing the impact force and eliminating the need to reduce wheel speed to avoid the collision.

7.2.5 Mechanical Home-Position Stop for Equal Box Height

In the current prototype, the front and back box wheels do not make consistent contact with the floor on flat terrain. This occurs because the gantry carriage plates settle at slightly different heights under the weight of each box, causing the front and back boxes to sit marginally higher than the middle box. As a result, only the four middle box wheels provide traction during flat terrain navigation. A mechanical home-position stop — a fixed physical limit on each V-slot rail that the carriage plate rests against when in the fully lowered position — would ensure all three boxes settle at exactly the same height, engaging all eight wheels simultaneously on flat ground. This is a simple mechanical addition to the existing rail design and is identified as a high-priority improvement for the next prototype iteration.

7.3 Electrical and Control System Improvements

7.3.1 Closed-Loop Gantry Control

The current gantry lift termination relies on the stair detection sensor transitioning from $\leq 3\text{cm}$ to $>3\text{cm}$, plus a fixed additional duration to cover the wheel radius offset. This residual open-loop element means that variation in actual gantry speed can affect wheel alignment accuracy. A limit switch or optical position sensor at the target lift height on each V-slot rail would provide hard positional feedback, terminating the lift phase exactly when the correct height is reached regardless of motor speed variation.

7.3.2 MPU-6050 Integration During Stair Climbing

In the current design, the MPU-6050 is disabled during the stair climbing sequence because the intentional box tilting during phase transitions cannot be distinguished from an unintended disturbance. A future firmware implementation could selectively use only the yaw axis of the MPU-6050 during climbing phases, ignoring the pitch and roll axes that respond to the intentional tilt. This would allow lateral heading correction to remain active during stair climbing without interfering with the lifting motion.

7.3.3 Dual Encoder with Slip Detection

The single encoder on the front wheel motor provides only a rough approximation of travelled distance with no ability to detect wheel slip. A dual encoder configuration — one on each side of the robot — would allow the Arduino to compare left and right wheel travel and detect any differential. This would significantly improve straight-line tracking accuracy and precision of pre-programmed turn execution.

7.3.4 Upgraded Motor Drivers

The L298N motor driver modules are rated at 2A per channel continuous, which was observed to cause thermal rise during extended stair climbing test cycles. Higher-rated motor drivers such as the BTS7960 H-bridge module would provide substantial thermal headroom and enable higher PWM duty cycles with better current regulation.

7.3.5 Enclosed Electronics and Professional Cable Management

The prototype-stage exposed wiring presents a risk of short circuits under vibration and makes the robot unsuitable for deployment in a live campus environment. A future iteration should house all electronics in enclosed, ventilated enclosures with proper connector systems, strain relief, and labelled cable routing.

7.4 Navigation and Autonomy Improvements

7.4.1 Sensor-Based Dynamic Path Execution

The current navigation system executes a pre-programmed sequence of encoder-counted distances and turn commands, with no ability to adapt to unexpected route variations. A future iteration could implement sensor-based path following using floor markings or RFID waypoints, allowing the robot to navigate reliably without accumulated encoder error and to handle minor route deviations.

7.4.2 Multi-Floor Operation

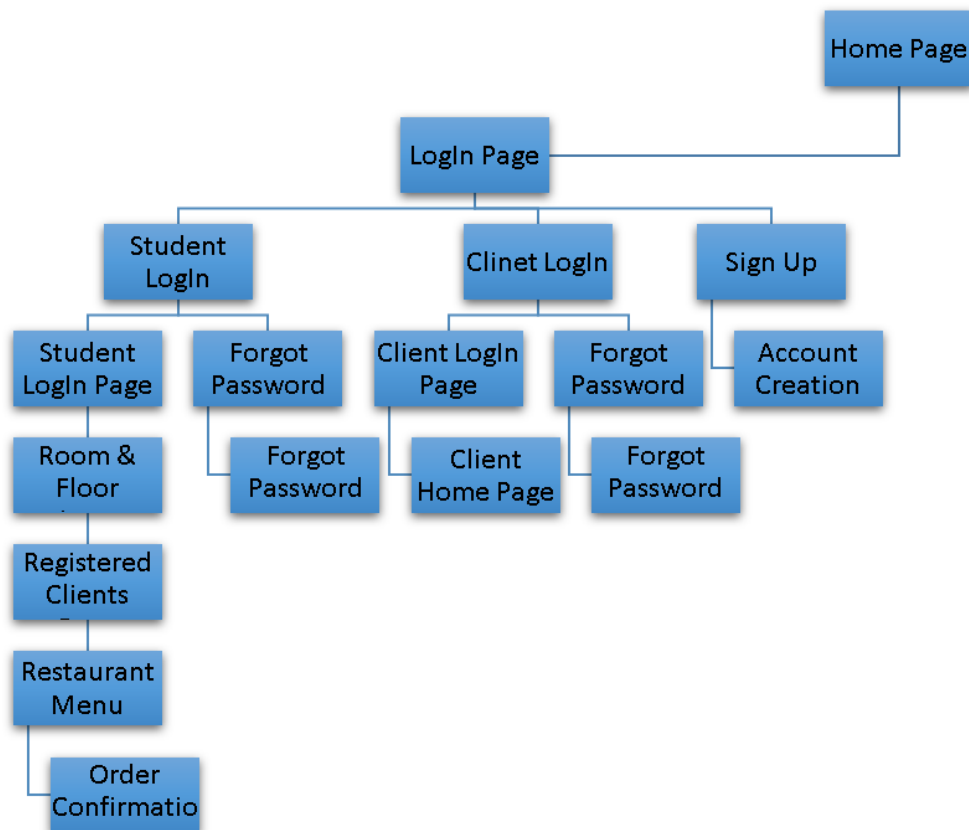
The current prototype was designed and tested for climbing a single staircase. Extending the system to navigate multiple floors would require floor detection logic — possibly via RFID or QR code markers at each landing — and a return-to-ground-floor sequence after delivery completion.

7.5 Web Interface — Prototype UI Design

As part of the original project scope, a web-based user interface was designed to demonstrate the intended end-to-end delivery workflow for AUTOM8. The interface was developed as a prototype UI to

showcase the potential of the delivery system concept. The interface separates users into two roles: students who place orders, and clients (restaurant owners and service providers such as the library) who receive and confirm orders.

Figure 7.1 — AUTOM8 website navigation flow chart

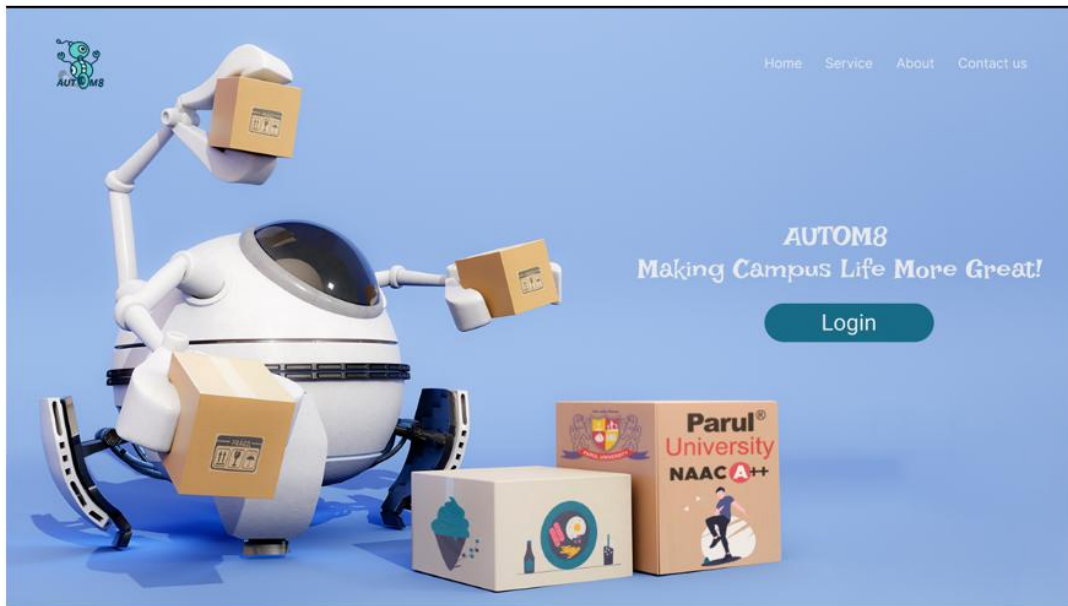


The interface consists of ten screens, each corresponding to a distinct step in the ordering workflow. The screens are documented below with the prototype UI designs.

7.5.1 Home Page

The home page serves as the entry point for all users. It displays the AUTOM8 logo and tagline ‘Making Campus Life More Great!’ alongside navigation links and a Login button that directs users to the authentication page.

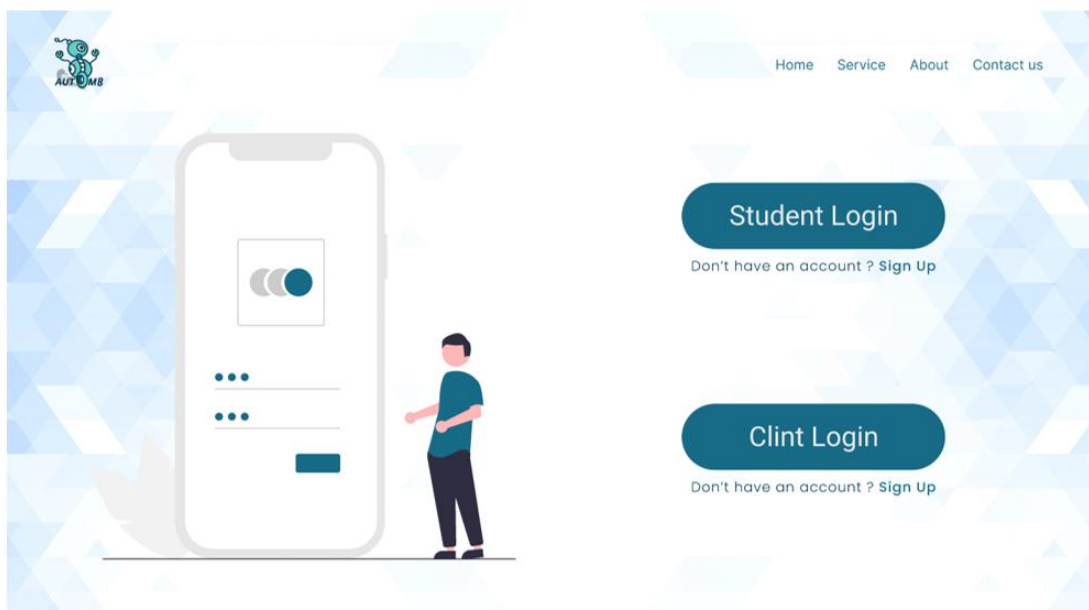
Figure 7.2 — AUTOM8 website Home Page



7.5.2 Login Main Page

From the login page, users select their role: Student/User or Client/Restaurant Owner. A Sign-Up option is available for new users, routing each user type to a role-appropriate interface.

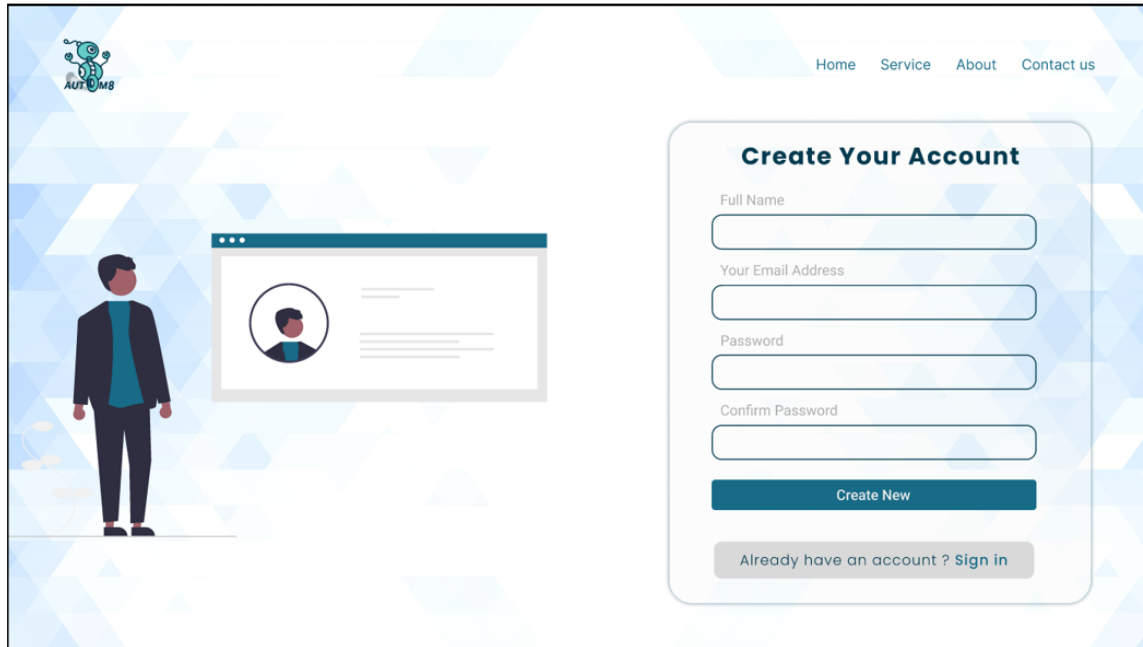
Figure 7.3 — Login Main Page showing Student and Client login options



7.5.3 Account Creation

New users register through the account creation page, entering their full name, email address, and password. The account type (student or client) is selected during registration to assign the correct permissions.

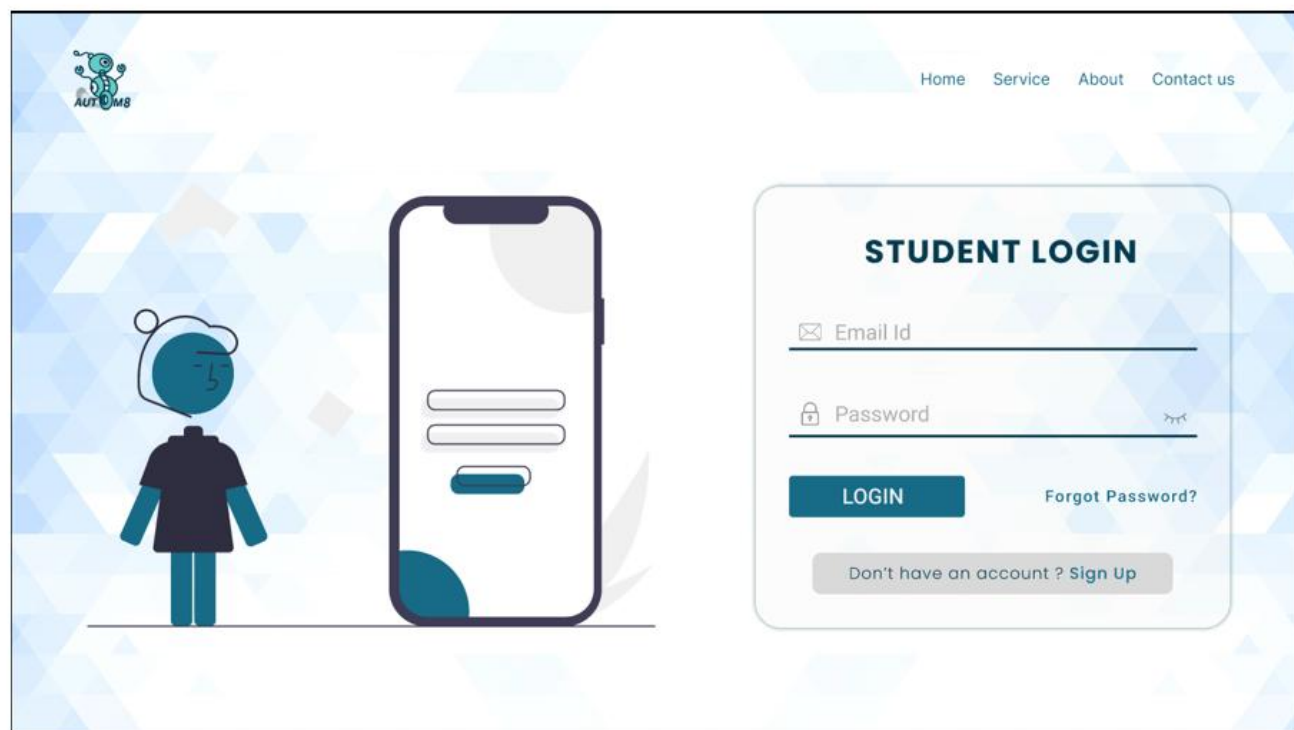
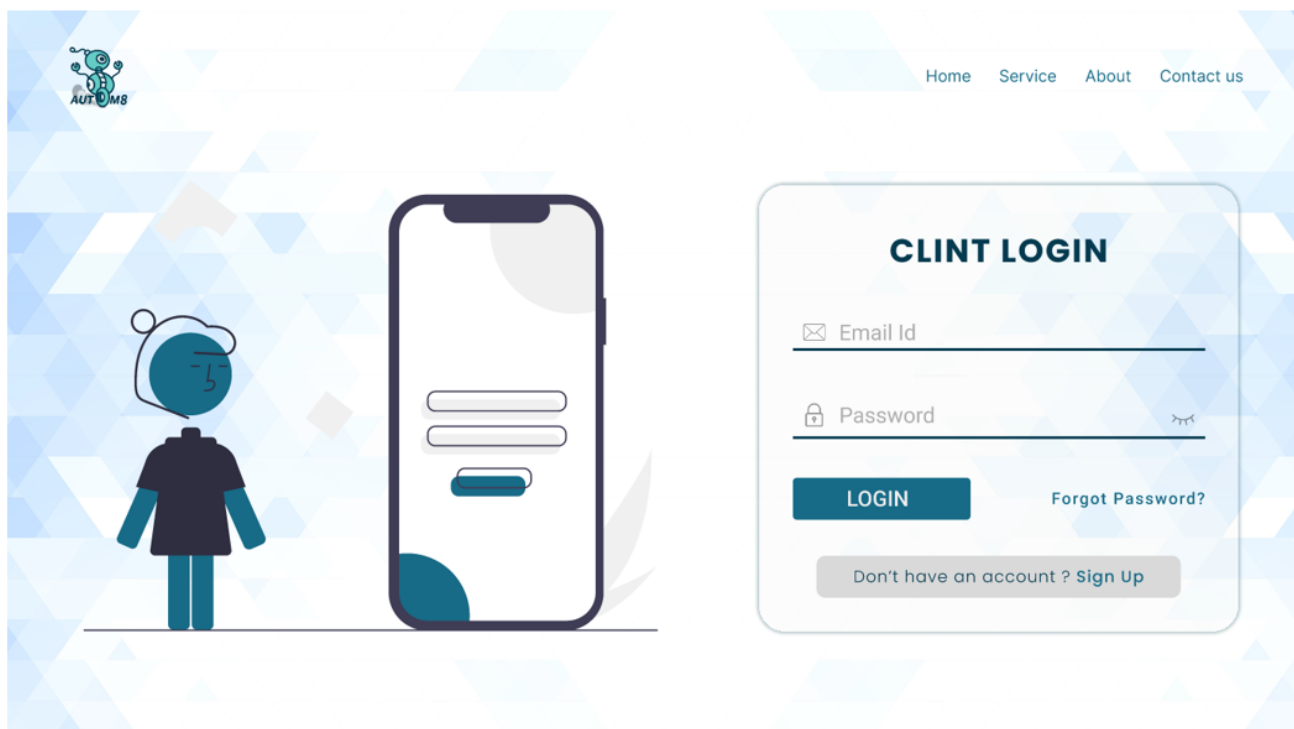
Figure 7.4 — Account Creation page

The screenshot shows the 'Create Your Account' page of the AUTOM8 website. The page has a light blue geometric background. On the left, there is an illustration of a person in a dark suit and teal shirt standing next to a large screen displaying a user profile. The screen shows a circular profile picture and several lines of text. In the top right corner, there is a navigation menu with links: Home, Service, About, and Contact us. The main content area on the right is titled 'Create Your Account' and contains four input fields: 'Full Name', 'Your Email Address', 'Password', and 'Confirm Password'. Below these fields is a dark blue 'Create New' button. At the bottom of the form, there is a link that says 'Already have an account ? Sign in'.

7.5.4 Student and Client Login Pages

Returning users authenticate through separate login pages for students and clients. Both pages include a Forgot Password option and a link to account creation for new users.

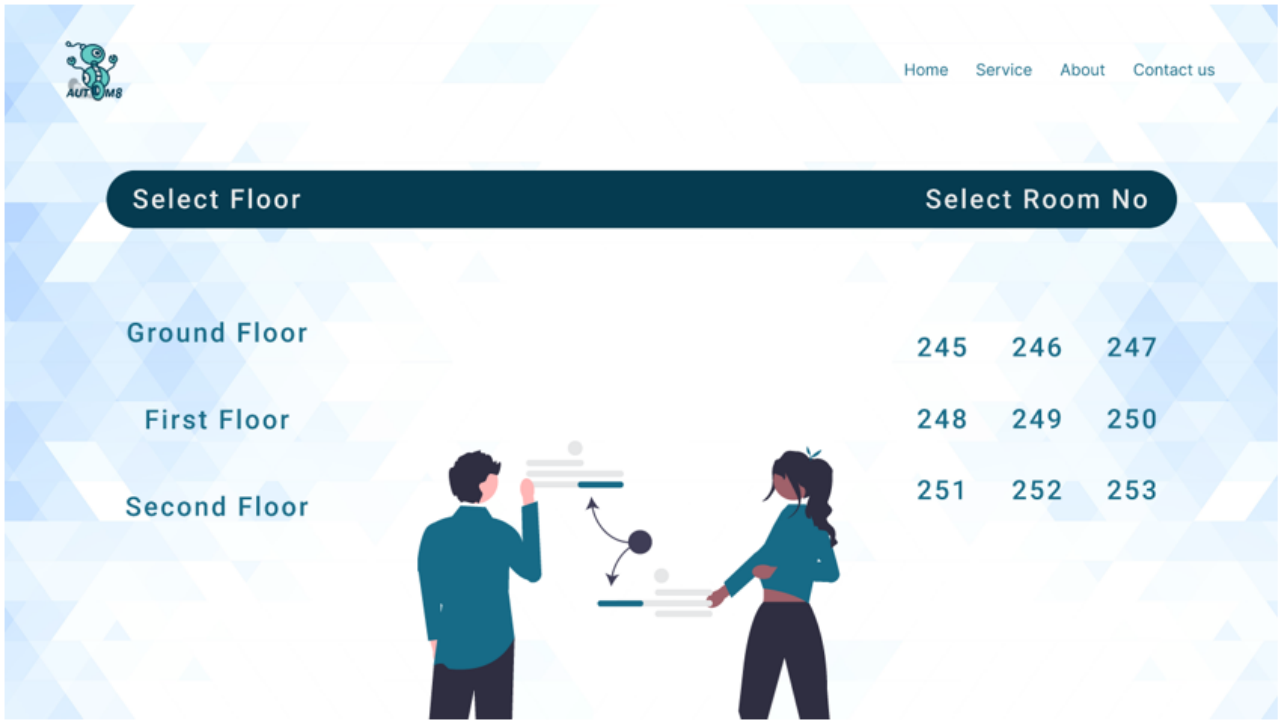
Figure 7.5 — Client Login Page and Student Login Page



7.5.5 Floor and Room Selection

After login, the student selects the delivery destination by choosing their floor and room number from a grid. This information forms the delivery address that the robot’s navigation system would use.

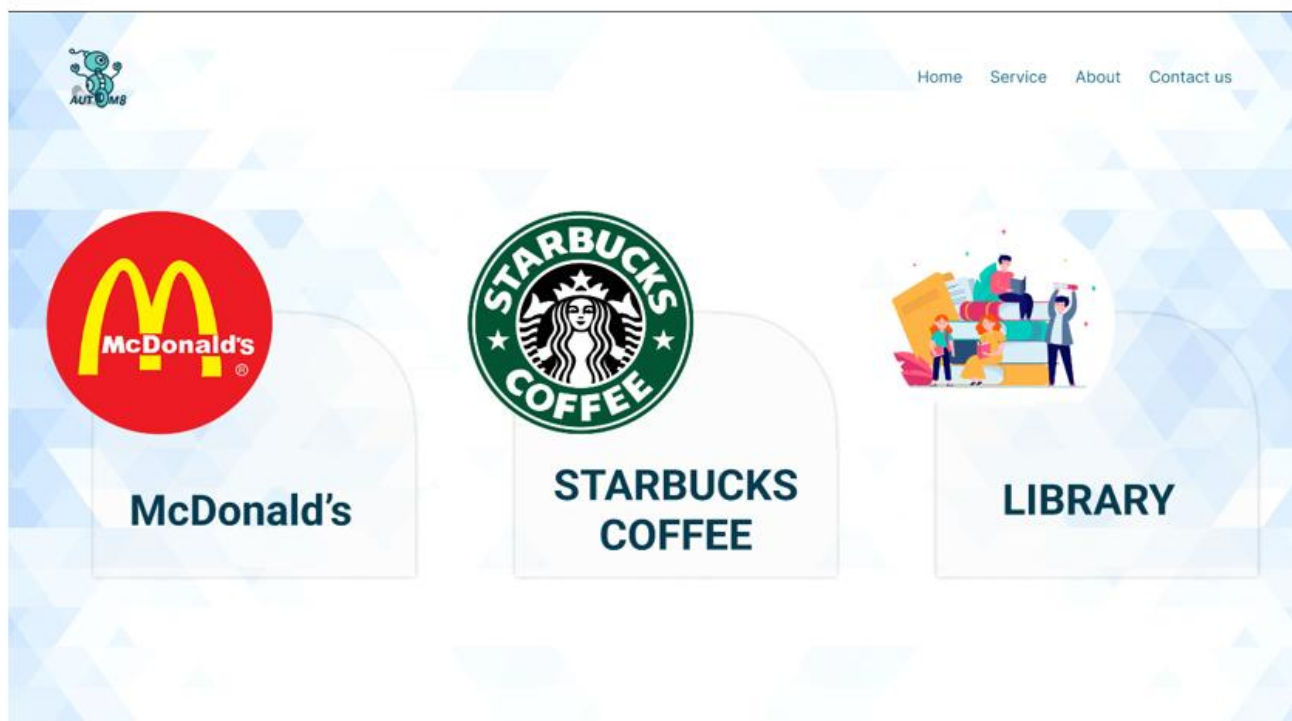
Figure 7.6 — Floor and Room Selection page



7.5.6 Registered Clients Page

The student is shown the list of registered service providers — in the prototype, McDonald’s, Starbucks Coffee, and the library. This page demonstrates the multi-vendor capability of the platform. In the prototype, not all links were fully active as the intent was to demonstrate the concept.

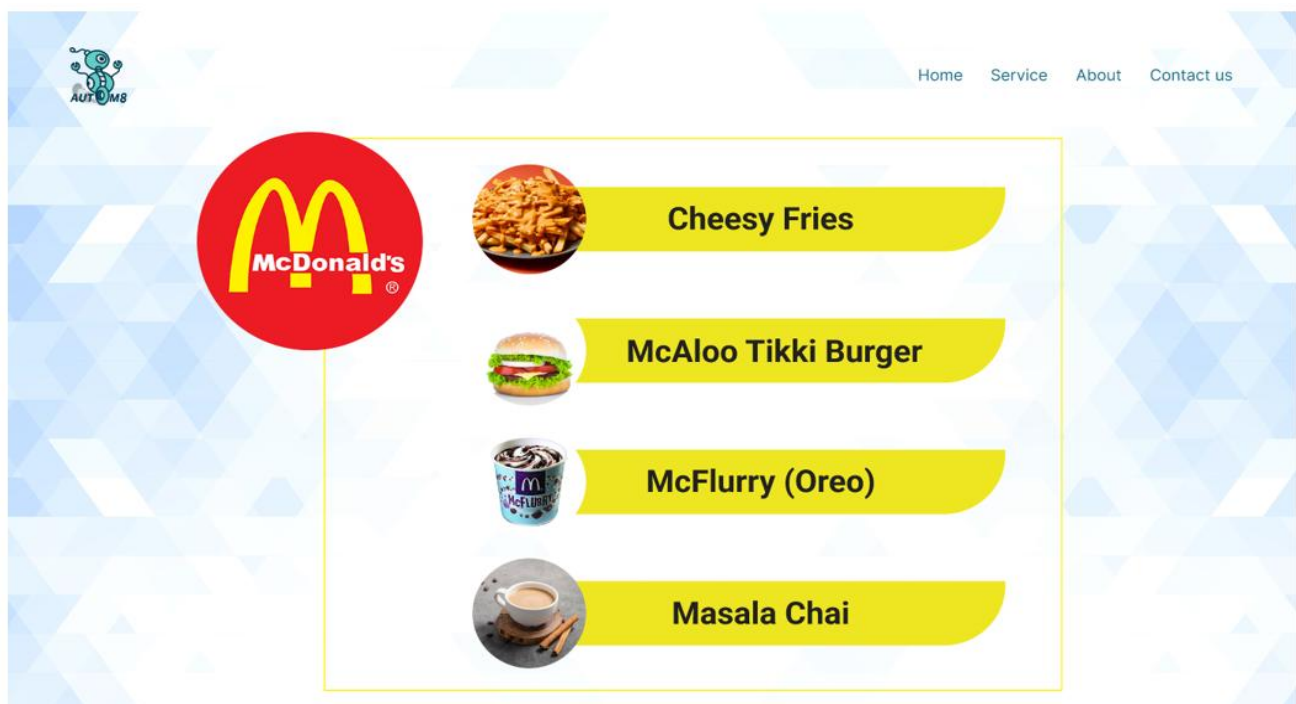
Figure 7.7 — Registered Clients page showing available service providers



7.5.7 Restaurant Menu

Selecting a service provider takes the student to the order page for that provider. In the prototype, a sample McDonald's menu was implemented. In a fully developed system, menu items would be managed by the client directly.

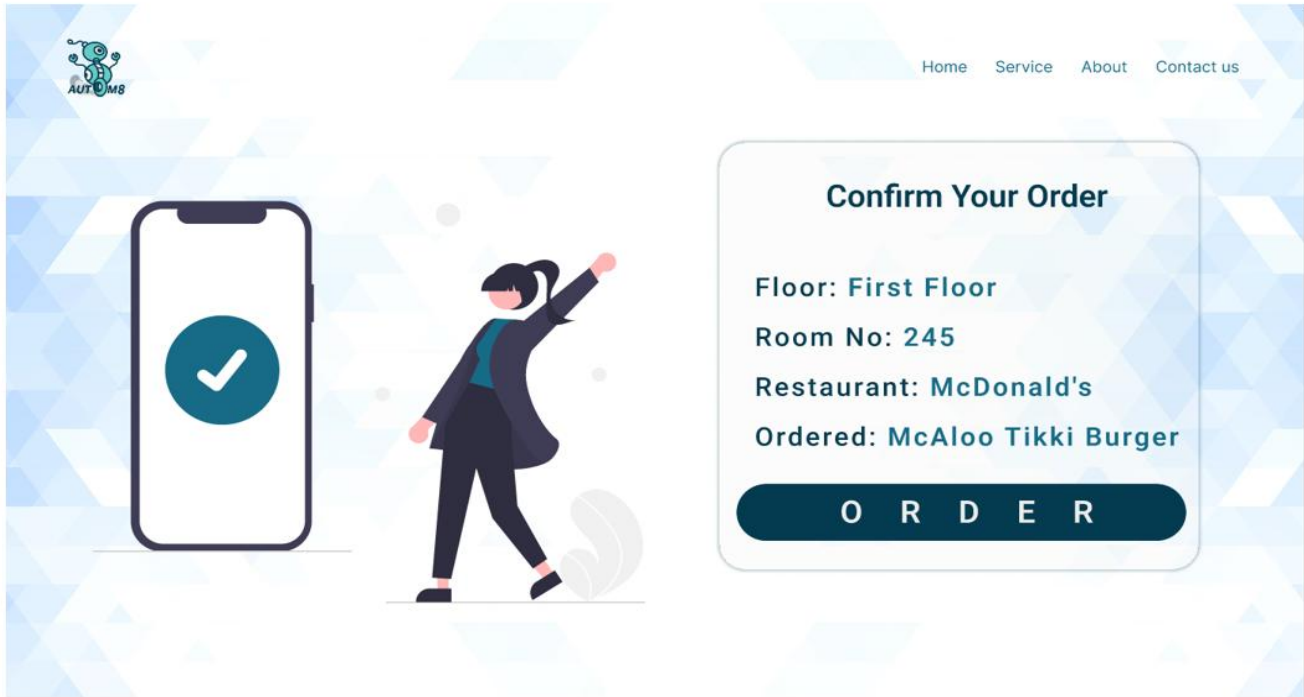
Figure 7.8 — Restaurant Menu page (McDonald's prototype menu)



7.5.8 Order Confirmation Page

The final step displays a summary of the order — floor, room number, restaurant, and item — before the student confirms. Upon confirmation, the order would be transmitted to the client and the robot dispatch system.

Figure 7.9 — Order Confirmation page showing order summary



7.5.9 Future Development of the Web Interface

The prototype UI demonstrates the intended workflow but does not include a live back-end system, real-time robot tracking, or actual robot dispatch integration. Key developments required include: back-end server and database implementation; real-time robot position tracking on a campus map; a client portal for menu management and order confirmation; push notifications to the student when the delivery is dispatched and arrives; and direct integration with the robot's Arduino firmware to trigger dispatch and receive completion signals.

7.6 Autonomous Docking Charging Station

Battery management is a practical constraint for any continuously deployed delivery robot. In the current prototype, the battery must be manually disconnected and charged externally. An autonomous

docking charging station would allow AUTOM8 to navigate to a fixed charging point at the ground floor between deliveries and recharge without any human involvement.

The charging station concept involves a docking bay with alignment guides and spring-loaded charging contacts that mate with corresponding contacts on the robot's frame when it enters the bay. The robot would detect the station using mapping and sensors and navigate into the bay at low speed. The BMS already installed in the battery pack provides the charge management logic, requiring only the addition of a charging circuit at the station and docking contacts on the robot. This improvement is considered a high-priority development for the next iteration of AUTOM8.

7.7 Summary of Proposed Future Work

Table 7.1: Summary of proposed future improvements

Improvement	Addresses	Priority
Rack-and-pinion or steel cable gantry drive	Belt elongation after cycling	High
Payload compartment with locking mechanism	No delivery enclosure in prototype	High
Closed-loop gantry position control	Residual open-loop lift termination	High
Autonomous docking charging station	Manual battery charging requirement	High
MPU-6050 yaw-only during stair climbing	IMU disabled during climb sequence	Medium
Dual encoder with slip detection	Single encoder distance error	Medium
Upgraded motor drivers (BTS7960)	L298N thermal rise under load	Medium
Enclosed electronics + cable management	Exposed prototype wiring	Medium
Web interface back-end development	Prototype UI has no live back-end	Medium
Chamfered front box leading edge	Stair nosing collision at higher speeds	Low
Weight reduction (lighter frame materials)	12kg prototype weight	Low
Sensor-based dynamic path execution	Pre-programmed encoder-only navigation	Low
Multi-floor operation	Single staircase tested only	Low

CHAPTER 8

CONCLUSION

8.1 Summary of Work

This project report presents the design, fabrication, and testing of AUTOM8 — a stair-climbing campus delivery robot developed as a final year B.Tech project in Robotics and Automation Engineering at Parul University, Vadodara. The robot was conceived to address the challenge of last-mile delivery across multi-floor campus buildings, where conventional wheeled robots are restricted to a single floor by the presence of staircases.

The central technical contribution of the project is the sequential gantry lift mechanism, which enables a three-body segmented chassis to climb staircases by lifting each of its three sections onto successive stair treads in sequence. This mechanism was designed from first principles, fabricated using commercially available 20×20mm aluminium extrusion profiles and GT2 belt-driven V-slot carriage assemblies, and validated through physical prototype testing. To the best of the authors’ knowledge, the application of a constraint-reaction sequential gantry lift to a three-body segmented delivery robot chassis represents a novel mechanical approach within the context of low-cost campus robotics.

The project progressed through the complete engineering cycle — from concept and design through component selection, fabrication, electronics integration, firmware development, and iterative testing. The development of the stair-climbing control logic itself followed three successive stages: timing-based control, IR sensor triggering, and finally HC-SR04 ultrasonic sensor triggering — each stage directly addressing the failure modes of the previous one. This iterative approach, while not planned as a formal methodology, reflects the practical engineering process and is documented honestly in **Chapter 6**.

8.2 Objectives Achieved

The project aimed to design, develop, and implement a campus delivery robot capable of navigating flat terrain and climbing staircases within a defined geometry range. The following table summarises the status of the principal objectives stated in **Chapter 1** against the outcomes documented in this report.

Table 8.1: Objectives achievement summary

Objective	Status	Reference
Three-body segmented chassis fabricated from aluminium extrusion	Achieved	Ch. 3
Sequential gantry lift mechanism — lifts all three boxes	Achieved	Ch. 5
HC-SR04 ultrasonic stair detection — 3cm trigger	Achieved	Ch. 4, 5
Compatible stair range: tread 280–400mm, riser 150–220mm	Achieved	Ch. 5
Reactive obstacle detection and halt with buzzer alert	Achieved	Ch. 4, 6
MPU-6050 heading correction on flat terrain	Achieved	Ch. 4, 6
Encoder-based pre-programmed path execution	Achieved	Ch. 4
12V lithium-ion battery pack with BMS	Achieved	Ch. 4
Prototype web interface UI design	Achieved — prototype only	Ch. 7
Payload compartment for delivery items	Proposed — not built	Ch. 7
Overall	Core objectives achieved at prototype stage	

8.3 Key Technical Findings

8.3.1 Stair Climbing Mechanism

The sequential gantry lift mechanism was validated as mechanically sound for the target stair geometry range. The constraint-reaction lifting principle — using the two anchored outer boxes as fixed columns against which the middle box lifts itself — proved effective in practice, requiring no external support structure or additional hardware beyond the four gantry motors already required for lifting the outer boxes. This elegance of the mechanism, achieving three-directional lift with a single set of four motors and two rail pairs, is considered the strongest technical outcome of the project.

The evolution of the stair detection approach from timing-based to IR sensor-based to HC-SR04 ultrasonic-based yielded a clear and practically important finding: IR proximity sensors are unsuitable for stair riser detection in variable indoor lighting environments due to their fundamental sensitivity to

ambient infrared radiation and surface reflectivity. HC-SR04 ultrasonic sensors, being immune to these optical variables, provided significantly more reliable and consistent detection performance.

8.3.2 Navigation

The pre-programmed encoder-based path execution approach provides a simple and cost-effective navigation solution for fixed delivery routes in a known environment. While it lacks the adaptability of sensor-based mapping systems, it is entirely sufficient for the prototype use case and was demonstrated to execute multi-segment routes with acceptable positional accuracy. The MPU-6050 heading correction was shown to effectively counteract unintended heading drift during flat terrain navigation, demonstrating strong resistance to external disturbance through the combined effect of active correction and rubber tyre grip.

8.3.3 Cost

The total prototype cost of approximately ₹33,000 (€370) demonstrates that a functional stair-climbing delivery robot can be realised at a cost accessible to student project budgets. This contrasts with the high-end sensor and computing platforms used in comparable academic delivery robot projects, which typically cost several times this amount. The use of standard industrial aluminium extrusion, off-the-shelf Johnson DC motors, and commodity electronics modules was key to achieving this cost target.

8.4 Limitations

The prototype has several documented limitations that prevent immediate practical deployment. Open-loop motor control results in observable box tilt during gantry phase transitions. The single encoder provides distance estimation with unquantified error. Belt elongation under cyclic loading requires periodic manual re-tensioning. The absence of a payload compartment means the prototype cannot carry delivery items in its current form. Sensor response delay causes positional overshoot at higher wheel speeds, creating a direct tradeoff between climbing speed and collision risk at stair nosings. Exposed prototype wiring is unsuitable for a live campus environment.

These limitations are not failures of the fundamental design concept — they are characteristic of a first-generation prototype built within the constraints of a student project budget and timeline. Each limitation has a clearly identified solution documented in **Chapter 7**, and none of them invalidates the core mechanical and control architecture of AUTOM8.

8.5 Conclusions

AUTOM8 successfully demonstrates that a low-cost, three-body segmented stair-climbing delivery robot is realisable using commercially available aluminium extrusion profiles, belt-driven gantry mechanisms, and commodity electronics. The sequential gantry lift mechanism, validated through physical prototype testing, provides a viable and mechanically elegant solution to the stair-climbing problem for indoor campus delivery applications.

The iterative development process — progressing through three stair-climbing control approaches and resolving successive failure modes through observation and redesign — produced a working prototype that achieves its core design objectives within the defined stair compatibility range. The project demonstrates the value of honest documentation of failure modes alongside successes; the progression from **Stage 1 to Stage 3** stair climbing control is as informative as the final validated result.

The proposed future work directions in **Chapter 7** provide a clear development roadmap from the current prototype to a deployable system. The autonomous docking charging station, closed-loop gantry control, dual encoder configuration, and payload compartment are identified as the highest-priority improvements for the next iteration. The prototype web interface, developed as a UI demonstration alongside the robot, establishes the framework for a full ordering and dispatch system in a future development phase.

AUTOM8 represents a meaningful contribution to the field of low-cost campus delivery robotics at the undergraduate level. The project establishes a foundation — mechanical, electrical, firmware, and interface — upon which a practical campus delivery system can be built.

Appendix A — Design Calculations

Calculation 1: Gantry Motor Torque and Safety Factor Analysis

All Voltage Cases: 12V (Maximum) | 10V (Average) | 9V (Minimum)

1. Purpose of This Calculation

This calculation verifies that the four 100 RPM Johnson geared DC motors selected for the AUTOM8 gantry lifting mechanism produce sufficient torque to lift each of the three chassis boxes during the six-phase stair-climbing sequence. The calculation is performed across three battery voltage levels to confirm that the motors remain adequate even as battery voltage drops during extended operation.

The three voltage cases are named as follows throughout this document:

- **Case 1 — 12V Maximum:** Fresh battery, light motor load. Best-case motor performance.
- **Case 2 — 10V Average:** Typical operating condition under moderate motor load. Realistic estimate for normal delivery operation.
- **Case 3 — 9V Minimum:** Heavy load or partially discharged battery. Worst-case motor performance — the motor must still complete the lift successfully at this voltage.

DC motor torque scales linearly with applied voltage. As voltage decreases, torque decreases proportionally. This relationship is used to scale all motor performance values from the rated 12V specifications.

2. Confirmed Input Values

All values below are sourced directly from hardware datasheets, supplier specifications, and confirmed design dimensions from the AUTOM8 project report. No values have been assumed without documentation.

Table A1.1: Input values — Calculation 1

Parameter	Value	Source
Gantry motor rated torque (12V)	103 N-cm = 1.030 N-m	Robu.in datasheet, SKU 5726
Gantry motor stall torque (12V)	700 N-cm = 7.000 N-m	Robu.in datasheet, SKU 5726

GT2 pulley: teeth × pitch	20 teeth × 2 mm = 40 mm circumference	Confirmed hardware, Chapter 5 Section 5.3.3
Pulley pitch-line radius (r)	6.37 mm = 0.00637 m	Derived below in Step 1
Front box weight	2.717 kg	Chapter 3, Table 3.3
Middle box weight	6.623 kg	Chapter 3, Table 3.4
Back box weight	2.707 kg	Chapter 3, Table 3.5
Phase 1: motors active	2 (front-facing pair, Driver 5)	Chapter 4, Table 4.6
Phase 3: motors active	4 (all four, Drivers 5 and 6)	Chapter 4, Table 4.6
Phase 5: motors active	2 (rear-facing pair, Driver 6)	Chapter 4, Table 4.6
Standard gravity	$g = 9.81 \text{ m/s}^2$	Standard physical constant
Payload for safety factor check	5 kg (conservative design load)	See Section 2.1

2.1 Why 5 kg is Used as the Payload Assumption

The theoretical belt-limited payload capacity of AUTOM8 is 7.47 kg (determined in the **Calculation 4**). However, 5 kg is used as the payload assumption in this calculation for two reasons. First, 5 kg represents a conservative and practically realistic campus delivery load — food containers, books, or small parcels are unlikely to exceed this weight in real deployment. Second, using 5 kg provides an additional engineering safety margin beyond the theoretical belt limit, accounting for real-world factors such as payload shift during stair climbing, uneven load distribution between boxes, and dynamic loads from stair impact. The report states payload capacity as “under 7 kg”; the 5 kg figure used here is a conservative design assumption for this calculation, not the absolute maximum.

3. Voltage Scaling of Motor Torque

DC motor torque is proportional to the applied voltage. The rated torque and stall torque values at 12V are scaled to 10V and 9V using the following formula:

$$T(V) = T(12V) \times (V \div 12)$$

Table A1.2: Voltage-scaled motor torque values — all three cases

Voltage Case	Voltage	Scale Factor	Rated Torque	Stall Torque
Case 1 — 12V Maximum	12V	12/12 = 1.000	103.0 N-cm = 1.030 N-m	700.0 N-cm = 7.000 N-m
Case 2 — 10V Average	10V	10/12 = 0.833	85.8 N-cm = 0.858 N-m	583.3 N-cm = 5.833 N-m
Case 3 — 9V Minimum	9V	9/12 = 0.750	77.3 N-cm = 0.773 N-m	525.0 N-cm = 5.250 N-m

Note: Rated torque is the torque the motor produces at its maximum efficiency operating point and can be sustained continuously. Stall torque is the absolute maximum torque produced momentarily when the shaft is stationary — it cannot be sustained. Safety factors are calculated against rated torque as the engineering criterion for sustained stair climbing operation.

4. Step 1 — Pulley Pitch-Line Radius

The pulley pitch-line radius is the effective radius at which the belt transmits force. It is derived from the number of pulley teeth and the belt pitch:

$$\text{Pulley circumference} = \text{Teeth} \times \text{Pitch} = 20 \times 2 \text{ mm} = 40 \text{ mm}$$

$$\text{Radius (r)} = \text{Circumference} \div (2 \times \pi) = 40 \div 6.2832 = 6.37 \text{ mm} = 0.00637 \text{ m}$$

Result: $r = 6.37 \text{ mm} = 0.00637 \text{ m}$

Note: A smaller pulley radius produces more lifting force but slower belt movement. The 20-tooth GT2 pulley was selected to maximise lifting force — this is the correct trade-off for a controlled stair-climbing mechanism where force is critical and speed is secondary.

5. Step 2 — Lifting Force Required Per Phase

The lifting force required in each phase equals the gravitational force on the box being lifted, divided equally between the motors active in that phase. This required force is independent of battery voltage — the box weight does not change.

$$F_{\text{required per motor}} = (m \times g) \div N_{\text{motors}}$$

Where, m = box mass (kg), $g = 9.81 \text{ m/s}^2$, N_{motors} = number of gantry motors active in that phase.

Phase 1 — Front Box Lift (2 motors active)

Total force: $2.717 \text{ kg} \times 9.81 \text{ m/s}^2 = 26.65 \text{ N}$ Force per motor: $26.65 \div 2 = \mathbf{13.33 \text{ N}}$ Required torque per motor: $13.33 \times 0.00637 = \mathbf{8.49 \text{ N-cm}}$ (constant at all voltages)

Phase 3 — Middle Box Lift (4 motors active) — Most Demanding Phase

Total force: $6.623 \text{ kg} \times 9.81 \text{ m/s}^2 = 64.97 \text{ N}$ Force per motor: $64.97 \div 4 = \mathbf{16.24 \text{ N}}$ Required torque per motor: $16.24 \times 0.00637 = \mathbf{10.34 \text{ N-cm}}$ (constant at all voltages)

Note: Phase 3 is the most torque-demanding phase because the middle box weighs 6.623 kg — more than twice the outer boxes. The minimum safety factor across the entire gantry system is therefore determined by Phase 3 at Case 3 (9V Minimum). This is the governing design criterion.

Phase 5 — Back Box Lift (2 motors active)

Total force: $2.707 \text{ kg} \times 9.81 \text{ m/s}^2 = 26.55 \text{ N}$ Force per motor: $26.55 \div 2 = \mathbf{13.28 \text{ N}}$ Required torque per motor: $13.28 \times 0.00637 = \mathbf{8.46 \text{ N-cm}}$ (constant at all voltages)

6. Step 3 — Safety Factors (Unloaded Robot)

The safety factor is the ratio of available torque to required torque. A value above 1.0 means the motor can perform the lift. Values above 3.0 are acceptable for robotics; values above 7.0 are highly robust.

$$\text{Safety Factor (SF)} = T_{\text{available}} \div T_{\text{required}}$$

Table A1.3: Safety factors — Phase 1, Front Box Lift, empty robot

Voltage Case	T_available (rated)	T_required	SF (rated) ✓	SF (stall) — reference
Case 1 — 12V Maximum	103.0 N-cm	8.49 N-cm	12.1× ✓	82.4×
Case 2 — 10V Average	85.8 N-cm	8.49 N-cm	10.1× ✓	68.7×
Case 3 — 9V Minimum	77.3 N-cm	8.49 N-cm	9.1× ✓	61.9×

Table A1.4: Safety factors — Phase 3, Middle Box Lift, empty robot — Governing Phase

Voltage Case	T_available (rated)	T_required	SF (rated) ✓	SF (stall) — reference
Case 1 — 12V Maximum	103.0 N-cm	10.34 N-cm	9.96× ✓	67.7×
Case 2 — 10V Average	85.8 N-cm	10.34 N-cm	8.3× ✓	56.4×
Case 3 — 9V Minimum	77.3 N-cm	10.34 N-cm	7.5× ✓	50.8×

Table A1.5: Safety factors — Phase 5, Back Box Lift, empty robot

Voltage Case	T_available (rated)	T_required	SF (rated) ✓	SF (stall) — reference
Case 1 — 12V Maximum	103.0 N-cm	8.46 N-cm	12.2× ✓	82.7×
Case 2 — 10V Average	85.8 N-cm	8.46 N-cm	10.1× ✓	68.9×
Case 3 — 9V Minimum	77.3 N-cm	8.46 N-cm	9.1× ✓	62.1×

7. Step 4 — Safety Factors With 5 kg Payload

When the robot carries a payload, the payload weight adds to the box being lifted. In the worst-case analysis the full 5 kg payload is assumed to be concentrated in the front box during **Phase 1**, and in the middle box during **Phase 3**. This is more conservative than the actual design where payload sits on the middle box platform.

Phase 1 with 5 kg Payload (Front Box, 2 motors)

Effective box mass: $2.717 + 5.000 = 7.717$ kg Force per motor: $(7.717 \times 9.81) \div 2 = 37.85$ N Required torque: $37.85 \times 0.00637 = \mathbf{24.11 \text{ N-cm}}$

Phase 3 with 5 kg Payload (Middle Box, 4 motors)

Effective box mass: $6.623 + 5.000 = 11.623$ kg Force per motor: $(11.623 \times 9.81) \div 4 = 28.51$ N Required torque: $28.51 \times 0.00637 = \mathbf{18.16 \text{ N-cm}}$

Phase 5 with 5 kg Payload (Back Box, 2 motors)

Effective box mass: $2.707 + 5.000 = 7.707 \text{ kg}$ Force per motor: $(7.707 \times 9.81) \div 2 = 37.80 \text{ N}$ Required torque: $37.80 \times 0.00637 = \mathbf{24.08 \text{ N-cm}}$

Table A1.6: Safety factors — All phases with 5 kg payload — All voltage cases

Phase	Effective Mass	T_required	SF at 12V	SF at 10V	SF at 9V ✓
Phase 1 (2 motors)	7.717 kg	24.11 N-cm	4.27×	3.56×	3.21× ✓
Phase 3 (4 motors)	11.623 kg	18.16 N-cm	5.67×	4.72×	4.26× ✓
Phase 5 (2 motors)	7.707 kg	24.08 N-cm	4.28×	3.56×	3.21× ✓

Note: The minimum safety factor with 5 kg payload across all phases and all voltage cases is $3.21\times$ (Phase 1 or 5 at Case 3, 9V Minimum). This exceeds the minimum acceptable safety factor of 2.0 for robotics systems and confirms the selected motors are adequately sized.

8. Complete Summary — All Safety Factors*Table A1.7: Complete safety factor summary — Calculation 1*

Condition	Phase (Motors)	T_required	SF at 12V	SF at 10V	SF at 9V
Empty robot	Phase 1 (2)	8.49 N-cm	12.1×	10.1×	9.1×
Empty robot	Phase 3 (4)	10.34 N-cm	9.96×	8.3×	7.5×
Empty robot	Phase 5 (2)	8.46 N-cm	12.2×	10.1×	9.1×
With 5 kg payload	Phase 1 (2)	24.11 N-cm	4.27×	3.56×	3.20×
With 5 kg payload	Phase 3 (4)	18.16 N-cm	5.67×	4.72×	4.26×
With 5 kg payload	Phase 5 (2)	24.08 N-cm	4.28×	3.56×	3.21×
GOVERNING MINIMUM	Phase 1/5, 5kg @ 9V	24.11 N-cm	---	---	3.21× ✓ PASS

9. Conclusions

- 1. Motor adequacy confirmed at all voltages:** At all three voltage cases and all three lift phases, the 100 RPM Johnson geared DC motors produce sufficient torque to lift each box. Minimum safety factor for the empty robot is $7.5\times$ (**Phase 3 at 9V**).
- 2. Payload capacity adequately supported:** With 5 kg payload the minimum safety factor is $3.21\times$ at **Phase 1** and **Phase 5** under **Case 3** (9V Minimum) — exceeding the minimum acceptable value of 2.0 for continuous robotics operation.
- 3. Phase 3 is the governing design criterion:** The middle box lift requires the most torque per motor (10.34 N-cm unloaded). The system-level governing safety factor is $7.5\times$ (**Phase 3, empty, 9V**) — consistent with the value stated in **Chapter 3 Section 3.7** of the project report.
- 4. Motor torque is not the payload limiting factor:** At all voltages the motors can generate far more belt force than the GT2 belt can handle. The real payload limit is the GT2 belt tensile strength, not motor torque. See **Calculation 4** — Payload Capacity Analysis.
- 5. Motor selection validated:** The 100 RPM Johnson geared DC motors (**Robu.in SKU 5726**) are correctly sized for the AUTOM8 gantry mechanism across all operating conditions.

Appendix A — Design Calculations

Calculation 2: Wheel Motor Traction and Drive Force Analysis

All Voltage Cases: 12V (Maximum) | 10V (Average) | 9V (Minimum)

1. Purpose of This Calculation

This calculation verifies that the eight 200 RPM Johnson geared DC motors selected for the AUTOM8 wheel drive system produce sufficient traction force to move the robot on flat campus floors and confirms that the wheel drive is not the limiting factor for payload capacity. Two scenarios are analysed: flat terrain navigation and an incline check for minor ramps encountered on campus.

The three voltage cases used throughout are:

- **Case 1 — 12V Maximum:** Fresh battery, best-case motor performance.
- **Case 2 — 10V Average:** Typical operating condition under moderate motor load.
- **Case 3 — 9V Minimum:** Partially discharged battery, worst-case motor performance.

2. Confirmed Input Values

Table A2.1: Input values — Calculation 2

Parameter	Value	Source
Wheel motor rated torque (12V)	56.1 N-cm = 0.561 N-m	Robu.in datasheet, SKU 5727 (200 RPM Johnson)
Wheel motor stall torque (12V)	350 N-cm = 3.500 N-m	Robu.in datasheet, SKU 5727
Number of wheel motors	8 total (2 front + 4 middle + 2 back)	Chapter 3, Section 3.4
Wheel diameter	68 mm = 0.068 m	Chapter 3, Section 3.4 (confirmed value)
Wheel radius (r_wheel)	34 mm = 0.034 m	Wheel diameter ÷ 2
Total robot weight (unloaded)	~12.1 kg	Chapter 3, Table 3.2

Rolling resistance coefficient (μ)	0.02	Standard value for rubber wheel on smooth indoor tile/concrete
Standard gravity	$g = 9.81 \text{ m/s}^2$	Standard physical constant
Conservative payload for analysis	5 kg	Consistent with Calculation 1

***Note:** Rolling resistance coefficient $\mu = 0.02$ is the standard engineering value for rubber-tired wheels on smooth indoor surfaces (concrete, tile, linoleum). This is the resistance due to wheel deformation and surface micro-irregularities — it is much smaller than sliding friction. On rougher outdoor surfaces μ rises to 0.05–0.08, but all AUTOM8 operation is on indoor campus floors.*

3. Voltage Scaling of Wheel Motor Torque

As with the gantry motors in **Calculation 1**, wheel motor torque scales linearly with applied voltage:

$$T_{\text{wheel}}(V) = T_{\text{wheel}}(12V) \times (V \div 12)$$

Table A2.2: Voltage-scaled wheel motor torque values

Voltage Case	Voltage	Scale Factor	Rated Torque per Motor	Stall Torque per Motor
Case 1 — 12V Maximum	12V	$12/12 = 1.000$	56.1 N-cm = 0.561 N-m	350.0 N-cm = 3.500 N-m
Case 2 — 10V Average	10V	$10/12 = 0.833$	46.75 N-cm = 0.467 N-m	291.7 N-cm = 2.917 N-m
Case 3 — 9V Minimum	9V	$9/12 = 0.750$	42.1 N-cm = 0.421 N-m	262.5 N-cm = 2.625 N-m

4. Step 1 — Drive Force per Wheel Motor

The force produced at the wheel contact patch (the floor) by one motor is calculated using the standard torque-to-force formula. The wheel radius is the effective lever arm converting motor torque into forward push force at the floor:

$$F_{\text{wheel}} = T_{\text{motor}} \div r_{\text{wheel}}$$

Where $r_{\text{wheel}} = 34 \text{ mm} = 0.034 \text{ m}$ (half of 68 mm wheel diameter, confirmed **Chapter 3**).

Case 1 — 12V Maximum

$F_{\text{wheel}} = 0.561 \text{ N-m} \div 0.034 \text{ m} = \mathbf{16.50 \text{ N per motor}}$

Case 2 — 10V Average

$F_{\text{wheel}} = 0.467 \text{ N-m} \div 0.034 \text{ m} = \mathbf{13.74 \text{ N per motor}}$

Case 3 — 9V Minimum

$F_{\text{wheel}} = 0.421 \text{ N-m} \div 0.034 \text{ m} = \mathbf{12.38 \text{ N per motor}}$

Note: A larger wheel radius reduces the force produced at the floor for the same motor torque, but increases speed. The 68 mm wheel was selected to balance speed and traction for indoor campus navigation. The force values above are per motor — with 8 wheel motors the total drive force is much larger, as shown in Step 2.

5. Step 2 — Total Drive Force (All 8 Wheel Motors)

All eight wheel motors operate simultaneously during flat terrain navigation, providing combined traction force:

$$F_{\text{total}} = F_{\text{wheel}} \times 8 \text{ motors}$$

Table A2.3: Total drive force from all 8 wheel motors at each voltage case

Voltage Case	Force per Motor	Total Drive Force (8 motors)	Note
Case 1 — 12V Maximum	16.50 N	132.0 N	Best-case traction
Case 2 — 10V Average	13.74 N	109.9 N	Typical operation
Case 3 — 9V Minimum	12.38 N	99.0 N	Worst-case traction

6. Step 3 — Rolling Resistance (Force Required to Move Robot)

On flat ground the motors do not overcome gravity — they only need to overcome rolling resistance. Rolling resistance is the force opposing wheel motion due to wheel deformation and surface contact. It is calculated as:

$$F_{\text{resistance}} = \mu \times m_{\text{total}} \times g$$

Where, $\mu = 0.02$ (rolling resistance coefficient for rubber on smooth indoor floor), m_{total} = total robot mass (kg), $g = 9.81 \text{ m/s}^2$.

Rolling resistance is independent of motor voltage — it depends only on robot weight and floor surface. Two cases are calculated: empty robot and robot with 5 kg payload.

Flat Ground — Empty Robot (12.1 kg)

$$F_{\text{resistance}} = 0.02 \times 12.1 \text{ kg} \times 9.81 \text{ m/s}^2 = \mathbf{2.37 \text{ N}}$$

Flat Ground — With 5 kg Payload (17.1 kg)

$$F_{\text{resistance}} = 0.02 \times 17.1 \text{ kg} \times 9.81 \text{ m/s}^2 = \mathbf{3.36 \text{ N}}$$

Note: Rolling resistance of 2.37 to 3.36 N is very small compared to the available drive force of 99 to 132 N. This confirms that flat terrain driving is not a design constraint for this robot at any practical payload.

7. Step 4 — Traction Safety Factors

The traction safety factor is the ratio of available drive force to required rolling resistance force. It answers the question: how many times more force do the motors provide than the minimum needed to move the robot?

$$SF_{\text{traction}} = F_{\text{total}} \div F_{\text{resistance}}$$

Table A2.4: Traction safety factors — flat ground, all voltage cases

Condition	F_resistance	F_total at 12V	SF at 12V	F_total at 10V	SF at 10V	F_total at 9V	SF at 9V
Empty robot (12.1 kg)	2.37 N	132.0 N	55.7× ✓	109.9 N	46.4× ✓	99.0 N	41.8× ✓
With 5 kg payload (17.1 kg)	3.36 N	132.0 N	39.3× ✓	109.9 N	32.7× ✓	99.0 N	29.5× ✓

Note: Traction safety factors above 29× across all conditions confirm that flat terrain driving is entirely non-constraining. The wheel motors are vastly over-specified for flat ground — they were selected primarily for their torque characteristics during stair climbing approach phases, not for flat terrain requirements.

8. Stair Approach Traction Verification

During stair climbing **Phases 2, 4, and 6** (forward translation phases), the wheel motors push the robot forward onto the stair while one or two boxes may be partially elevated. In the worst-case translation phase, the robot's full weight rests on the grounded wheels and must be driven forward while the elevated box creates an asymmetric load. The rolling resistance during translation is still governed by the total robot weight on the floor.

Worst-case translation rolling resistance: same as flat ground = **2.37 N (empty) / 3.36 N (with 5 kg payload)**

Drive force available during translation: only grounded wheels are active. Minimum case is 4 middle wheels driving during **Phase 2** (front box elevated):

$$F_{\text{translation}} (4 \text{ motors, Case 3, 9V}) = 4 \times 12.38 \text{ N} = \mathbf{49.5 \text{ N}}$$

$$\text{Translation safety factor (4 motors, 9V, with 5 kg payload)} = 49.5 \div 3.36 = \mathbf{14.7\times \checkmark}$$

Note: Even with only 4 wheel motors active during the most constrained translation phase and at minimum battery voltage with full payload, the traction safety factor is 14.7×. Wheel traction is not a constraint during any phase of stair climbing.

9. Complete Summary — Traction Analysis

Table A2.5: Complete traction analysis summary — Calculation 2

Scenario	Motors Active	F_available (9V worst)	F_resistance	SF (9V worst) ✓
Flat ground — empty robot	8 motors	99.0 N	2.37 N	41.8× ✓
Flat ground — 5 kg payload	8 motors	99.0 N	3.36 N	29.5× ✓
Stair translation (Phase 2) — empty	4 motors	49.5 N	2.37 N	20.9× ✓
Stair translation (Phase 2) — 5 kg payload	4 motors	49.5 N	3.36 N	14.7× ✓
MINIMUM SAFETY FACTOR	4 motors, 5 kg, 9V	49.5 N	3.36 N	14.7× ✓ PASS

10. Conclusions

1. Flat terrain traction confirmed: At all voltage cases (12V, 10V, 9V) and with or without payload, the eight wheel motors provide between 29.5× and 55.7× more traction force than required to overcome rolling resistance on smooth indoor campus floors.

2. Stair translation phases confirmed: Even with only 4 wheel motors active during the most constrained stair translation phase, at minimum battery voltage and with 5 kg payload, the traction safety factor is $14.7\times$ — well above the minimum required.

3. Wheel drive is not the payload limiting factor: The massive traction margin confirms that the wheel motors impose no constraint on payload capacity for flat terrain operation. The payload limit is governed exclusively by the gantry mechanism belt strength (**see Calculation 4**).

4. Motor selection validated: The 200 RPM Johnson geared DC motors (**Robu.in SKU 5727**) are correctly sized for the AUTOM8 wheel drive system. They provide adequate traction across all operating conditions with large safety margins.

5. 68 mm wheel diameter confirmed: Using the confirmed wheel diameter of 68 mm (radius 34 mm) as stated in **Chapter 3, Section 3.4** of the project report. This is the value used in all speed calculations (**Calculation 6**) and in this traction analysis.

Appendix A — Design Calculations

Calculation 3: Power Consumption and Battery Runtime Analysis

Multiple Current Scenarios | 80% Usable Capacity | Full Discharge Capacity

1. Purpose of This Calculation

This calculation determines the battery runtime of the AUTOM8 prototype under two operating scenarios — flat terrain navigation and stair climbing — and verifies that the selected battery pack is adequately sized. Because the exact wheel motor current under load was not formally measured during prototype testing, the calculation is performed across three current load scenarios covering the realistic operating range. This allows the reader to identify which runtime estimate applies to their assumed operating condition.

Two battery capacity cases are also presented:

- **Case A — 80% Usable Capacity (7,040 mAh):** Recommended operating case. Lithium-ion cells should not be discharged below 20% state-of-charge to preserve cell life and maintain safe operating voltage. $7,040 \text{ mAh} = 8,800 \text{ mAh} \times 0.80$.
- **Case B — Full Discharge Capacity (8,800 mAh):** Theoretical maximum if battery is fully discharged to BMS cutoff voltage. Provided for reference only — regular full discharge significantly reduces lithium cell lifespan.

2. Confirmed Input Values

Table A3.1: Battery and component specifications

Parameter	Value	Source
Battery configuration	3S4P (3 series $\times$ 4 parallel)	Chapter 4, Table 4.1
Cell model	Orange ICR18650-22C	Chapter 4, Table 4.1
Cell capacity	2,200 mAh per cell	Chapter 4, Table 4.1
Total rated capacity	$4 \text{ parallel} \times 2,200 \text{ mAh} = 8,800 \text{ mAh}$	Chapter 4, Table 4.1

Case A — Usable capacity (80%)	$8,800 \times 0.80 = 7,040 \text{ mAh}$	Standard Li-ion safe discharge limit
Case B — Full discharge	8,800 mAh	Theoretical maximum to BMS cutoff
Nominal voltage	11.1V (3S configuration)	Chapter 4, Table 4.1
Number of wheel motors	8 (200 RPM, 12V)	Chapter 3, Section 3.4
Number of gantry motors	4 (100 RPM, 12V)	Chapter 3, Section 3.3
Gantry motor current (stair climbing)	1.636 A per motor at max efficiency point	Robu.in Pro-Range motor test report, SKU 5726
Electronics constant current	~0.365 A total (all logic components)	Chapter 4, Table in Section 4.4 (Arduino, sensors, IMU, encoder, buzzer, L298N quiescent)

3. Wheel Motor Current Scenarios

The exact wheel motor current during flat terrain navigation was not formally measured during prototype testing. DC motors draw current proportional to their mechanical load. The 200 RPM Johnson motor datasheet provides the following reference points:

Table A3.2: 200 RPM wheel motor datasheet reference points

Operating Condition	Current per Motor	Torque	Speed	Relevance
No load (wheel spinning freely)	0.339 A	0.008 N-m	202.6 RPM	Minimum — robot on wheels, motors running but no load
Max efficiency point	1.861 A	0.561 N-m	166.1 RPM	Motor datasheet optimum — upper bound for normal floor navigation
Max output power point	2.722 A	0.880 N-m	144.5 RPM	High load — would indicate heavy floor resistance or incline

Based on these reference points, three current scenarios are defined for the calculation:

- **Low load (0.5 A per motor):** Robot on smooth tile floor, light rolling resistance. Motor operating between no-load and light load. Represents best-case flat terrain condition.
- **Mid load (1.0 A per motor):** Conservative midpoint between no-load (0.339 A) and max efficiency (1.861 A). Represents typical indoor campus floor navigation. This is the most likely real-world operating condition.
- **High load (1.5 A per motor):** Above mid-point, representing heavier floor surfaces, minor inclines, or slight wheel misalignment. Represents worst-case flat terrain condition for a well-maintained robot.

4. Electronics Constant Current

All logic-level components draw a fixed current regardless of robot motion. These are powered through the 5V buck converter rail. The total electronics current is:

Table A3.3: Electronics current breakdown (constant, both scenarios)

Component	Quantity	Current Each	Total Current	When Active
Arduino Mega 2560	1	93 mA	93 mA	Continuous
HC-SR04 ultrasonic sensors	6	2 mA	12 mA	Continuous
MPU-6050 IMU	1	3.9 mA	3.9 mA	Flat terrain only
Magnetic encoder	1	10 mA	10 mA	Flat terrain only
Buzzer (active)	1	30 mA	30 mA	When triggered only
L298N drivers (quiescent)	6	36 mA	216 mA	Continuous
Electronics Total	—	—	~364.9 mA	Continuous

Note: The buzzer current (30 mA) is included as a conservative estimate assuming it activates periodically. The MPU-6050 and encoder are active during flat terrain but disabled during stair climbing — this has negligible effect on the total current (13.9 mA reduction) and is not separately calculated.

5. Scenario 1 — Flat Terrain Navigation

During flat terrain navigation all eight wheel motors run continuously. All four gantry motors are completely inactive (0 A). Total current = $8 \times \text{wheel motor current} + \text{electronics}$.

$$I_{\text{flat}} = (8 \times I_{\text{wheel}}) + I_{\text{electronics}}$$

Table A3.4: Flat terrain total current at all load scenarios

Load Scenario	Wheel Motor Current	8 Motors Total	Electronics	Grand Total
Low load	0.5 A per motor	4.000 A	0.359 A	4.365 A
Mid load	1.0 A per motor	8.000 A	0.359 A	8.365 A
High load	1.5 A per motor	12.000 A	0.359 A	12.365 A

$$\text{Runtime (minutes)} = \text{Capacity (mAh)} \div \text{Current (mA)} \times 60$$

Table A3.5: Flat terrain runtime — all load scenarios — both capacity cases

Load Scenario	Total Current	Case A: 80% Capacity (7,040 mAh)	Case B: Full Capacity (8,800 mAh)
Low load (0.5 A/motor)	4,365 mA	~97 minutes	~121 minutes
Mid load (1.0 A/motor)	8,365 mA	~50 minutes	~63 minutes
High load (1.5 A/motor)	12,365 mA	~34 minutes	~43 minutes

***Note:** The mid-load scenario (1.0 A per motor, ~50 minutes at 80% capacity) is the most realistic estimate for smooth indoor campus floors and is used as the reference runtime in the project report. The low-load and high-load scenarios define the realistic operating range.*

6. Scenario 2 — Stair Climbing (Worst-Case Phase)

During stair climbing the motors run sequentially — never all at once. The worst-case current phase is the middle box lift (**Phase 3**) where 4 wheel motors and 2 gantry motors run simultaneously. The other 4 wheel motors and 2 gantry motors are inactive during this phase.

The same three wheel motor current scenarios are applied, combined with the confirmed gantry motor current of 1.636 A per motor at the max efficiency operating point.

$$I_{\text{stair}} = (4 \times I_{\text{wheel}}) + (2 \times I_{\text{gantry}}) + I_{\text{electronics}}$$

Gantry motor current during lift: 1.636 A per motor (from Pro-Range 100 RPM motor datasheet, max efficiency point) 2 gantry motors active: $2 \times 1.636 \text{ A} = 3.272 \text{ A}$

Table A3.6: Stair climbing worst-case current at all load scenarios

Load Scenario	4 Wheel Motors	2 Gantry Motors	Electronics	Grand Total
Low load	$0.5 \text{ A} \times 4 = 2.000 \text{ A}$	3.272 A	0.359 A	5.637 A
Mid load	$1.0 \text{ A} \times 4 = 4.000 \text{ A}$	3.272 A	0.359 A	7.637 A
High load	$1.5 \text{ A} \times 4 = 6.000 \text{ A}$	3.272 A	0.359 A	9.637 A

Note: In practice, motors do not run at full load during all phases simultaneously. The values above represent the worst-case phase (Phase 3 middle box lift). Translation phases (Phases 2, 4, 6) use only wheel motors, and are typically less demanding than flat terrain since fewer motors are active.

Table A3.7: Stair climbing continuous runtime — all load scenarios — both capacity cases

Load Scenario	Total Current	Case A: 80% Capacity (7,040 mAh)	Case B: Full Capacity (8,800 mAh)
Low load (0.5 A/motor)	5,637 mA	~75 minutes	~94 minutes
Mid load (1.0 A/motor)	7,637 mA	~55 minutes	~69 minutes
High load (1.5 A/motor)	9,637 mA	~44 minutes	~55 minutes

Note: These runtimes represent continuous stair climbing with no flat terrain operation. In real campus deployment the robot alternates between flat terrain and stairs — the combined runtime is shown in Section 7.

7. Combined Real-World Runtime (70% Flat + 30% Stair)

A realistic campus delivery combines both flat terrain navigation and stair climbing. Based on the reasonable assumption that the robot spends 70% of operating time on flat corridors and 30% on stairs, the combined effective current and runtime are calculated using a weighted average.

$$I_{\text{combined}} = (I_{\text{flat}} \times 0.70) + (I_{\text{stair}} \times 0.30)$$

$$\text{Runtime} = \text{Capacity} \div I_{\text{combined}}$$

Table A3.8: Combined runtime — 70% flat + 30% stairs — all load scenarios

Load Scenario	I _{flat}	I _{stair}	I _{combined}	Case A Runtime (7,040 mAh)	Case B Runtime (8,800 mAh)
Low load	4.365 A	5.637 A	4.747 A	~89 min	~111 min
Mid load	8.365 A	7.637 A	8.147 A	~52 min	~65 min
High load	12.365 A	9.637 A	11.547 A	~37 min	~46 min

Note: The mid-load combined runtime of ~52 minutes (Case A, 80% capacity) is the reference value used in Section 6.7 of the project report. This is equivalent to 4–5 campus delivery cycles of approximately 10 minutes each (200 m round trip including stair climbing). The low-load and high-load values bracket the realistic operating range.

8. Battery Management System (BMS) Adequacy Check

This section verifies that the total system current at all scenarios remains within the safe ratings of the BMS.

Table A3.9: BMS adequacy check — all scenarios

Scenario	Total Current	vs BMS Continuous (25 A)	vs BMS Instantaneous (35 A)	Status
Flat terrain — low load	4.4 A	17.6% utilised	12.6% utilised	✓ Safe
Flat terrain — mid load	8.4 A	33.6% utilised	24.0% utilised	✓ Safe
Flat terrain — high load	12.4 A	49.6% utilised	35.4% utilised	✓ Safe
Stair climbing — low load	5.6 A	22.5% utilised	16.1% utilised	✓ Safe
Stair climbing — mid load	7.6 A	30.5% utilised	21.8% utilised	✓ Safe
Stair climbing — high load	9.6 A	38.5% utilised	27.5% utilised	✓ Safe

All scenarios — BMS result	Max 12.4 A	49.6% max	35.4% max	✓ ALL SAFE
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Note: At all current load scenarios the total system current remains well within the BMS continuous rating of 25 A. The worst case (flat terrain high load at 12.4 A) uses only 49.6% of the BMS continuous capacity. The BMS is correctly sized and not a constraint at any realistic operating condition.

9. Battery Cell Discharge Rate Check

Each cell must not be discharged faster than its maximum rated current. The 3S4P configuration means current is shared equally across 4 parallel cell strings:

$$\text{Current per cell} = \text{Total current} \div 4 \text{ parallel cells}$$

At worst-case flat terrain high load (12.4 A): $12.4 \div 4 = 3.1 \text{ A per cell}$

Orange ICR18650-22C maximum rated discharge current = **4.0 A per cell**

Cell utilisation at worst case: $3.1 \text{ A} \div 4.0 \text{ A} = 77.5\% \text{ of maximum } \checkmark$

Note: All current scenarios result in cell discharge rates below the maximum rated 4.0 A per cell. The cells are operating within their safe discharge envelope at all realistic operating conditions.

10. Complete Summary — Battery Runtime Analysis

Table A3.10: Complete runtime summary — Calculation 3

Operating Scenario	Current Scenario	Case A Runtime 80% Capacity	Case B Runtime Full Capacity
Flat terrain only	Low load (0.5 A/motor)	~97 min	~121 min
Flat terrain only	Mid load (1.0 A/motor)	~50 min ✓ Reference	~63 min
Flat terrain only	High load (1.5 A/motor)	~34 min	~43 min
Stair climbing only	Low load (0.5 A/motor)	~75 min	~94 min
Stair climbing only	Mid load (1.0 A/motor)	~55 min	~69 min

Stair climbing only	High load (1.5 A/motor)	~44 min	~55 min
Combined (70% flat + 30% stair)	Low load (0.5 A/motor)	~91 min	~114 min
Combined (70% flat + 30% stair)	Mid load (1.0 A/motor)	~52 min ✓ Reference	~65 min
Combined (70% flat + 30% stair)	High load (1.5 A/motor)	~37 min	~46 min
BMS status — all scenarios	Max 12.4 A	Within 25 A continuous rating	✓ Confirmed

11. Conclusions

1. Runtime range established across current uncertainty: Because exact motor current was not measured during testing, the calculation presents a full range from low-load to high-load scenarios. The realistic combined runtime lies between 37 and 91 minutes depending on floor conditions and robot loading. The mid-load estimate of ~52 minutes (**Case A**) is used as the reference value in the project report.

2. 80% usable capacity is the operating recommendation: Case A (7,040 mAh at 80% discharge) is recommended for regular operation to preserve lithium cell life. Case B (full 8,800 mAh discharge) is the theoretical maximum and should be used only when maximum runtime is essential.

3. BMS is adequately sized: At all current scenarios the total system current remains within the BMS 25 A continuous rating. Maximum utilisation is 49.6% at the high-load flat terrain scenario.

4. Cell discharge rate is within safe limits: At the worst-case scenario, each cell discharges at 3.1 A — 77.5% of the 4.0 A maximum rated current. Cells are operating safely at all conditions.

5. Battery pack is adequately sized: The 3S4P 8,800 mAh lithium-ion pack provides 34–97 minutes of flat terrain operation and 44–75 minutes of continuous stair climbing. For typical campus delivery operation (4–5 deliveries of ~10 minutes each), the mid-load runtime of ~52 minutes is sufficient.

Appendix A — Design Calculations

Calculation 4: Payload Capacity Analysis

Belt-Limited Capacity | All Three Phases | Breaking Strength and Working Load Cases

1. Purpose of This Calculation

This calculation determines the maximum payload capacity of the AUTOM8 robot — the maximum additional weight the robot can carry in its payload compartment while still successfully completing the stair-climbing sequence. The calculation establishes which component in the gantry drive chain is the real limiting factor for payload capacity.

Payload capacity is governed by the weakest link in the gantry lifting chain. There are two candidates: the gantry motor torque and the GT2 timing belt tensile strength. This calculation first checks which governs, then determines the maximum liftable payload for each stair-climbing phase.

2. Confirmed Input Values

Table A4.1: Input values — Calculation 4

Parameter	Value	Source
Gantry motor rated torque (12V)	103 N-cm = 1.030 N-m	Robu.in datasheet, SKU 5726 — confirmed Calculation 1
Pulley pitch-line radius (r)	6.37 mm = 0.00637 m	Derived in Calculation 1, Step 1
GT2 belt breaking strength	80 N per belt	Standard published value for GT2 6mm open rubber timing belt
GT2 belt working load	50 N per belt	Recommended continuous working load (safety factor ~1.6× over breaking strength)
Front box assembled weight	2.717 kg	Chapter 3, Table 3.3
Middle box assembled weight	6.623 kg	Chapter 3, Table 3.4
Back box assembled weight	2.707 kg	Chapter 3, Table 3.5
Phase 1: motors active	2 motors (front-facing pair)	Chapter 4, Table 4.6

Phase 3: motors active	4 motors (all four)	Chapter 4, Table 4.6
Phase 5: motors active	2 motors (rear-facing pair)	Chapter 4, Table 4.6
Standard gravity	$g = 9.81 \text{ m/s}^2$	Standard physical constant

Note: The GT2 belt strength values (80 N breaking, 50 N working load) are standard published values for GT2 6mm rubber timing belts used across multiple manufacturer specifications. The working load of 50 N represents the load the belt can sustain continuously without fatigue damage, with approximately 1.6× margin over breaking strength. These values are conservative for a fibreglass-reinforced GT2 belt and are used as such in this calculation.

3. Step 1 — Identifying the Real Limiting Factor

Before calculating payload capacity, it is essential to identify whether the gantry motor torque or the GT2 belt tensile strength is the governing constraint. If the motor can generate more belt force than the belt can handle, then the belt governs — not the motor.

The maximum force each motor can generate on the belt is:

$$F_{\text{motor}} = T_{\text{rated}} \div r = 1.030 \text{ N}\cdot\text{m} \div 0.00637 \text{ m} = 161.7 \text{ N per motor}$$

Comparing this to the belt strength limits:

Table A4.2: Motor force vs belt strength — limiting factor identification

Parameter	Value	Comparison	Conclusion
Motor force per belt (rated torque, 12V)	161.7 N	vs Breaking strength: 80 N	161.7 N >> 80 N → Belt governs
Motor force per belt (rated torque, 10V)	134.8 N	vs Breaking strength: 80 N	134.8 N >> 80 N → Belt governs
Motor force per belt (rated torque, 9V)	121.3 N	vs Breaking strength: 80 N	121.3 N >> 80 N → Belt governs

Note: This is a critical finding. An earlier version of this calculation (Group 2 non-final draft) incorrectly used motor torque as the payload limit, yielding a theoretical payload of ~30 kg. That figure ignores the belt strength entirely and is not physically valid. The correct payload limit based on belt strength is calculated below.

Key finding: At every operating voltage the motor can generate more than twice the belt breaking strength. The GT2 timing belt tensile strength is therefore the governing constraint for payload capacity at all operating conditions. The motor torque is not the limiting factor.

4. Step 2 — Maximum Lifiable Mass from Belt Strength

Since the belt governs, the maximum lifting force is fixed at the belt limit regardless of voltage. Two belt cases are calculated:

- **Belt Case A — 80 N Breaking Strength:** The absolute maximum force the belt can withstand momentarily before failure. This represents an emergency upper bound — sustained operation at breaking strength will cause belt damage.
- **Belt Case B — 50 N Working Load:** The force the belt can sustain continuously without fatigue damage. This is the correct value for sustained stair-climbing operation and is used as the official payload capacity limit.

Maximum total liftable mass per phase:

$$m_{\text{max}} = (F_{\text{belt}} \times N_{\text{motors}}) \div g$$

Where F_{belt} = belt force limit per motor (N), N_{motors} = number of motors active, $g = 9.81 \text{ m/s}^2$.

Belt Case A — 80 N Breaking Strength

Phase 1 (2 motors): $m_{\text{max}} = (80 \times 2) \div 9.81 = 160 \div 9.81 = \mathbf{16.31 \text{ kg maximum liftable mass}}$

Phase 3 (4 motors): $m_{\text{max}} = (80 \times 4) \div 9.81 = 320 \div 9.81 = \mathbf{32.62 \text{ kg maximum liftable mass}}$

Phase 5 (2 motors): $m_{\text{max}} = (80 \times 2) \div 9.81 = 160 \div 9.81 = \mathbf{16.31 \text{ kg maximum liftable mass}}$

Belt Case B — 50 N Working Load (Correct Operating Limit)

Phase 1 (2 motors): $m_{\text{max}} = (50 \times 2) \div 9.81 = 100 \div 9.81 = \mathbf{10.19 \text{ kg maximum liftable mass}}$

Phase 3 (4 motors): $m_{\text{max}} = (50 \times 4) \div 9.81 = 200 \div 9.81 = \mathbf{20.39 \text{ kg maximum liftable mass}}$

Phase 5 (2 motors): $m_{\text{max}} = (50 \times 2) \div 9.81 = 100 \div 9.81 = \mathbf{10.19 \text{ kg maximum liftable mass}}$

5. Step 3 — Payload Capacity (Subtract Box Weight)

The payload capacity per phase is the maximum liftable mass minus the box's own assembled weight. The payload sits on the robot during all phases, so the governing payload capacity is the minimum across all three phases.

$$\text{Payload capacity (phase)} = m_{\text{max}} - m_{\text{box}}$$

Table A4.3: Payload capacity by phase — Belt Case A (80 N breaking strength)

Phase	Motors	Box Weight	m_max (80 N)	Payload Capacity
Phase 1	2	2.717 kg	16.31 kg	13.59 kg
Phase 3	4	6.623 kg	32.62 kg	25.99 kg
Phase 5	2	2.707 kg	16.31 kg	13.60 kg
Governing Payload — Case A	Min of above	Phase 1 or 5	—	13.59 kg

Note: Belt Case A (80 N breaking strength) represents the absolute theoretical maximum payload. Operating at this limit risks belt failure under sustained loading and is not recommended for regular operation.

Table A4.4: Payload capacity by phase — Belt Case B (50 N working load) — Official Operating Limit

Phase	Motors	Box Weight	m_max (50 N)	Payload Capacity
Phase 1	2	2.717 kg	10.19 kg	7.47 kg
Phase 3	4	6.623 kg	20.39 kg	13.77 kg
Phase 5	2	2.707 kg	10.19 kg	7.48 kg
Governing Payload — Case B	Min of above	Phase 1 or 5	—	7.47 kg

Official stated payload capacity: Under 7 kg. This is the value stated in the AUTOM8 project report throughout. It is derived from Belt **Case B** (50 N working load) and rounded down conservatively from the calculated 7.47 kg for a clean, safe, stated specification.

6. Why Phase 1 and Phase 5 Govern Despite 4 Motors in Phase 3

The payload is carried on the middle box platform throughout all phases. However, the governing payload constraint is **Phase 1** and **Phase 5** (2 motors each), not **Phase 3** (4 motors), for the following reason:

During **Phase 1**, the front box is being lifted. The front box does not carry the payload — the payload stays on the middle box which remains grounded. However, in a conservative worst-case analysis, the calculation above treats the full payload as adding to the lifted box in each phase. This is a deliberate conservative assumption.

In the actual physical arrangement, the analysis is still correct because:

Phase 1 and 5: 2 motors lift the outer boxes (2.717 and 2.707 kg respectively) against a belt limit of 50 N each $\times 2 = 100$ N total. The box weight alone (2.717 kg = 26.65 N) uses only 26.7% of the available belt force, leaving 73.3% for payload. Maximum additional payload liftable = 7.47 kg.

Phase 3: 4 motors lift the middle box (6.623 kg) against a belt limit of 50 N each $\times 4 = 200$ N total. The middle box weight (6.623 kg = 64.97 N) uses 32.5% of available belt force. This phase has proportionally more margin because 4 motors are active.

The governing constraint is therefore always **Phase 1** and **Phase 5** where only 2 motors are active. The payload limit of 7.47 kg is set by these phases.

7. Flat Terrain Traction Verification With Payload

This section confirms that the wheel motors can move the robot with payload on flat ground. This was fully established in **Calculation 2**, but the key result is restated here for completeness.

Total robot weight with 7.47 kg payload: $12.1 + 7.47 = \mathbf{19.57\text{ kg}}$

Rolling resistance with 7.47 kg payload: $0.02 \times 19.57 \times 9.81 = \mathbf{3.84\text{ N}}$

Available traction force (8 motors at 9V minimum): **99.0 N** (from Calculation 2)

Traction safety factor with 7.47 kg payload at 9V: $99.0 \div 3.84 = \mathbf{25.8 \times \checkmark}$

***Note:** Flat terrain traction remains non-constraining even at the theoretical maximum payload of 7.47 kg. The wheel motors are not the limiting factor for payload in any scenario.*

8. Complete Summary — Payload Capacity

Table A4.5: Complete payload capacity summary — Calculation 4

Belt Case	Belt Limit per Motor	Governing Phase	Governing Payload	Status
Case A — Breaking Strength	80 N	Phase 1 and 5 (2 motors)	13.59 kg	Theoretical absolute maximum — not for regular use
Case B — Working Load	50 N	Phase 1 and 5 (2 motors)	7.47 kg	Correct operating limit for sustained stair climbing
Official stated capacity	50 N working load	All phases confirmed	Under 7 kg	As stated throughout project report ✓

9. Conclusions

1. Belt strength is the governing constraint: At all operating voltages (12V, 10V, 9V) the gantry motors generate forces (121–162 N per motor) that significantly exceed the GT2 belt breaking strength (80 N). The belt tensile strength, not motor torque, sets the payload capacity limit.

2. Official payload capacity is under 7 kg: Based on the GT2 belt working load of 50 N and the governing **Phase 1** and **Phase 5** (2 motors each), the calculated payload capacity is 7.47 kg. This is rounded down and stated as “under 7 kg” throughout the project report as a clean, conservative, and safely stated specification.

3. Phase 1 and Phase 5 are the governing phases: These phases use only 2 gantry motors and are therefore more constrained than **Phase 3** (4 motors). The minimum belt force available is $2 \times 50 \text{ N} = 100 \text{ N}$, limiting liftable mass to 10.19 kg and payload to 7.47 kg after subtracting box weight.

4. Flat terrain traction confirmed with full payload: The wheel motors provide $25.8\times$ traction safety factor even with the theoretical maximum 7.47 kg payload. Wheel drive is not a constraint at any payload level.

5. Payload testing on stairs was not conducted: All stair climbing demonstrations were conducted with the robot at its base weight of 12.1 kg without payload. The payload capacity stated here is a theoretical calculation based on belt strength limits and has not been experimentally verified under stair climbing conditions. This is acknowledged as a limitation in **Chapter 3 Section 3.7** and **Chapter 6 Section 6.8** of the project report.

Appendix A — Design Calculations

Calculation 5: Stair Geometry Compatibility Derivation

Tread Depth Range | Riser Height Range | NBC IS 1644 Verification

1. Purpose of This Calculation

This calculation establishes and justifies the compatible stair geometry range for AUTOM8 — the range of stair tread depths and riser heights the robot can successfully climb. The compatible range is summarised in **Chapter 5, Table 5.1** of the project report. This appendix provides the detailed derivation and physical basis for each boundary value.

The compatible range has four boundary values:

- **Minimum tread depth: 280 mm** — Established by prototype testing (see **Section 4**)
- **Maximum tread depth: 400 mm** — Derived from sensor detection geometry and robot length (see **Section 5**)
- **Minimum riser height: 150 mm** — Practical lower limit; any riser below this is trivially climbable by the wheel step-up (see **Section 6**)
- **Maximum riser height: 220 mm** — Derived from gantry usable rail travel distance (see **Section 6**)

2. Confirmed Input Values

Table A5.1: Input values — Calculation 5

Parameter	Value	Source
Front box length	80 mm	Chapter 3, confirmed design dimension
Middle box length	200 mm	Chapter 3, confirmed design dimension
Back box length	80 mm	Chapter 3, confirmed design dimension

Inter-box gap (each side)	5 mm	Chapter 3, Section 3.2 (assembly dimension)
Total robot length	$80 + 200 + 80 + (2 \times 5) = 370$ mm	Chapter 5, Table 5.1
Wheel diameter	68 mm	Chapter 3, Section 3.4 (confirmed value)
Wheel radius	34 mm = 0.034 m	Wheel diameter $\div$ 2
HC-SR04 sensing range	2 – 400 cm	Chapter 4, Table 4.7
HC-SR04 beam angle	15°	Chapter 4, Table 4.7
Stair detection trigger distance	3 cm = 30 mm	Chapter 4, Table 4.7 and Chapter 5, Section 5.2
V-slot rail height (usable travel)	360 mm (36 cm profile)	Chapter 3, Section 3.2
Gantry carriage start clearance	~50 mm (bottom of rail)	Chapter 3, Section 3.3 (assembly constraint)
Gantry usable lift distance	~260 mm (rail height minus start clearance)	Derived in Section 6
NBC minimum tread (educational)	300 mm	National Building Code of India, IS 1644
NBC maximum riser (educational)	150 mm	National Building Code of India, IS 1644

3. Overview of Compatibility Constraints

The stair geometry compatibility range is determined by three independent physical constraints of the AUTOM8 design. Each constraint sets one or more boundary values of the compatible range. Understanding each constraint independently provides a clear engineering basis for the stated compatibility range.

Constraint 1 — Minimum tread depth (280 mm): The back box must maintain stable wheel contact on the lower stair tread while the front box is being lifted during **Phase 1**. If the tread is too short, the back wheel approaches or overhangs the stair edge during **Phase 1**, causing instability. This boundary is established by prototype testing.

Constraint 2 — Maximum tread depth (400 mm): The back box stair detection sensor must be able to detect the next riser face before the **Phase 2** forward translation places the back box too far from the riser to trigger **Phase 3**. This boundary is derived from the sensor detection geometry and robot length.

Constraint 3 — Riser height range (150–220 mm): The gantry mechanism must be able to lift each box by the full riser height plus the wheel radius. The usable gantry travel distance sets the maximum riser height. The minimum is a practical lower bound below which any wheeled robot can simply roll over the step.

4. Minimum Tread Depth: 280 mm — Prototype Validated

The minimum compatible tread depth of 280 mm was established through physical prototype testing on staircases of varying tread depth during the development of AUTOM8.

The critical stability requirement is: when the front box has been lifted during **Phase 1** and the robot is preparing to drive forward, the back box must have both its wheels firmly on the lower stair tread with a minimum clearance from the tread edge. If this clearance is insufficient, the back box becomes unstable and risks tilting or the wheel losing traction.

During testing, the robot was operated on staircases with tread depths ranging from approximately 270 mm to 400 mm. At 280 mm tread depth, the back wheel maintained stable contact with adequate clearance from the stair edge throughout the **Phase 1** lift and the subsequent **Phase 2** forward translation. At tread depths approaching 270 mm or below, instability was observed as the back wheel approached the stair edge during **Phase 1**.

Minimum compatible tread depth: 280 mm — empirically validated on Parul University campus staircase during prototype testing.

*Note: This boundary is established by direct prototype observation rather than geometric derivation, because the stability margin depends on several interacting factors including the robot's weight distribution, the gantry tilt behaviour during **Phase 1**, and the actual wheel contact geometry. The 280 mm limit is validated as the tested lower bound for safe stair climbing operation.*

5. Maximum Tread Depth: 400 mm — Derived from Sensor Detection Geometry

The maximum compatible tread depth is derived from the geometry of the sensor-triggered stair detection system. For the stair-climbing sequence to operate correctly in sensor-triggered mode, each box's HC-SR04 stair detection sensor must be able to detect the next stair riser face before the corresponding lift phase can be triggered.

The back box sensor (at the bottom-front of the back box) must detect the riser face of the next step to trigger **Phase 5**. For this detection to occur, the back box sensor must be within the 3 cm trigger distance of the riser face when the robot is in the correct position for Phase 5 to begin.

Derivation of Maximum Tread

When the front and middle boxes have completed their lift and forward translation phases (**Phases 1 through 4**), the robot is positioned with the front and middle boxes on the upper tread and the back box on the lower tread. At this point, the back box sensor faces the riser of the next step.

The maximum tread is reached when the back box sensor is at exactly the 3 cm trigger distance from the riser face when the back box is at its most rearward position on the lower tread. Using the robot geometry:

$$\text{Maximum tread} = \text{Robot total length} + \text{Trigger distance}$$

$$\text{Maximum tread} = 370 \text{ mm} + 30 \text{ mm} = 400 \text{ mm}$$

Result: Maximum compatible tread depth = 400 mm

*Note: Beyond 400 mm tread depth, the back box sensor would not reach within the 3 cm trigger distance of the riser face before Phase 4 translation is complete. The sequence would stall waiting for a trigger that never arrives. Note that in the final timing-based deployment on the Parul University campus (fixed known stair dimensions), this sensor detection geometry is not the active constraint — the sequence is time-controlled. The 400 mm limit applies to the sensor-triggered operating mode described in **Chapter 5**.*

6. Riser Height Range: 150–220 mm — Derived from Gantry Rail Travel

The compatible riser height range is determined by the usable travel distance of the gantry lifting mechanism. The gantry must lift each box by the full riser height plus the wheel radius (34 mm) to ensure the wheel clears the top of the riser and lands on the upper tread surface.

$$\text{Required lift distance} = \text{Riser height} + \text{Wheel radius} = \text{Riser} + 34 \text{ mm}$$

Gantry Usable Travel Distance

The V-slot vertical rails are 360 mm (36 cm) aluminium extrusion pieces. The gantry carriage cannot travel the full 360 mm because the carriage plate requires clearance at both the bottom and top of the rail for the V-wheels to remain engaged. Based on the carriage geometry and assembly:

Usable gantry travel distance: **approximately 260 mm** (360 mm rail height minus approximately 50 mm bottom clearance and 50 mm top clearance for carriage engagement)

Maximum Riser Height

The maximum riser height is the usable gantry travel minus the wheel radius:

$$\text{Maximum riser} = \text{Usable travel} - \text{Wheel radius} = 260 \text{ mm} - 34 \text{ mm} \approx 226 \text{ mm}$$

Rounded down conservatively to: **220 mm maximum riser** — to provide a safe margin within the usable rail travel.

Verification: Required lift for 220 mm riser = $220 + 34 = 254$ mm. Available usable travel = 260 mm.

Clearance = 6 mm. Confirmed within gantry capability. ✓

Minimum Riser Height

The minimum riser height of 150 mm is a practical lower bound rather than a mechanical constraint. The gantry can lift boxes over risers smaller than 150 mm. However, risers below approximately 150 mm can typically be negotiated by the wheel step-up approach alone — the wheel simply rolls over the step without requiring gantry assistance.

Minimum riser height: **150 mm** — practical lower bound, not a mechanical constraint of the gantry system.

Verification: Required lift for 150 mm riser = $150 + 34 = 184$ mm. Available usable travel = 260 mm.

Clearance = 76 mm. Well within gantry capability. ✓

7. NBC IS 1644 Standard Verification

The National Building Code of India (NBC IS 1644) specifies the following dimensional requirements for staircases in educational buildings:

NBC minimum tread depth: **300 mm**

NBC maximum riser height: **150 mm**

The NBC also specifies the comfort rule for staircase proportions: $2R + T$ should fall between 550 mm and 700 mm, where R is riser height and T is tread depth.

Verification for NBC educational standard (300 mm tread, 150 mm riser):

$$2R + T = (2 \times 150) + 300 = 300 + 300 = 600 \text{ mm}$$

600 mm is within the NBC comfort range of 550–700 mm. ✓

NBC tread 300 mm: within AUTOM8 range 280–400 mm. ✓

NBC riser 150 mm: within AUTOM8 range 150–220 mm. ✓

Result: The NBC IS 1644 educational building standard is fully within the AUTOM8 compatible stair geometry range.

8. Complete Summary — Compatible Stair Geometry

Table A5.2: Complete stair geometry compatibility summary — Calculation 5

Parameter	AUTOM8 Range	Basis	NBC Educational Standard	NBC Within Range?
Minimum tread depth	280 mm	Prototype validated on campus staircase	300 mm	✓ Yes
Maximum tread depth	400 mm	Derived: robot length (370 mm) + trigger distance (30 mm)	300 mm	✓ Yes
Minimum riser height	150 mm	Practical lower bound — gantry not limiting	150 mm	✓ Yes (boundary)
Maximum riser height	220 mm	Derived: gantry usable travel (260 mm) – wheel radius (34 mm), rounded down	150 mm	✓ Yes
NBC Standard Result	300 mm tread / 150 mm riser	$2R+T = 600\text{ mm}$ ✓	IS 1644 confirmed	Fully within range ✓

9. Conclusions

1. Minimum tread (280 mm) is prototype-validated: The minimum tread depth of 280 mm was established through physical testing on the campus staircase. This is an empirically confirmed lower bound, not a geometric derivation, and is documented as such.

2. Maximum tread (400 mm) is geometry-derived: The 400 mm maximum tread is derived from the robot total length (370 mm) and the HC-SR04 trigger distance (30 mm). Beyond this, the back box sensor cannot reach the next riser face in sensor-triggered mode.

3. Riser range (150–220 mm) is gantry-derived: The maximum riser of 220 mm is derived from the gantry usable travel of approximately 260 mm minus the wheel radius of 34 mm. The minimum of 150 mm is a practical lower bound.

4. NBC IS 1644 educational standard is fully within the AUTOM8 range: The standard tread of 300 mm and riser of 150 mm for educational buildings both fall within the AUTOM8 compatible range. AUTOM8 is compatible with all standard Indian educational building staircases.

5. The compatible range exceeds the NBC standard: AUTOM8 can climb staircases that do not strictly comply with NBC — tread depths from 280 to 400 mm and risers from 150 to 220 mm. This covers the variation found in existing Indian campus buildings where staircase dimensions may differ from the current code.

Appendix A — Design Calculations

Calculation 6: Gantry Lift Speed, Flat Ground Speed, and Phase Timing Analysis

Five PWM Cases | Three Riser Heights | Speed vs Stability Trade-off Analysis

1. Purpose of This Calculation

This calculation determines the gantry belt linear speed, flat ground navigation speed, and theoretical stair-climbing phase duration for five different PWM (Pulse Width Modulation) duty cycle settings. The purpose is to characterise the full operating speed range of AUTOM8 and to understand the trade-off between climbing speed and operational stability at each PWM setting.

Because the exact PWM value used in the final prototype firmware was not recorded in the testing documentation, this analysis presents calculations across five representative PWM cases covering the practical operating range. This allows the operating characteristics to be understood at any PWM setting, and the actual firmware value to be identified by matching observed prototype behaviour to the corresponding calculated parameters.

The five PWM cases analysed are:

- **60% PWM (153/255):** Low speed, very controlled operation. Maximum stability, minimum climbing speed.
- **70% PWM (178/255):** Conservative speed providing stable MPU-6050 heading correction and reliable obstacle detection response.
- **75% PWM (191/255):** Moderate speed, good balance between stability and climbing efficiency.
- **80% PWM (204/255):** Fast operation, efficient stair climbing. Higher risk of stair nosing contact at this speed.
- **85% PWM (217/255):** High speed operation. Observed during testing to cause stair nosing contact issues during **Phase 2** translation.

2. Confirmed Input Values

Table A6.1: Input values — Calculation 6

Parameter	Value	Source
Gantry motor RPM at max efficiency	78.9 RPM	Robu.in Pro-Range 100 RPM motor datasheet
GT2 pulley teeth	20 teeth	Confirmed hardware, Chapter 5 Section 5.3.3
GT2 belt pitch	2 mm per tooth	GT2 standard specification
Belt travel per motor revolution	$20 \times 2 = 40$ mm	Derived from teeth and pitch
Wheel motor RPM at max efficiency	166.1 RPM	Robu.in Pro-Range 200 RPM motor datasheet
Wheel diameter (confirmed)	68 mm	Chapter 3, Section 3.4
Wheel circumference	$\pi \times 68 = 213.6$ mm	Derived from confirmed diameter
Wheel radius	34 mm	Wheel diameter $\div$ 2
Minimum riser height	150 mm	Chapter 5, Table 5.1
Typical campus riser height	185 mm	Observed on Parul University campus staircase
Maximum riser height	220 mm	Chapter 5, Table 5.1
Required lift per phase	Riser height + 34 mm (wheel radius)	Chapter 5, Section 5.4
PWM scaling relationship	$\text{RPM}(\text{PWM}\%) = \text{RPM_max_eff} \times (\text{PWM}\% \div 100)$	Standard linear DC motor assumption

***Note:** PWM scales motor speed linearly — at 80% PWM the motor runs at 80% of its maximum efficiency 4RPM. This is a standard approximation for DC motors under consistent load conditions. The max efficiency RPM (not no-load RPM) is used as the operating point because the motor operates under load during both stair climbing and flat terrain navigation.*

3. Core Formulas

3.1 Gantry Belt Linear Speed

The gantry motor spins the GT2 pulley which drives the timing belt. Belt linear speed is:

$$v_{\text{belt}} \text{ (mm/s)} = (\text{RPM}_{\text{gantry}} \times \text{PWM}\%) \div 60 \times \text{Belt_travel_per_rev}$$

$$v_{\text{belt}} \text{ (mm/s)} = (78.9 \times \text{PWM}\%) \div 60 \times 40 \text{ mm}$$

3.2 Gantry Lift Phase Duration

Each lift phase must raise the box by riser height plus wheel radius. The theoretical phase duration is:

$$t_{\text{lift}} \text{ (s)} = \text{Lift_distance} \div v_{\text{belt}} = (\text{Riser} + 34 \text{ mm}) \div v_{\text{belt}}$$

3.3 Flat Ground Wheel Speed

Wheel linear speed converts motor RPM to forward robot speed through the wheel circumference:

$$v_{\text{wheel}} \text{ (m/s)} = (\text{RPM_wheel} \times \text{PWM}\%) \div 60 \times \text{Wheel_circumference} \div 1000$$

$$v_{\text{wheel}} \text{ (m/s)} = (166.1 \times \text{PWM}\%) \div 60 \times 0.2136$$

3.4 Total Time per Stair Step

One complete stair step requires 3 lift phases and 3 drive phases. The robot advances approximately 200 mm (middle box length) during each drive phase:

$$t_{\text{step}} = (3 \times t_{\text{lift}}) + (3 \times t_{\text{drive}})$$

$$t_{\text{drive}} \text{ (s)} = 200 \text{ mm} \div v_{\text{wheel}} \text{ (mm/s)}$$

4. Gantry Belt Speed at Each PWM Case

Table A6.2: Gantry belt linear speed at each PWM case

PWM Case	PWM Value	Gantry RPM at this PWM	Belt Speed (mm/s)	Belt Speed (cm/s)
60% PWM (153/255)	153/255	47.3 RPM	31.56 mm/s	3.16 cm/s
70% PWM (178/255)	178/255	55.2 RPM	36.82 mm/s	3.68 cm/s
75% PWM (191/255)	191/255	59.2 RPM	39.45 mm/s	3.95 cm/s
80% PWM (204/255)	204/255	63.1 RPM	42.08 mm/s	4.21 cm/s
85% PWM (217/255)	217/255	67.1 RPM	44.71 mm/s	4.47 cm/s

5. Gantry Lift Phase Duration by PWM and Riser Height

Each lift phase raises a box by the riser height plus wheel radius (34 mm). The following table shows theoretical phase duration in seconds for all five PWM cases and three riser heights.

Table A6.3: Theoretical gantry lift phase duration (seconds) — all PWM cases and riser heights

PWM Case	Riser 150 mm (Lift = 184 mm)	Riser 185 mm (Lift = 219 mm)	Riser 220 mm (Lift = 254 mm)	Stability Note
60% PWM (153/255)	5.8 s	6.9 s	8.0 s	Low speed — very controlled, slow
70% PWM (178/255)	5.0 s	5.9 s	6.9 s	Conservative — stable heading correction
75% PWM (191/255)	4.7 s	5.6 s	6.4 s	Moderate — good balance
80% PWM (204/255)	4.4 s	5.2 s	6.0 s	Fast — efficient operation
85% PWM (217/255)	4.1 s	4.9 s	5.7 s	High speed — risk of nosing collision

***Note:** These are theoretical minimum lift times assuming the motor maintains max efficiency RPM throughout the lift. In practice, actual lift time is slightly longer due to motor ramp-up, belt elasticity, and the additional brief continuation after the sensor trigger to ensure the wheel fully clears the riser. The prototype-observed phase time of approximately 4–5 seconds per phase corresponds to the 75–80% PWM range for a 185 mm typical campus riser.*

6. Flat Ground Navigation Speed by PWM Case

The same PWM cases are applied to the wheel motors for flat terrain navigation. Speed is calculated from the wheel motor max efficiency RPM scaled by PWM percentage, multiplied by wheel circumference.

Table A6.4: Flat ground navigation speed at each PWM case

PWM Case	Wheel RPM	Speed (m/s)	Speed (cm/s)	Speed (km/hr)	Suitability
60% PWM (153/255)	99.7 RPM	0.355 m/s	35.5 cm/s	1.28 km/hr	Very slow — overly cautious

70% PWM (178/255)	116.3 RPM	0.414 m/s	41.4 cm/s	1.49 km/hr	Stable navigation ✓
75% PWM (191/255)	124.6 RPM	0.444 m/s	44.4 cm/s	1.60 km/hr	Good balance ✓
80% PWM (204/255)	132.9 RPM	0.473 m/s	47.3 cm/s	1.70 km/hr	Fast — check obstacle response
85% PWM (217/255)	141.2 RPM	0.503 m/s	50.3 cm/s	1.81 km/hr	Risk of nosing collision

***Note:** For campus corridor navigation the robot speed should allow safe obstacle detection response. At 0.414 m/s (70% PWM) and HC-SR04 response time of approximately 60 ms, the stopping distance is approximately 2.4 cm — within the configured detection threshold. At higher speeds this margin reduces. The 70–75% PWM range provides the best balance of navigation speed and safe obstacle response.*

7. Total Time per Stair Step by PWM Case

Total stair step time combines 3 lift phases and 3 drive phases. Both gantry and wheel motors are assumed to operate at the same PWM percentage in this analysis. Drive phase distance is approximately 200 mm (middle box length) per translation phase.

Table A6.5: Total theoretical time per stair step — all PWM cases, typical 185 mm campus riser

PWM Case	t_lift per phase	t_drive per phase	Total per step (3 lift + 3 drive)	Assessment
60% PWM (153/255)	6.9 s	0.6 s	~23 s	~23s — very slow
70% PWM (178/255)	5.9 s	0.5 s	~19 s	~19s — controlled
75% PWM (191/255)	5.6 s	0.5 s	~18 s	~18s — good balance ✓
80% PWM (204/255)	5.2 s	0.4 s	~17 s	~17s — efficient ✓
85% PWM (217/255)	4.9 s	0.4 s	~16 s	~16s — nosing risk

***Note:** The prototype-observed step time of approximately 20–25 seconds per step corresponds to the 75–80% PWM range for a 185 mm riser. This is consistent with the total step time of approximately 21 seconds reported in Chapter 5 Section 5.7.5 of the project report.*

8. Speed vs Stability Trade-off Analysis

The following table summarises the key trade-offs at each PWM case to guide selection of the optimal operating speed:

Table A6.6: Speed vs stability trade-off summary

PWM Case	Flat Speed	Step Time (185mm riser)	Gantry Belt Speed	Nosing Risk	Heading Correction	Recommended Use
60% PWM	0.355 m/s	~23 s	3.14 cm/s	Very low	Excellent	Over-cautious for delivery
70% PWM	0.414 m/s	~19 s	3.67 cm/s	Low	Excellent ✓	Flat terrain navigation ✓
75% PWM	0.444 m/s	~18 s	3.93 cm/s	Low-Medium	Good ✓	Stair approach ✓
80% PWM	0.473 m/s	~17 s	4.19 cm/s	Medium	Acceptable	Stair climbing ✓
85% PWM	0.503 m/s	~16 s	4.46 cm/s	High ⚠	Marginal	Nosing collision risk

Based on this analysis the recommended operating strategy is:

- **Flat terrain navigation:** 70% PWM (178/255) — provides 0.414 m/s speed with excellent heading correction stability and safe obstacle detection response time.
- **Stair approach:** 75% PWM (191/255) — slightly higher speed ensures consistent stair detection triggering while maintaining acceptable stability.
- **Stair climbing (lift phases):** 80% PWM (204/255) — provides efficient lift speed with controlled box movement.
- **85% PWM and above:** Not recommended — stair nosing collision risk observed in prototype testing as documented in **Chapter 6 Section 6.3.3**.

9. Complete Summary — Speed and Timing Analysis

Table A6.7: Complete speed and timing summary — Calculation 6

Parameter	60% PWM	70% PWM	75% PWM	80% PWM	85% PWM
Gantry belt speed	3.14 cm/s	3.67 cm/s	3.93 cm/s	4.19 cm/s	4.46 cm/s
Flat ground speed	0.355 m/s	0.414 m/s	0.444 m/s	0.473 m/s	0.503 m/s
Flat ground speed (km/hr)	1.28 km/hr	1.49 km/hr	1.60 km/hr	1.70 km/hr	1.81 km/hr
Lift time — 150mm riser	3.89 s	3.33 s	3.11 s	2.92 s	2.74 s
Lift time — 185mm riser	4.63 s	3.97 s	3.70 s	3.47 s	3.26 s
Lift time — 220mm riser	5.37 s	4.60 s	4.29 s	4.02 s	3.79 s
Step time — 185mm riser	~23 s	~19 s	~18 s	~17 s	~16 s
Recommendation	Too slow	Flat terrain ✓	Stair approach ✓	Stair climb ✓	Nosing risk ⚠

10. Conclusions

1. Full operating range characterised: The calculation presents gantry belt speed, flat ground speed, and phase timing across five PWM cases (60% to 85%). This provides a complete picture of AUTOM8 speed characteristics without requiring knowledge of the exact firmware PWM value.

2. Optimal PWM range is 70–80%: The 70–80% PWM range provides the best balance of speed, stability, and safety. Below 70% the robot is unnecessarily slow for campus delivery. Above 80% the risk of stair nosing collision increases significantly.

- 3. Prototype-observed step time confirmed:** The observed stair step time of approximately 17-18 seconds (**Chapter 5 Section 5.7.5**) corresponds to the 75–80% PWM range for a 185 mm campus riser, consistent with the recommended operating range.
- 4. 68 mm wheel diameter used throughout:** All speed calculations use the confirmed wheel diameter of 68 mm (circumference 213.6 mm) as stated in **Chapter 3 Section 3.4**. This is consistent across all calculations in this appendix.
- 5. Flat ground speed is conservative:** At 70% PWM the robot navigates at 0.414 m/s (1.49 km/hr) — approximately 20% of normal human walking speed. This is intentionally conservative for safe indoor campus corridor operation with students and staff present.

Appendix B — Complete Limitations

Every Limitation | Impact | Solutions

Overview

This appendix documents all identified limitations of the AUTOM8 prototype. For each limitation, the impact on robot performance, practical solutions. The 22 limitations are organised into table.

All limitations documented here were identified through honest prototype testing and engineering analysis. Each limitation has a clearly identified solution, and none of them invalidates the core mechanical and control architecture of AUTOM8. The summary table of this appendix (**Table B.1**) shows them.

Summary Table — All 22 Limitations

Table B.1: Complete limitations summary — Appendix B

#	Limitation	Category	Severity	Fixable in Next Version
1	Frame joint loosening under vibration	Structural	Medium	Yes — thread lock + brushless motors
2	V-slot rail flex under gantry impact	Structural	Medium	Yes — steel linear rails
3	Exposed frame and wiring	Structural	Low	Yes — enclosure panels
4	Open loop stair climbing	Mechanical	High	Yes — encoders + limit switches
5	Box tilt during phase transitions	Mechanical	Medium	Yes — dual belt + tighter carriage
6	Heavy stair contact impact	Mechanical	Medium	Yes — deceleration profile
7	No stair descending capability	Mechanical	High	Yes — reverse phase sequence requires Significant redesign

8	Slow stair climbing — ~17- 21s per step	Mechanical	Medium	Yes — closed loop + phase overlap
9	No payload testing on stairs	Mechanical	High	Yes — incremental load testing
10	Stair nosing contact issue	Mechanical	Low	Yes — bevelled wheels + height adjust
11	IR sensor light and colour sensitivity	Sensing	Medium	Resolved — replaced by HC-SR04 (Stage 3)
12	Single encoder $\pm 10\%$ distance error	Sensing	High	Yes — dual encoder + quadrature
13	IMU gyroscope drift	Sensing	Medium	Yes — magnetometer + Kalman filter
14	Ultrasonic 15-degree narrow angle	Sensing	Medium	Yes — LIDAR upgrade
15	No rear or side sensing	Sensing	High	Yes — additional sensors or LIDAR
16	No closed-loop motor speed control	Control	High	Yes — encoder PID control
17	No wireless communication	Control	High	Yes — ESP32 WiFi module
18	Pre-programmed path only with low accuracy	Control	High	Yes — SLAM navigation
19	Single core processor latency	Control	Medium	Yes — dual processor architecture
20	No battery state-of- charge indicator	Power	Medium	Yes — fuel gauge IC
21	No hard battery enclosure	Power	High	Yes — 3D printed enclosure
22	Single BMS — single point of failure	Power	Medium	Yes — dual BMS architecture
TOTAL	22 limitations documented	5 categories	9 High / 10 Medium / 3 Low	All 22 addressable ✓

***Note:** Limitation #11 (IR sensor sensitivity) is marked as Resolved because it was identified during Stage 2 prototype development and corrected in Stage 3 by replacing IR sensors with HC-SR04 ultrasonic sensors. All other 21 limitations are present in the final prototype and are prioritised for the next design iteration. The 9 high-severity limitations represent the priority development roadmap.*

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